UNIVERSIDADE DE SÃO PAULO
Instituto de Ciências Matemáticas e de Computação

Asymptotics of Partition Functions of Coulomb Gases

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Dissertação de Mestrado do Programa de Pós-Graduação em Matemática (PPG-Mat)

ICMC USP
SÃO CARLOS

SERVIÇO DE PÓS-GRADUAÇÃO DO ICMC-USP

Data de Depósito:

Assinatura: ______________________

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Asymptotics of Partition Functions of Coulomb Gases

Master dissertation submitted to the Institute of Mathematics and Computer Sciences – ICMC-USP, in partial fulfillment of the requirements for the degree of the Master Program in Mathematics. *EXAMINATION BOARD PRESENTATION COPY*

Concentration Area: Mathematics

Advisor: Prof. Dr. Guilherme Lima Ferreira Silva

USP – São Carlos
July de 2026

Ficha catalográfica elaborada pela Biblioteca Prof. Achille Bassi
e Seção Técnica de Informática, ICMC/USP,
com os dados fornecidos pelo(a) autor(a)

Antonelli, Humberto Lidio
A634m Modelo de teses e dissertações em LaTeX do ICMC / Humberto Lidio Antonelli; orientadora Renata Pontin de Mattos Fortes. -- São Carlos, 2017.
79 p.

Tese (Doutorado - Programa de Pós-Graduação em Ciências de Computação e Matemática Computacional) -- Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, 2017.

1. Modelo. 2. Monografia de qualificação. 3. Dissertação. 4. Tese. 5. Latex. I. Fortes, Renata Pontin de Mattos, orient. II. Título.

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"En remontant chez moi pour y passer la soirée à travailler de mon mieux,
je me disais que le monde n'est pas construit pour l'équilibre.
Le monde est désordre.
L'équilibre n'est pas la règle, c'est l'exception.
Et je faisais le serment de travailler pour l'ordre et l'équilibre.
Que de serments! Tu vas sourire."
G.Duhamel, Maitres, 1937

ACKNOWLEDGEMENTS

For all of those who were before and are now:
Your courage, patience, daring, and caring has brought me here.
To hoping that I become deserving of such act, for I feel loved and dear; of immense luck.
For all of those who will be after:
To promise that I can for you as it was done for me.

"João, é bom ver que você aprendeu e fez tantas coisas bonitas!
Continue assim, muito feliz! Brinque e aprenda muito."
Te amo muito, Papai.

RESUMO

JOÃO V. A. PIMENTA. **Assintóticas de Funções Partição em Gases de Coulomb**. 2026. 97 p. Dissertação (Mestrado em Ciências – Matemática) – Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos – SP, 2026.

Este trabalho é um breve modelo para a escrita de monografias de qualificação, dissertações e teses utilizando o ambiente LaTeX, de acordo com as normas exigidas pelo Instituto de Ciências Matemáticas e de Computação (ICMC), da Universidade de São Paulo (USP). Para a confecção deste modelo foi utilizado a última versão (1.9.6) do pacote de classes *abnTeX2* que segue as normas da Associação Brasileira de Normas Técnicas. A elaboração de uma monografia, dissertação ou tese pode ser feita sobrescrevendo o conteúdo deste modelo.

Palavras-chave: Modelo, Monografia de qualificação, Dissertação, Tese, Latex.

ABSTRACT

JOÃO V. A. PIMENTA. **Asymptotics of Partition Functions of Coulomb Gases** . 2026. 97 p. Dissertação (Mestrado em Ciências – Matemática) – Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos – SP, 2026.

Random Matrices can be though of in two ways: As a measurable function from a probability space into a set of matrices or, more concretely, as a matrix whose entries are random variables with some joint distribution. A set of such matrices with a given arbitrary distribution is called an ensemble. One can, for ensembles of interest, study statistic such as the eigenvectors and eigenvalue distributions. The symmetries of chosen set will reflect on what models such statistics can model. For example, distances of the characteristic values of a symmetric matrix with random coefficients model distances between successive levels of energies of heavy nuclei. More interestingly, there is some universality in the statistics found among classes of ensembles, suggesting the relevance of the theory in studying the universality in general modeled phenomena. Particularly important is the study of such matrices when they grow infinitely large, taking asymptotically its statistical behavior. We focus on a technique for deriving asymptotics for the so-called partition function of some interesting ensembles of random matrices.

Keywords: Random Matrix Theory, Planar Matrix Ensembles, Asymptotics.

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LIST OF SYMBOLS

He — Hermite Polynomial

$\mathbb{1}$ — Identity Function

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INTRODUCTION

Some initial words on the the use of random matrix and place of the dissertation.

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RANDOM MATRIX THEORY

The chapter that follows is heavily based (to be very generous with my writing) on the incredibly organized and written Thesis (RAHMAN, 2023). To the author, my best wishes. The idea of this chapter is to motivate and introduce the concepts we will touch on later for the main part of the work. Also, if it is of any worth, this chapter serves as a good recap for any reader and a disclaimer of all the notation and conventions we will use in this work. What follows is my best attempt in introducing the part of Random Matrices that are pertinent to my scope of work and are of general interested, albeit inside the limitations of time, effort and experience.

Random Matrices can be though of in two ways: As a measurable function from a probability space into a set of matrices or, more concretely, as a matrix whose entries are random variables with some joint distribution. In some cases there is an explicit density formula for the eigenvalues of such a random matrix.

2.1 Random Matrix

Borrowing some concepts from physics, an *ensemble* can be though as a virtual collection of all possible states that a system can be in, with more probable states appearing more often in the collection. There is a analogy to be made with ensembles of thermodynamic: When considering sets of matrices with equal probability - in some sense to be more clear soon - we get the Gibbs weight for the probability distribution as we would in the *Canonical Ensemble* of constant temperature. More concretely, in the context of Random Matrix Theory (RMT) we have the following definition.

> **Definition 1.** *A **random matrix ensemble** $\mathcal{E} = (\mathcal{S}, \mathcal{P})$ is a set of matrices $\mathcal{S}$ of fixed size equipped with probability density function (p.d.f) $\mathcal{P} : \mathcal{S} \to [0, \infty)$. If we say that a matrix $\hat{X}$ is drawn from the random ensemble $\mathcal{E}$, we are saying that it is a random variable assuming values in $\mathcal{S}$ with p.d.f. $\mathcal{P}$.*

Example 1. *The $n \times n$ circular unitary ensemble (CUE) is the unitary group $U(n)$ with its Haar measure. If $U \in$ CUE(n), then the symmetric unitary matrix $U^T U$ represents the $n \times n$ circular orthogonal ensemble (COE); we have $\mathcal{S} \cong U(n)/O(n)$. Note that if $U \in U(n)$ and $O \in O(n)$, we have that $U^T U = (OU)^T(OU)$, that is, U and OU are in the same co-class.*

To study random matrices, we need to determine which quantities related to the matrix we want to measure. This may be the expectation of some function on the matrices or even on the eigenvalues and eigenvector, if they exist. Assume a matrix is such that it is determined by a set of eigenvalues and eigenvectors. While eigenvectors may be of interest in some cases, eigenvalue statistics have been considerably more explored. For eigenvalues $\{\lambda\}_i \in \mathrm{spec}\{\hat{M}\}$, one could talk about the expected characteristic polynomial

$$P_n(x) = \mathbb{E}[\det\left(\lambda \hat{I}_n - \hat{M}\right)], \quad n \in \mathbb{N}. \tag{2.1}$$

Naturally, the characteristic polynomial should have roots coinciding with the eigenvalues. For some ensembles this goes further: Taken the limit for the size of the matrix - consequently the order of the expected characteristic polynomial - one should expect the measure on the roots of the polynomial to coincide with the one on the eigenvalues. This is true for ensembles with real eigenvalues. Taking the asymptotic on the polynomials reveal something about the nature of the distribution of eigenvalues! This is not necessarily true for ensembles for non-real roots.

Example 2. *(Expected characteristic polynomial for Gaussian Matrices) Hermite Polynomials are the average characteristic polynomials of the Gaussian Ensemble. That is, if $\hat{M}$ is an order N matrix which is hermitian and with joint density given by*

$$\frac{1}{\mathcal{Z}_N} \mathrm{e}^{-\mathrm{Tr}\,\hat{M}^2}\, \mathrm{d}M \tag{2.2}$$

then the average characteristic polynomial satisfies

$$P_n(x) = xP_{n-1}(x) - \frac{n-1}{2} P_{n-2}(x). \tag{2.3}$$

If one notices that $P_0(x) = 1$ and $P_1(x) = x$, then one can also see that the recurrence equation give us the Hermite Polynomials. One can show numerically that their concentrations coincide.

We are naturally interested in the eigenvalue j.p.d.f. $\mathrm{p}(\lambda_1, \lambda_2, \ldots, \lambda_n)$ of the eigenvalues $\{\lambda_i\}_{i=1}^n$ of $\hat{M}$. The eigenvalue j.p.d.f. is defined to be such that

$$\int_{[a_1,b_1]\times\cdots\times[a_n,b_n]} \mathrm{p}(\lambda_1, \lambda_2, \ldots, \lambda_n) \mathrm{d}\lambda_1 \cdots \mathrm{d}\lambda_n \tag{2.4}$$

is the probability that $a_i < \lambda_i < b_i$ for every i.

Example 3. *(The complex case for Characteristic Polynomial) On the complex plane one could write an ensemble with measure*

$$\mathrm{dp}_N^{(\beta)}(z_1,\cdots,z_N) := \frac{1}{\mathcal{Z}_N^{(\beta)}} \prod_{j>k=1}^{N} |z_j - z_k|^\beta \prod_{j=1}^{N} \mathrm{e}^{-\frac{\beta N}{2} Q(z_j)}\, \mathrm{d}A(z_j) \tag{2.5}$$

This is the expression, if $\beta = 2$ for the measure of the random normal matrix ensemble. Setting $Q(z) = |z|^2$ this would be the so-called complex Ginibre ensemble. Consider, however, $Q(z) = q(|z|)$ for some $q : \mathbb{R} \mapsto \mathbb{R}$, i.e., Q is a radially symmetric potential. The theory says that the roots of such ensemble should be distributed in a disk or annulus. However, $\{z^j\}_j$ are the family of expected characteristic polynomials. Clearly its roots, which concentrate at zero, could not agree with the distribution os eigenvalues.

Example 4. *The eigenvalue j.p.d.f. of the $n \times n$ circular ensembles is*

$$\mathrm{p}(\mathrm{e}^{\mathrm{i}\theta_1},\cdots,\mathrm{e}^{\mathrm{i}\theta_n};\beta) = \frac{1}{\mathcal{Z}_N^{(\beta)}} |\Delta_n(\mathrm{e}^{\mathrm{i}\theta})|^\beta \tag{2.6}$$

where $Z_{n,\beta}^{(C)}$ is a normalization constant and

$$\Delta_n(\lambda) := \det\left[\lambda_i^{j-1}\right]_{i,j=1}^{n} = \prod_{1\le i<j\le n} (\lambda_i - \lambda_j) \tag{2.7}$$

is the Vandermonde determinant, and $\beta = 1,2,4$ correspond to the COE, CUE and CSE, respectively.

Related do the eigenvalue j.p.d.f. is the univariate eigenvalue density defined by

$$\mathfrak{p}(\lambda_1;n) := n\int_{\mathbb{R}^{n-1}} \mathrm{p}(\lambda_1,\lambda_2,\ldots,\lambda_n)\mathrm{d}\lambda_2\cdots\mathrm{d}\lambda_n, \tag{2.8}$$

which is normalized such that the expect number of eigenvalues lying between a and b is given by $\int_a^b \mathfrak{p}(\lambda)\mathrm{d}\lambda$. For most applications there are three ideologies that stand out for defining ensembles: By universality, uniformity and compositions. The third won't be considered in this work, but consists of ensembles of matrices defined by products of other random matrices (POZNIAK; ZYCZKOWSKI; KUS, 1998). The first approach, on the other hand, will be important and consists on assuming very little on the system and to focus on the universal results. This lead us to the study of the *Wigner matrix ensembles*: Let $\hat{X}$ be drawn from either the independently distributed real symmetric, complex hermitian or quaternionic skew-symmetric $N \times N$ matrix ensemble. Then its entries $X_{i,j}$ are independently distributed with mean zero and bounded moments. Also, its diagonal entries are equally distributed as are its upper triangular entries. Many results hold for independently and identically distributed centered random variables - and those are the advantages of such approach. Take the matrix $\hat{Y} = \{Y_N\}_{i,j}$ to be such that $Y_{i,j} = Y_{j,i}^\dagger$ and satisfying the following properties that $\{Y_{i,j}\}_{1\le i\le j\le n}$ are independent; $\{Y_{i,j}\}_{i\neq j}$

are identically distributed and also $\{Y_{i,i}\}$ are identically distributed; $\mathbb{E}(Y_{i,j}) = 0$ and $\mathbb{E}(Y_{i,j}^k) < \infty$, i.e, $r_k = \max\left(\mathbb{E}(Y_{i,j}^k), \mathbb{E}(Y_{i,j}^k)\right) < \infty$. Then we can define the following ensemble, central to this approach.

Definition 2. *Let every $\hat{Y}$ be as before and set $r_2 < \infty$ with $\mathbb{E}(Y_{1,2}^2) > 0$. Then, we say $\hat{Y}$ is a* ***Wigner Matrix****. Furthermore, if $\hat{X}_n = n^{-\frac{1}{2}}\hat{Y}_n$, $\hat{X}$ is what we call a* ***normalized Wigner Matrix****.*

We give the name of semicircle distribution to the measure $\mathfrak{p}^{(SC)}$ defined by

$$\mathfrak{p}^{(SC)}(\mathrm{d}x) := \frac{1}{2\pi}\sqrt{(4-x^2)_+}\mathrm{d}x, \tag{2.9}$$

where the support is given by $\operatorname{supp}\mathfrak{p}^{(SC)} = [-2,2]$. Is is not hard to notice why the distribution take such name; it is shaped such as the upper half of a circle. More interestingly, this distribution has a very important role when discussing Wigner Matrices: It is an universal distribution. This means that, for Wigner Matrices, however they are constructed, when taken to the limit, all of them must converge to this limiar shape. This is better structured in *Wigner's Semicircle Law*.

Theorem 1. *(Wigner's Semicircle Law) The averaged eigenvalue distribution $\mathfrak{p}_N$ for Wigner matrices converges weakly to the semicircle distribution $\mathfrak{p}^{(SC)}$.*

Example 5. *(Numerical analysis of Semicircle Law) We can numerically see the convergence theorem in action. If we take , then we expect to see, with $N \longrightarrow \infty$, the eigenvalue measure $\mathfrak{p}_N$ approximating the semicircle distribution $\mathfrak{p}^{SC}\ldots$*

This type of result is named *universality* in the theory and can be better explained when we discuss the theory of *point processes*. The second ideology is assuming equal probability on each state os the system - this is a common assumption in thermodynamics. So this approach is to assume that every matrix is equally likely.

Theorem 2. *(Haar, compact case). Given a compact group $\mathcal{S}$, let $U \in \mathcal{S}$ be a generic variable. There exists a unique (up to normalization measure)* $\mathrm{dp}(U)$ *on $\mathcal{S}$, called the Haar measure such that*

$$\int_{\mathcal{S}} \mathrm{dp}(U) < \infty \tag{2.10}$$

and for any fixed $U_0 \in \mathcal{S}$,

$$\mathrm{dp}(U_0 U) = \mathrm{dp}(U) = \mathrm{dp}(U U_0),$$

that is, finite for compact sets and invariant by left and right actions. If $\mathrm{dp}(U)$ *integrates to unity in $\mathcal{S}$ it is referred as the Haar probability measure.*

Example 6. *The $n \times n$ circular unitary ensemble (CUE) is the unitary group $U(n)$ with its Haar measure. If $U \in$ CUE(n), then the symmetric unitary matrix $U^T U$ represents the $n \times n$ circular orthogonal ensemble (COE); we have $\mathcal{S} \cong U(n)/O(n)$. Note that if $U \in U(n)$ and $O \in O(n)$, we have that $U^T U = (OU)^T(OU)$, that is, U and OU are in the same co-class.*

Interestingly, there is only one ensemble in the intersection of both of these approaches: The Gaussian Ensembles. No other ensemble of Wigner matrices are also invariant. For the purpose of this work, we will mainly discuss the latter approach.

2.2 Invariant Ensembles

Invariant ensembles of random matrices are probability spaces which are invariant under conjugation of the appropriate matrix group - these ensembles only depend on their eigenvalue distributions while their eigenvectors become independent and are Haar distributed. Conjugation invariance is a physically natural assumption and simply signifies that all coordinate systems are equivalent. This also implies that the distributions of matrices is uniform conditional on its eigenvalues. To properly understand this concept, let's make precise the definition of three infamous ensembles: Unitary, Orthogonal and Symplectic ensembles. Other ensembles many times can be of interest, but their theory may be less straightforward.

1. *Orthogonal Ensembles* consist of $n \times n$ real symmetric matrices $\hat{M} = \overline{\hat{M}} = \hat{M}^T$ together with a probability distribution that is invariant under orthogonal conjugation $\hat{M} \mapsto \hat{O}\hat{M}\hat{O}^T$, $\hat{O}$ orthogonal, i.e., $\hat{O}\hat{O}^T = \hat{O}^T\hat{O} = \hat{I}$.

2. *Unitary Ensembles* consists of $n \times n$ Hermitian matrices $\hat{M} = \hat{M}^*$ together with a probability distribution that is invariant under unitary conjugation $\hat{M} \mapsto \hat{U}\hat{M}\hat{U}^*$, $\hat{U}$ unitary, i.e., $\hat{U}\hat{U}^* = \hat{U}^*\hat{U} = \hat{I}$.

3. *Symplectic Ensembles* consists of $2m \times 2m$ Hermitian self-dual matrices $\hat{M} = \hat{M}^* = \hat{J}\hat{M}^T\hat{J}^T$ where $\hat{J} = \mathrm{diag}(\sigma, \cdots, \sigma)$ with
$$\sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$
together with a probability distribution that is invariant under unitary-symplectic conjugation $\hat{M} \mapsto \hat{U}\hat{M}\hat{U}$, with $\hat{U}$ unitary and symplectic, i.e. $\hat{U}\hat{U}^* = \hat{U}^*\hat{U} = \hat{I}$ and $\hat{U}\hat{J}\hat{U}^T = \hat{J}$.

Examples of such invariant ensembles are the Gaussian Orthogonal, Unitary and Symplectic ensembles - *GOE*, *GUE*, *GSE*. These take Gaussian distributed and independent entries. We may also choose to include the Ginibre Ensembles in the types of invariance we will work with. The three types of Ginibre Emsembles are: Ginibre Orthogonal, Unitary and Symplectic are matrices with independent, identically distributed standard complex Gaussian entries. In this case, the eigenvalues are distributed in the complex plane.

Let's make it precise. Given a group $\mathcal{G}$ and a group action $\mathcal{A}:\mathcal{G}\times\mathcal{S}\to\mathcal{S}$, a random matrix ensemble $\mathcal{E}=(\mathcal{S},\mathcal{P})$ is said to be $\mathcal{A}$-invariant if the probability measure $\mathrm{d}\mathcal{P}(\hat{X}) := \mathcal{P}(\hat{X})\mathrm{d}\hat{X}$ satisfies

$$\mathrm{d}\mathcal{P}(\mathcal{A}(g,\hat{X})) = \mathrm{d}\mathcal{P}(\hat{X}) \tag{2.11}$$

for every $g\in\mathcal{G}$. The term *invariant matrix ensemble* usually refers to invariance under the mapping $\hat{X}\mapsto g\hat{X}g^{-1}$ for g drawn from the appropriate compact group $\mathcal{G}$; when a matrix ensemble is considered to be irreducible and its underlying set of matrix is $S\subseteq\mathbb{M}_{n,n}(\mathbb{R}),\mathbb{M}_{n,n}(\mathbb{C}),\mathbb{M}_{n,n}(\mathbb{H})$, the group $\mathcal{G}$ must be one of $O(n),U(n),Sp(2n)$. Then, we say the ensemble is orthogonal, unitary or symplectic.

Definition 3. *(Invariant Ensemble) We call $(\mathcal{S},\mathcal{P})$ an* ***Invariant Ensemble*** *if $\mathcal{S}$ is the set of real, complex or quaternionic matrices ($\beta=1,2,4$) with the probability density funciton*

$$\mathcal{P}(\hat{M}) = \frac{1}{\mathcal{Z}_{N,\beta}}\prod_{1\le j<k\le n}^{N}|\lambda_j-\lambda_k|^{\beta}\prod_{j=1}^{N}\mathrm{e}^{-\beta NV(\lambda_j)}. \tag{2.12}$$

We can also determine the probability measure $\mathrm{d}\mathcal{P}(\hat{X})$ to be invariant under the adjoint action of a group $\mathcal{G}$ is we establish that both the p.d.f. $\mathcal{P}(\hat{X})$ and the Lebesgue measure $\mathrm{d}\hat{X}$ are invariant under this action. The invariance of the Lebesgue measure is more involved and depends on whether or not the associated matrix set $\mathcal{S}$ consists of purely of self-adjoint matrices. Because of that, we start with the case of the Gaussian, Laguerre and Jacobi ensembles.

Proposition 1. *Given an $n\times n$ real ($\beta=1$), complex ($\beta=2$) or quaternionic ($\beta=4$) self-adjoint matrix variable $\hat{X}$, diagonalize it as $\hat{X}=\hat{U}\hat{\Lambda}\hat{U}^{\dagger}$ where $\hat{\Lambda}=\mathrm{diag}(\lambda_1,\lambda_2,\dots,\lambda_n)$ is a diagonal matrix of eigenvalues, and $\hat{U}$ is corresponding matrix of eigenvectors; the spectral theorem dictates that $\hat{U}$ is an element of $O(n)$ in the real case, $U(n)$ in the complex case, and $Sp(2n)$ in the quaternionic case. To make the mapping $\hat{X}\mapsto\hat{U}\hat{\Lambda}\hat{U}^{\dagger}$ bijective, we insist that the eigenvalues are listed in increasing order and the first row of $\hat{U}$ is non-negative real. The canonical Lebesgue measure*

$$\mathrm{d}\hat{X} = \prod_{i=1}^{n}\mathrm{d}X_{ii}\prod_{1\le j<k\le n}\prod_{s=1}^{\beta}\mathrm{d}X_{jk}^{s} \tag{2.13}$$

on the set of adjoint matrices decomposes as

$$\mathrm{d}\hat{X} = |\Delta_n|^{\beta}\mathrm{d}\hat{\Lambda}\mathrm{d}\mathfrak{p}_{Haar}(\hat{U}) \tag{2.14}$$

where $\mathrm{d}\hat{\Lambda}=\prod_{i=1}^{n}\mathrm{d}\lambda_i$ is the Lebesgue measure on the Weyl Chamber $\lambda_1<\dots<\lambda_n\subset\mathbb{R}^n$ and

$$\mathrm{d}\mathfrak{p}_{Haar}(\hat{U}) = \bigwedge_{1\le l<k\le n}\bigwedge_{s=1}^{\beta}\left(\hat{U}^{\dagger}[\mathrm{d}U_{ij}]_{i,j=1}^{n}\right)_{lk}^{(s)} \tag{2.15}$$

is the unnormalized Haar measure on the quotient space $O(n)/O(1)^n$ ($\beta = 1$), $U(n)/U(1)^n$ ($\beta = 2$), or $Sp(2n)/Sp(2)^n$ ($\beta = 4$), given by the exterior product of the independent real components of the entries of the matrix of differentials $\hat{U}^\dagger[\mathrm{d}U_{ij}]_{i,j=1}^n$.

At this point, one can check (albeit with some non-trivial computations) that the Lebesgue measure is indeed invariant in these cases for the appropriate group action.

The situation concerning the $n \times n$ Ginibre Ensemble is more complicated. In the 1965 work, Ginibre extended the proposition above and its arguments starting with the diagonalization formula $\hat{G} = \hat{P}\hat{\Lambda}\hat{P}^{-1}$ - note that, with $\hat{\Lambda}$ diagonal of eigenvalues and $\hat{P}$ of eigenvectors, this formula is valid with probability one since the subset of matrices with repeated eigenvalues has measure zero. However, $\hat{P}$, diagonalizing matrix, is not necessarily orthogonal, unitary, nor symplectic. Nonetheless, QR decomposition can be used to write $\hat{P} = \hat{U}\hat{T}$ where $\hat{U}$ is as before and $\hat{T}$ is an upper triangular matrix with real positive diagonal entries. Then,

$$\mathrm{d}\hat{G} = J(\lambda)\mathrm{d}\hat{\Lambda}\mathrm{d}\mathfrak{p}_{Tri}(\hat{T})\mathrm{d}\mathfrak{p}_{Haar}(\hat{U}) \tag{2.16}$$

where $\mathrm{d}\hat{\Lambda}$ is the Lebesgue measure on the space housing the eigenvalues, $\mathrm{d}\mathfrak{p}_{Haar}(\hat{U})$ is the Haar measure defined before, $J(\lambda)$ is a Jacobian consisting of products of Vandermonde determinants involving the eigenvalues, and $\mathfrak{p}_{Tri}(\hat{T})$ is some measure dependent on the entries of T. From here, it follows that the Lebesgue measure on $\hat{G}$ is invariant under the desired conjugation action.

2.2.1 Orthogonal Polynomial Ensembles

The subclass of invariant matrix ensembles this section refers to are those self-adjoint matrices $\hat{X}$ with p.d.f.s $\mathcal{P}(\hat{X})$ such that one has the decomposition

$$\mathcal{P}(\hat{X}) = \prod_{i=1}^{n} w(\lambda_i), \tag{2.17}$$

where $\{\lambda_i\}$ are the indistinguishable eigenvalues of $\hat{X}$ and $w(\lambda)$ is some continuous weight function that is real valued on $\mathbb{R}$ and decays fast enough as $|\lambda| \to \infty$. This is the same as removing the constrain on the weight function of the classical ensembles. We are interested in the probability distribution of the eigenvalues $\lambda_1, \cdots, \lambda_m$ of $\hat{X}$. Those eigenvalues must be simple.

Proposition 2. *The set of Hermitian matrices with simple spectra is open and dense in $\mathcal{M}$ and has full measure, that is, generally, hermitian matrices have simple eigenvalues.*

Proof. Let the Hermitian matrix $\hat{M}$ have the spectral decomposition

$$\hat{M} = \hat{U}\hat{\Lambda}\hat{U}^*. \tag{2.18}$$

There exists $\epsilon_1, \epsilon_2, \ldots, \epsilon_n$ real, with $\sum_{i=1}^n |\epsilon_i| \to 0$, such that

$$\hat{M}_\epsilon = \hat{U} \begin{bmatrix} \lambda_1 + \epsilon_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n + \epsilon_n \end{bmatrix} \hat{U}^* = \hat{M}_\epsilon^* \tag{2.19}$$

has a simple spectrum. As $\hat{M} \to \hat{M}$, matrices with simple spectra are dense in $\mathcal{M}$ the space of hermitian matrices. □

Because the matrices are self-adjoint, there exists $\hat{U} \in U(n)$ such that

$$\hat{U}^* \hat{X} \hat{U} = \hat{\Lambda} = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n). \tag{2.20}$$

We make such a transformation

$$\hat{X} \mapsto (\hat{U}, \hat{\Lambda}) \tag{2.21}$$

and then integrate the eigenvector element away, so that we are left with the part depending on the eigenvalues. This can be done using Weyls formula.

Theorem 3. *(Weyl) Suppose f is an integrable class function on $\mathbb{S}_n$, i.e., $f(\hat{U}\hat{X}\hat{U}^*) = f(M)$ for all $\hat{U} \in U(n)$. Then we have*

$$\int_{\mathrm{He}_n} f(\hat{X}) \mathrm{d}\hat{X} = c_n \int_{R^n} f(\lambda_1, \lambda_2, \ldots, \lambda_n) \prod_{i<j} (\lambda_j - \lambda_i)^\beta \mathrm{d}\lambda_1 \cdots \mathrm{d}\lambda_n, \tag{2.22}$$

where $\mathbb{S}_n$ are the self-adjoint matrices of order n and

$$c_n = \frac{\pi^{n(n-1)/2}}{\prod_{j=1}^n j!}. \tag{2.23}$$

More generally,

$$\mathrm{d}M = c_n \prod_{i<j} (\lambda_i - \lambda_j)^\beta \mathrm{d}\lambda_1 \cdots \mathrm{d}\lambda_n \mathrm{d}\hat{U}, \tag{2.24}$$

where $\mathrm{d}\hat{U}$ is the Haar measure on $U(n)$

Because of this, we arrive at the following definition.

Definition 4. *Let $\beta = 1,2,4$, fix $n \in \mathbb{N}$ and let $w(\lambda)$ be a continuous, non-negative, real-valued function such that $w(\lambda) = o(\lambda^{-\beta(n-1)-1})$. Then, the j.p.d.f.*

$$\mathrm{p}^{(w)}(\lambda_1, \lambda_2, \ldots, \lambda_n; \beta) = \frac{1}{\mathcal{N}_{n,\beta}^{(w)}} \prod_{i=1}^n w(\lambda_i) |\Delta_n(\lambda)|^\beta \tag{2.25}$$

*describes the n-sized **orthogonal polynomial ensemble** (OPE) when $\beta = 2$, **skew-orthogonal polynomial ensemble of real type** ($\mathbb{R}$-SOPE) when $\beta = 1$, and the **skew-orthogonal polynomial ensemble of quaternion type** ($\mathbb{H}$-SOPE) when $\beta = 4$.*

The condition on the weight function guarantees that the j.p.d.f. has a well-defined normalization constant. If one wishes to integrate the determinantal and Pfaffian structures given by the Jacobian against the product of weight functions, it is beneficial to use monic polynomials $\{q_j(\lambda)\}_{j=0}^{\infty}$ that are skew-orthogonal with respect to the weight $w(\lambda)$.

Definition 5. *With $w(\lambda)$ a suitable function, define the following inner products on polynomial space:*

$$\langle f,g\rangle^{(2)} := \int_{\mathbb{R}} f(x)g(x)w(x)\mathrm{d}x \tag{2.26}$$

For $\beta = 2$, let $\{q_j^{(\beta)}\}_{j=0}^{\infty}$ be a family of monic polynomials such that each $q_j^{(\beta)}(\lambda)$ is of degree j. Impose also that exists finite positive constants $\{r_j^{(\beta)}\}_{j=0}^{\infty}$ s.t.

$$\langle q_j^{(2)}, q_k^{(2)}\rangle^{(2)} = r_j^{(2)}\chi_{j=k} \tag{2.27}$$

for any $j,k>0$. Then, we say that the family $\{q_j^{(2)}(\lambda)\}_{j=0}^{\infty}$ is orthogonal in respect to $w(\lambda)$.

We can now express the eigenvalue j.p.d.f. $\mathrm{p}^{(w)}(\lambda_1,\lambda_2,\ldots,\lambda_n;\beta)$ as a determinant. For other values of β, i.e. $\beta = 1,4$ this could also have a *Pfaffian* structure with a very similar construction.

Theorem 4. *Let $\{q_j^{(\beta)}(x)\}_{j=0}^{\infty}, \{r_j^{(\beta)}\}_{j=0}^{\infty}$, be as in the definition above, and let $n\in\mathbb{N}$. The kernel corresponding respectively to the complex case is*

$$K_n^{(2)}(x,y) := \sqrt{w(x)w(y)}\sum_{k=0}^{N-1}\frac{1}{r_k^{(2)}}q_k^{(2)}(x)q_k^{(2)}(y). \tag{2.28}$$

Then, the eigenvalue j.p.d.f. of Definition 4 is given by

$$p^{(w)}(\lambda_1,\lambda_2,\ldots,\lambda_n;\beta) = \det\left[K_n^{(2)}(\lambda_i,\lambda_j)\right]_{i,j=1}^{n}, \quad \beta=2. \tag{2.29}$$

The significance of this theorem is that, for $\beta = 2$, as for other values, the kernels $K_n^{(\beta)}(x,y)$ have *Reproducing properties* such as

$$\int_{\mathbb{R}} \det\left[K_n^{(2)}(\lambda_i,\lambda_j)\right]_{i,j=1}^{k+1}\mathrm{d}\lambda_{k+1} = (n-k)\det\left[K_n^{(2)}(\lambda_i,\lambda_j)\right]_{i,j=1}^{k} \tag{2.30}$$

Thus, one may integrate the j.p.d.f. over $\mathrm{d}\lambda_{k+1}\cdots\mathrm{d}\lambda_n$ to obtain the *k-point correlation function*

$$\mathrm{p}_k^{(w)}(\lambda_1,\lambda_2,\ldots,\lambda_k;n;\beta) := \frac{n!}{(n-k)!}\int_{\mathbb{R}^{n-k}} \mathrm{p}^{(w)}(\lambda_1,\lambda_2,\ldots,\lambda_n;\beta)\mathrm{d}\lambda_{k+1}\cdots\mathrm{d}\lambda_n \tag{2.31}$$

$$= \det\left[K_n^{(2)}(\lambda_i,\lambda_j)\right]_{i,j=1}^{k}, \quad \beta=2. \tag{2.32}$$

This implies that he eigenvalue density $\mathfrak{p}^{(w)} = \mathrm{p}_1^{(w)}(\lambda)$ is given by the formula

$$\mathfrak{p}^{(w)}(\lambda;\beta) = K_n^{(2)}(\lambda,\lambda), \quad \beta = 2. \tag{2.33}$$

Most properties of a determinantal point process - such as the eigenvalues of a random matrix from a unitary ensemble - follow from properties of the Kernel. We will come back to this results on Chapter 3.

CHAPTER

3

POINT PROCESSES

The definitions given in this section are a heavily adaptation from (SERFOZO, 1990), (JOHANSSON, 2006) and (GIROTTI, 2014).

There are many instances where one might want to describe the behavior of a set of random points in time or in space. This could come from an investigation on system of diffusing particles or occurrences in time of some type of phenomena. Point processes are models to describe such random objects: The number of occurrence of events in a given time-frame or, alternatively, the number of points in regions of some space. For this work, we mainly focus on the latter application: density of points in regions of space. Those can be associated later with measure for eigenvalues of random matrices and the theory developed on those processes, especially the so-called determinantal point processes, will help us studying distributions for these matrices eigenvalues and associated quantities.

3.1 Point Processes

Suppose there are m points at the random locations in some region Λ a complete separable metric space, given by some configuration $\chi_m = \{x_1, x_2, \ldots, x_m\}$. We can define a quantity $\xi(A)$ that measures the amount of points ($\#\{\chi_m \cap A\}$) for any sub-region $A \subset \Lambda$ in the following way:

$$\xi(A) = \sum_{k=1}^{m} \mathbb{1}_A(x_k). \tag{3.1}$$

Moreover, let $\mathcal{I}$ denote all intervals, or rectangles, in Λ, $\mathcal{E}$ denote the class of Borel sets and $\mathcal{B}$ the class of bounded Borel sets. One can notice that this function ξ defines a *random counting measure* in the usual sense, i.e., a mapping $\mathfrak{p} : \mathcal{E} \to \{0, 1, \cdots, \infty\}$ such that

$$\mathfrak{p}(\varnothing) = 0, \quad \mathfrak{p}(B) < \infty, \text{ for } B \in \mathcal{B}, \tag{3.2}$$

and that, for disjoint $B_1, B_2, \cdots$ in $\mathcal{B}$,

$$\mathfrak{p}\left(\bigcup_k B_k\right) = \sum_k \mathfrak{p}(B_k). \tag{3.3}$$

We also say that ξ is boundedly finite. Let $\mathcal{N}(\Lambda)$ denote the space of all counting measures ξ on Λ. If $B \in \mathcal{B}$, then $\xi(B)$ is finite and we can write

$$\xi|_B = \sum_{i=1}^{\xi(B)} \delta_{x_i}, \tag{3.4}$$

for some $\chi = \{x_1, x_2, \ldots, x_{\xi(B)}\}$, $x_i \in \Lambda \forall 1 \leq i \leq \xi(B)$. Finally, we can explicitly define point processes.

Definition 6. *(Point Processes) Let Λ be a complete separable metric space. A **point process** $\mathbb{P}$ on Λ is a probability measure on the space of all configuration of points χ or counting measures $\mathcal{N}(\Lambda)$. We say $\mathbb{P}$ to be a n-point process if $\mathbb{P}(\#\chi = n) = 1$ or, equivalently, $\mathbb{P}(\xi(\Lambda) = n) = 1$. We say that the counting measure ξ is simple if $\xi(\{x\}) \leq 1$ for all $x \in \Lambda$, then, the point process is simple if $\mathbb{P}(\xi$ is simple$) = 1$.*

An important note is that ξ is the same for any ordering of the subscripts on $x_1, x_2, \ldots, x_n$. This lead us the the conclusion that if $u_n(x_1, x_2, \ldots, x_n)$ is a probability function on $\mathbb{R}^n$ such that it is invariant under permutations, i.e.

$$u_n(x_{\sigma(1)}, \cdots, x_{\sigma(n)}) = u_n(x_1, x_2, \ldots, x_n), \quad \forall \sigma \in S_n, \tag{3.5}$$

then u_n defines naturally a point process.

Example 7. *(The j.p.d.f. of eigenvalues for GUE) Consider the measure for the hermitian matrices $\hat{M}$,*

$$\mathrm{p}^{(G)}(\hat{H}) = \frac{1}{\mathcal{Z}_{N,\beta}^{(G)}} \mathrm{e}^{-\mathrm{Tr}(\hat{H}^2)} = \frac{1}{\mathcal{Z}_{N,\beta}^{(G)}} \prod_{j>k}^{N} |\lambda_j - \lambda_k|^\beta \prod_{j=1}^{N} \mathrm{e}^{-\lambda_j^2} = \mathrm{p}_N(\lambda_1, \lambda_2, \ldots, \lambda_N). \tag{3.6}$$

This is, of course, a probability functions on the space of configuration of eigenvalues in $\mathbb{R}$ if $\mathcal{Z}_{n,\beta}^{(G)}$, finite, is a normalization. Moreover, it is simple, as configurations with repeated eigenvalues must have probability 0. Note also that the function p is invariant under permutations: This is the origin of the name Gaussian Unitary, Orthogonal and Symplectic Ensembles, the invariance by those respective transformations (equivalent to taking $\beta = 1, 2, 4$ and real, complex and quaternionic entries).

The mapping $\mathcal{E} \ni A \mapsto \mathbb{E}[\#(\chi \cap A)]$, that assigns to the the Borel set A the expected value of the number of points in itself under the configuration χ, is a measure on $\mathbb{R}$.

Definition 7. *The* ***mean measure****, or* ***first moment****, of a point process on Λ is defined by*

$$\mathfrak{p}(A) = \mathbb{E}[\xi(A)] := \mathbb{E}[\#(\chi \cap A)], \quad A \in \mathcal{E}. \tag{3.7}$$

This $\mathfrak{p}$ is a measure on Λ in the standard sense that $\mathfrak{p} : \to [0,\infty]$ satisfies $\mathfrak{p}(\emptyset)) = 0$ and, for disjoint $\Lambda_1, \Lambda_2, \cdots$ in $\mathcal{E}$,

$$\mathfrak{p}\left(\bigcup_k \Lambda_k\right) = \sum_k \mathfrak{p}(\Lambda_k). \tag{3.8}$$

Assuming there exists a density $\rho_1 : \mathcal{E} \ni A \to \mathbb{R}_+$ with respect to the Lebesgue measure (1**-point correlation function**), there is also a way to define the mean measure as the integral of such density:

$$\mathfrak{p}(A) = \int_A \rho_1(x)\mathrm{d}x. \tag{3.9}$$

This can be called the *rate* or *intensity* of ξ and is the infinitesimal probability of a point at x, i.e. $\mathbb{P}(\xi(\mathrm{d}x) = 1) = \rho_1(x)\mathrm{d}x$. In this case $\mathfrak{p}$ is written as $\mathfrak{p}(\mathrm{d}x) = \rho_1(x)\mathrm{d}x$. This mean measure shows up in expressions of expectations for functionals of ξ.

Definition 8. *More generally, the* ***k-th moment*** $\mathfrak{p}_k$ *of* ξ *, written*

$$\mathfrak{p}_k(A_1 \times A_2 \times \cdots \times A_k) := \mathbb{E}\left[\xi_1(A)\xi_2(A)\cdots\xi_k(A)\right], \quad A_1, A_2, \ldots, A_k \in \mathcal{E}, \tag{3.10}$$

can be expressed in respect to the ***k-th correlation function*** $\rho_k : A^k \to \mathbb{R}_+$ *in the following way*

$$\mathfrak{p}_k(A_1 \times A_2 \times \cdots \times A_k) = \mathbb{E}\left[\prod_{j=1}^{k}(\#\chi \cap A_j)\right] = \int_{A_1} \cdots \int_{A_k} \rho_k(x_1, \cdots, x_k)\mathrm{d}x_1 \cdots \mathrm{d}x_k. \tag{3.11}$$

This variable stand for the expected number of k-tuples $\chi_k = \{x_1, x_2, \ldots, x_k\} \in \chi^k$ such that $x_j \in A_j \ \forall j$. For example, if $u_n(x_1, x_2, \ldots, x_n)$ is a probability density function on Λ^n, invariant under permutation of coordinates, then we can build an n-point process on Λ with correlation functions

$$\rho_k(x_1, x_2, \ldots, x_k) := \frac{n!}{(n-k)!} \int_{\Lambda^{n-k}} u_n(x_1, x_2, \ldots, x_n)\mathrm{d}x_{k+1} \cdots \mathrm{d}x_n. \tag{3.12}$$

Example 8. *Let $\mathcal{H}_N$ be the space of all $N \times N$ Hermitian matrices. This space can be identified naturally with $\mathbb{R}^{N^2}$. If $\mathfrak{p}_N$ is a probability measure on $\mathcal{H}_N$ and $\{\lambda_1(M), \cdots, \lambda_N(M)\}$ denotes the set of eigenvalues of $M \in \mathcal{H}_N$, then*

$$M \to \sum_{j=1}^{N} \delta_{\lambda_j(M)} \tag{3.13}$$

maps $\mathfrak{p}_N$ *to a finite point process* P *on* $\mathbb{R}$*. If* $\mathrm{d}M$ *is the Lebesgue measure on* $\mathcal{H}_N$*, then*

$$\mathrm{d}\mathfrak{p}_N(M) = \frac{1}{\mathcal{Z}_N} \mathrm{e}^{-\operatorname{Tr} M^2} \mathrm{d}M \tag{3.14}$$

is a Gaussian probability measure on $\mathcal{H}_N$*, called the GUE. It can be shown that for any symmetric, continuous function* f *on* $\mathbb{R}^N$ *with compact support*

$$\int_{\mathcal{H}_N} f(\lambda_1(M), \cdots, \lambda_N(M)) \mathrm{d}\mathfrak{p}_N(M) = \int_{\mathbb{R}^N} f(x_1, x_2, \ldots, x_N) u_N(x_1, x_2, \ldots, x_N) \mathrm{d}^N x \tag{3.15}$$

where

$$u_N(x_1, x_2, \ldots, x_N) = \frac{1}{\mathcal{Z}_N N!} \prod_{1 \le i < j \le j}^{N} (x_i - x_j)^2 \prod_{j=1}^{N} \mathrm{e}^{-x_j^2} \tag{3.16}$$

is the induced eigenvalue measure on $\mathbb{R}^N$*. Hence the point process* P *on* $\mathbb{R}$ *defined by this induced measure has correlation functions*

$$\mathfrak{p}_k(x_1, x_2, \ldots, x_k) = \frac{N!}{(N-k)!} \int_{\mathbb{R}^{n-k}} \frac{1}{\mathcal{Z}_N N!} \prod_{1 \le i < j \le j}^{N} (x_i - x_j)^2 \prod_{j=1}^{N} \mathrm{e}^{-x_j^2} \mathrm{d}x_{k+1} \cdots \mathrm{d}x_n, \tag{3.17}$$

as given by (3.12).

For simple points processes on $\mathbb{R}$ there is a direct way to understand correlation functions. Let $A_i = [x_i, x_i + \Delta x_i], 1 \le i \le k$, be disjoint intervals. If the Δx_i are small enough, then we should expect either one or zero particles in each interval A_i. Hence, the product $\xi(A_1) \cdots \xi(A_k)$ is typically either 1 or 0 depending on whether we have one particle in each interval. From (3.11), we should expect

$$\rho_k(\lambda_1, \lambda_2, \ldots, \lambda_k) = \lim_{\Delta x_i \to 0} \frac{\mathbb{P}(\text{one particle in each } A_i)}{\Delta x_1 \Delta x_2 \ldots \Delta x_k}. \tag{3.18}$$

This means that $\rho_1(\lambda)$ is the density of particles at λ, but since we have many particles, the events of finding particles at their respective points are not disjoint events, even for distinct points. It follows from the definition that if we have a simple point process, then $\rho_k(x_1, x_2, \ldots, x_k)$ is exactly the probability of finding particles at $x_1, x_2, \ldots, x_n$.

3.2 Determinantal Point Processes

Repulsive random processes are, in general, intractable, that is, they are usually difficult of even impossible to, for example, compute marginals and conditional probabilities. In these cases, their theoretical analysis is a major challenge. However, there is a class of processes, even repulsive ones, that are an rare exception to this rule: Determinantal Point Processes (DDP); or Fermionic point processes in the physics literature, are random processes that have the property of being both repulsive and tractable (TREMBLAY, 2025). This property comes from the fact that their correlation function can be expressed as a certain determinantal form. Let's make this precise.

Definition 9. *Consider a point process P on a complete separable metric space Λ, with reference measure p_N, all of whose correlation functions ρ_k exist. If there is a function $K : \Lambda \times \Lambda \to \mathbb{C}$ such that*

$$\rho_k(x_1, x_2, \ldots, x_k) = \det(K(x_i, x_j))_{i,j=1}^k \tag{3.19}$$

for all $x_1, x_2, \ldots, x_k \in \Lambda$, $k \geq 1$, then we say that P is a ***determinantal point process****. We call K the* ***correlation kernel*** *of the process.*

Example 9. *(Eigenvalue for orthogonal polynomial ensembles) We have shown in subsection 2.2.1 that*

$$\mathrm{p}_k^{(w)}(\lambda_1, \lambda_2, \ldots, \lambda_k; n; \beta = 2) = \det\left[K_n^{(2)}(\lambda_i, \lambda_j)\right]_{i,j=1}^k, \tag{3.20}$$

for a subclass of the invariant ensembles called orthogonal polynomial ensembles. This is the main connection for the theory of random matrices and determinantal point process; and one that we take major advantage of. For $\beta = 2$, we should have

$$\mathrm{p}_n^{(w)}(\lambda_1, \lambda_2, \ldots, \lambda_n; 2) \propto \prod_{i=1}^n w(\lambda_i)|\Delta_n(\lambda)|^2 = \det\left(w(\lambda_i) q_{j-1}^{(2)}(\lambda_i)\right)_{n\times n}^2 \tag{3.21}$$

where $w(\lambda_i)$ is a weight function and $|\Delta_n(\lambda)|$ is the Vandermonde determinant. Then, we can write

$$\mathrm{p}_n^{(w)}(\lambda_1, \lambda_2, \ldots, \lambda_n; 2) \propto \det\left(\sqrt{w(\lambda_i)w(\lambda_k)} \sum_{j=1}^n q_{j-1}^{(2)}(\lambda_i) q_{j-1}^{(2)}(\lambda_k)\right)_{i,k=1}^n. \tag{3.22}$$

$$\propto \det\left[K_n^{(2)}(\lambda_i, \lambda_k)\right]_{i,k=1}^n. \tag{3.23}$$

The orthogonal polynomials $\{q_j^{(2)}(\lambda)\}_{j=0}^\infty$ that compose the kernel are know to satisfy the three-term recurrence

$$q_j^{(2)}(\lambda) = (\lambda + a_j) q_{j-1}^{(2)}(\lambda) - \frac{r_{j-1}^{(2)}}{r_{j-2}^{(2)}} q_{j-2}^{(2)}(\lambda) \tag{3.24}$$

where the coefficients depend on the choice of weight.

All important information is contained in the correlation kernel for these processes and, therefore, all quantities of interest can be expresses i terms of the same K. The kernel K can be seen as an integral kernel of an operator K on $L^2(\Lambda, \lambda)$,

$$Kf(x) = \int_\Lambda K(x,y)f(y)\mathrm{d}\lambda(y) \tag{3.25}$$

as long as the right and side is well defined. Important examples of DDP's are constructed by using the following result

Theorem 5. *(Construction for DDP (GIROTTI, 2014)) Consider a kernel $K : \mathbb{R}\times\mathbb{R} \to \mathbb{R}_+$ with the following properties:*

- *Trace-class:* $\mathrm{Tr}(K) = \int_\mathbb{R} K(x,x)\mathrm{d}x = n < +\infty$;
- *Positivity:* $\det[K(x_i,x_j)]_{i,j=1}^n$ *is non-negative for every* $x_1, x_2, \ldots, x_n \in \mathbb{R}$;
- *Reproducing property:* $\forall x, y \in \mathbb{R}$

$$K(x,y) = \int_R K(x,s)K(s,y)\mathrm{d}s. \tag{3.26}$$

Then,

$$u_n(x_1, x_2, \ldots, x_n) := \frac{1}{n!}\det[K(x_i,x_j)]_{i,j=1}^n \tag{3.27}$$

is a probability measure on $\mathbb{R}^n$, invariant under coordinated permutations. The associated point process is a determinantal point process with K as a correlation kernel.

Example 10. *content...*

3.3 Limit and Transformations of Determinantal Point Processes

Let $f \in C^1(\mathbb{R})$ be a bijection , with continuously differentiable inverse $f^{-1} = g \in \mathbb{C}^1(\mathbb{R})$. Then, f maps configurations on $\mathbb{R}$ to configuration on $\mathbb{R}$. The push-forward of a point process $\mathbb{P}$ under such mapping is $f(\mathbb{P})$ and it is again a point process.

Proposition 3. *(GIROTTI, 2014) If K is the kernel of a DDP $\mathbb{P}$, then $f(\mathbb{P})$ is also a DDP with kernel*

$$\hat{K}(x,y) = \sqrt{g'(x)g'(y)}K(g(x), g(y)). \tag{3.28}$$

where $f^{-1} = g$.

In particular, if we consider a linear function $f(x) = \alpha(x - x^*)$, for $\alpha > 0$ and $x^* \in \mathbb{R}$, then the transformed point process $f(\mathbb{P})$ is a scaled and translated version of the original point process $\mathbb{P}$. The new correlation kernel is

$$\hat{K}(x,y) = \frac{1}{\alpha}K\left(x^* + \frac{x}{\alpha}, x^* + \frac{y}{\alpha}\right). \tag{3.29}$$

Others transformations are also possible. Consider what happens when one fixes $x^* \in \mathbb{R}$ and, given that $x^* \in \chi$, removes x^* from the configuration. The resulting process $\chi \setminus \{x^*\}$ is a new point process.

Proposition 4. *(GIROTTI, 2014) Let K be the correlation kernel of a DDP and let x^* be a point such that $0 < K(x^*,x^*) < +\infty$. Then, the point process obtained by removing x^* is determinantal with kernel*

$$\hat{K}(x,y) = K(x,y) - \frac{K(x,x^*)K(x^*,x)}{K(x^*,x^*)}. \tag{3.30}$$

We now want to consider sequences of finite determinantal point processes $\mathbb{P}_n$ with correlation kernel K_n and understand how the limit behaves.

Proposition 5. *(GIROTTI, 2014) Let $\mathbb{P}$ and $\mathbb{P}_n$ be determinantal point processes with kernels K and K_n respectively. Let K_n converge pointwise to K*

$$\lim_{n\to\infty} K_n(x,y) = K(x,y) \tag{3.31}$$

uniformly in x, y over compact subsets of $\mathbb{R}$. Then, the point processes $\mathbb{P}_n$ converge to $\mathbb{P}$ weakly.

A very important case studied here is of the limit of an scaled and centered determinantal point process. Suppose a sequence of kernels K_n and fixed point x^* of interest. We first perform a centering and rescaling of the form

$$x \mapsto Cn^{\alpha}(x - x^*) \tag{3.32}$$

for some suitable value of $C, \gamma > 0$. Then the scaled kernels have a limit

$$\lim_{n\to\infty} \frac{1}{Cn^{\alpha}} K_n\left(x^* + \frac{x}{Cn^{\alpha}}, x^* + \frac{y}{Cn^{\alpha}}\right) = K(x,y) \tag{3.33}$$

Therefore, the scaling limit K is a kernel that corresponds to a determinantal point process with an infinite number of points. Interestingly, in many situations, the same scaling limit K may appear. The phenomenon in known as **universality** in RMT. Examples of such kernels are the sine kernel

$$K_{\sin} = \frac{\sin \pi(x-y)}{x-y} \tag{3.34}$$

and the Airy kernel

$$K_{Ai} = \frac{Ai(x)Ai'(y) - Ai'(x)Ai(y)}{x-y} \tag{3.35}$$

where Ai is the Airy function. Both these kernels can be identified as local laws on the bulk and edge of invariant ensembles, as posed in Equation 6.

Text that associates eigenvalues with point process

Theorem 6. *If*

$$\lim_{n\to\infty} \frac{1}{n} K_n(x,x) = \phi_V(x), \quad x \in \mathbb{R}, \tag{3.36}$$

then the asymptotic distribution of the points in the point process (eigenvalues of $\hat{M}_n$) is given by the measure with density ϕ_V:

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=1}^{n} f(x_j) = \int f(x)\phi_V(x)\mathrm{d}x. \tag{3.37}$$

Moreover, suppose ϕ_V to be supported on one interval $[a,b]$, then for $x^* \in (a,b)$ with $\phi_V(x^*) > 0$, one has

$$\lim_{n\to\infty} \frac{1}{\phi_V(x^*)n} K_n\left(x^* + \frac{x}{\phi_V(x^*)n}, x^* + \frac{y}{\phi_V(x^*)n}\right) = \frac{\sin \pi(x-y)}{\pi(x-y)}. \tag{3.38}$$

Finally, if near the endpoint b

$$\phi_V(x) = \frac{c}{\pi}(b-x)^{\frac{1}{2}}(1+o(1)), \quad x \to b, \tag{3.39}$$

then

$$\lim_{n\to\infty} \frac{1}{(cn)^{\frac{2}{3}}} K_n\left(b + \frac{x}{(cn)^{\frac{2}{3}}}, b + \frac{y}{(cn)^{\frac{2}{3}}}\right) = \frac{Ai(x)Ai'(y) - Ai'(x)Ai(y)}{x-y}, \tag{3.40}$$

where Ai is the Airy function.

CHAPTER

4

EQUILIBRIUM

4.1 Coulomb Gases

We start by describing a familiar distribution or physicists: The Boltzmann-Gibbs distribution. This is a probability distribution that represents the probability of a given state for a system of particles as a function of the state's *energy* ($\mathcal{H}$) and the *temperature* ($1/(\beta N^2)$) of the system. This is the distribution that maximizes the entropy: lower energy states have higher probability of being occupied than states with higher energy. Under the appropriate conditions, we name Coulomb Gas p_N the Boltzmann-Gibbs probability measure given in $(\mathbb{R}^d)^N$. The measure p_N models an coulomb-interacting gas of charged indistinguishable particles under some external field Q. Particles are interpreted to be at positions $x_1, x_2, \ldots, x_N \in \mathbb{S}$ of dimension d, or some configuration $\xi_N = (x_1, x_2, \ldots, x_N)$, embedded in some space $\mathbb{R}^n$. We write

$$\mathrm{dp}_N(x_1, x_2, \ldots, x_N) = \frac{e^{-\beta N^2 \mathcal{H}_N(\xi_N)}}{Z_N^\beta} \mathrm{d}x_1 \mathrm{d}x_2 \ldots \mathrm{d}x_N \tag{4.1}$$

to be te measure of such system, where

$$\mathcal{H}_N(\xi_N) = \frac{1}{N} \sum_{i=1}^{N} Q(x_i) + \frac{1}{2N^2} \sum_{i \neq j}^{N} \mathfrak{g}(x_i - x_j) \tag{4.2}$$

is usually called the Hamiltonian, or more concretely, the energy of the system. The function $Q\colon \mathbb{S} \mapsto \mathbb{R}$ is the external potential. If one wishes p_N to be a probability measure, then one must ensure that Q is such that the normalizing constant $Z_{N,\beta}$ is finite for all N and the measure support is compact. This is all to say that the potential should be confining: As the number of particles increases, it must counter the repulsion effect and keep limited the region where particles move around. The function $\mathfrak{g}\colon\ \mathbb{S} \mapsto (-\infty, \infty]$ is the *Coulomb Interacting Kernel*, solution to the Poisson equation given by $-\nabla g(\vec{x}) = c_n \delta_0$.

Recall the expression for invariant ensembles in Equation 3, note that, rearranging the terms in Equation 4.1 and considering a logarithm interaction kernel, we should get

$$\mathrm{d}\mathfrak{p}_N(x_1, x_2, \ldots, x_N) = \frac{1}{Z_N^\beta} \prod_{1 \leq j < k \leq n}^{N} |x_j - x_k|^\beta \prod_{j=1}^{N} \mathrm{e}^{-\beta N Q(x_j)}. \tag{4.3}$$

This is to say that for invariant ensembles, their eigenvalues behave as a Coulomb gas. This opens the way for interpretations on the distribution of eigenvalues but also to computations using simulation methods for particle dynamics (CHAFAï; FERRé, 2018).

Example 11. *For some $Q : \mathbb{R} \to \mathbb{R}$, we can take the appropriate dimensions for $d = 1$ and $n = 2$ to recover the, for example, GUE ensemble density*

$$\mathfrak{p}_N(\vec{x}) = \frac{e^{-\beta N^2 \mathcal{H}_N(\vec{x})}}{Z_{N,\beta}}, \quad \mathcal{H}_N(\vec{x}) = \frac{1}{N} \sum_{i=1}^{N} \frac{x_i^2}{2} + \frac{1}{N^2} \sum_{i<j}^{N} \ln \frac{1}{|x_i - x_j|}. \tag{4.4}$$

In this case, we would be dealing with particles on the plane confined to move only on the real line. However, the gases measure accepts a natural extension for any potential that we call acceptable - that obeys some growth and smoothness conditions. Together with the dimensions, we can recover other invariant random matrix models.

4.2 Equilibrium Measure

The description of the eigenvalues as a Coulomb gas and as a point process lead us to the idea of minimizing some type of energy of the system by finding equilibrium configurations. Suppose we are dealing with the measure

$$\mathfrak{p}_n(\xi_n) = \frac{e^{-\beta n^2 \mathcal{H}_n(\vec{x})}}{Z_{n,\beta}}, \quad \mathcal{H}_n(\vec{x}) = \frac{2}{n} \sum_{i=1}^{n} Q(x_i) + \frac{1}{n^2} \sum_{i<j}^{n} \ln \frac{1}{|x_i - x_j|}. \tag{4.5}$$

Then, for any n-configuration $\xi_n = x_1, x_2, \ldots, x_n$, if we let μ_x be the normalized counting measure on $\mathbb{C}$,

$$\mu_x = \frac{1}{n} \sum_{i=1}^{n} \delta_{x_i}, \quad \int_{\mathbb{R}} \mathrm{d}\mu_x = 1, \tag{4.6}$$

then

$$\frac{1}{n^2} \sum_{i \neq j} \ln |x_i - x_j|^{-1} + \frac{2}{n} \sum_{i=1}^{n} Q(x_i) \tag{4.7}$$

may be written in the form

$$\int \int \ln |t - s|^{-1} \mathrm{d}\mu_x(t) \mathrm{d}\mu_x(s) + 2 \int Q(x_i) \mathrm{d}\mu_x(s). \tag{4.8}$$

This lead us to the energy minimization problem

$$E^{(Q)} = \inf_{\mu\in\mathcal{M}^1(\mathbb{R})}\left[\int\int \ln|t-s|^{-1}\mathrm{d}\mu_x(t)\mathrm{d}\mu_x(s) + 2\int Q(s)\mathrm{d}\mu_x(s)\right] \tag{4.9}$$

$$:= \inf_{\mu\in\mathcal{M}^1(\mathbb{R})}\left[I_Q(\mu_x)\right] \tag{4.10}$$

where

$$\mathcal{M}^1(\mathbb{R}) = \left\{\mu \in \text{Borel measures on } \mathbb{R} \;\middle|\; \int \mathrm{d}\mu = 1\right\}. \tag{4.11}$$

The quantity $E^{(Q)}$ is the *equilibrium energy* of the system and $I_Q(\mu_x)$ is the *energy integral*. However, we must step backwards so that we can advance.

The classical case for such minimization problems corresponds to the cases when the conductor Σ is compact and the external field Q is zero. Under some mild conditions on the external field, however, the equilibrium measure μ_Q - the measure that minimizes such problem - has a unique solution which is a measure of compact support. For it to be compact, Q the confining potential, must increase sufficiently fast around infinity. Let $\Sigma \subseteq \mathbb{C}$ be a compact subset of the complex plane and $\mathcal{M}(\Sigma)$ be the collection of all positive Borel measure of compact support in Σ. Let $\mu \in \mathcal{M}(\Sigma)$ be such a measure in the complex plane $\mathbb{C}$. The associated *logarithmic potential* is defined by

$$U^{\mu}(z) := \int \ln\frac{1}{|z-t|}\mathrm{d}\mu(t). \tag{4.12}$$

Also, $I(\mu)$, called the *logarithm energy* of a $\mu \in \mathcal{M}(\Sigma)$ is defined

$$I(\mu) := \int\int \ln\frac{1}{|z-t|}\mathrm{d}\mu(z)\mathrm{d}\mu(t), \tag{4.13}$$

and the energy V of Σ

$$V := \inf\{I(\mu)|\mu \in \mathcal{M}(\Sigma)\}. \tag{4.14}$$

We can say that

Proposition 6. *(SAFF; TOTIK, 2024) The logarithmic potential are super-harmonic on the whole plane, that is, they have the following properties.*

1. $U^{\mu}(z) > -\infty$ *for all* z,
2. U^{μ} *is lower semi-continuous,*
3. $U^{\mu}(z) \geq \frac{1}{2\pi}\int_{-\pi}^{\pi} U^{\mu}(z)(x + r\,\mathrm{e}^{\mathrm{i}\theta})\mathrm{d}\theta$ *for all* z *and all* $r \geq 0$.

Proof. Firstly, property (1) follows directly by the properties of μ, especially the compact support. The property (2) can be proven noticing that

$$U^{\mu}(z) = \lim_{M\to\infty}\int \min\left(M, \ln\frac{1}{|z-t|}\right)\mathrm{d}\mu(t). \tag{4.15}$$

The functions on the right-hand side are continuous and increasing with M, such limit of an increasing sequence of functions must be lower semi-continuous. To prove (3), note that the kernel $\ln(1/|z-t|)$ is super-harmonic in $\mathbb{C}$ for each t. If some F is analytic in a domain D and $\rho > 0$, then $|F(z)|^\rho$ is sub-harmonic in D; furthermore, $\ln|F(z)|$ is sub-harmonic in D provided F is not identically zero. Thus,

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} \ln\frac{1}{|z-r\,\mathrm{e}^{\mathrm{i}\theta}-t|}\mathrm{d}\theta \le \ln\frac{1}{|z-t|}, \quad z,t \in \mathbb{C}. \tag{4.16}$$

Using the Fubini-Toneli theorem and the inequality above

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} U^\mu(z+r\,\mathrm{e}^{\mathrm{i}\theta})\mathrm{d}\theta = \int \frac{1}{2\pi}\int_{-\pi}^{\pi} \ln\frac{1}{|z-r\,\mathrm{e}^{\mathrm{i}\theta}-t|}\mathrm{d}\theta\mathrm{d}\mu(t), \tag{4.17}$$

$$\le \int \ln\frac{1}{|z-t|}\mathrm{d}\mu(t) = U^\mu(z) \tag{4.18}$$

□

Also, a result that will become useful soon but we do not prove.

> **Lemma 1.** *(SAFF; TOTIK, 2024) Let $\mu = \mu_1 - \mu_2$ be a signed Borel measure with compact support and total mass $\mu(\mathbb{C}) = 0$. Suppose further that each of the positive measures μ_1 and μ_2 has finite logarithmic energy. Then the logarithmic energy of μ is nonnegative:*
>
> $$I(\mu) := \int\int \ln\frac{1}{|z-t|}\mathrm{d}\mu(z)\mathrm{d}\mu(t), \tag{4.19}$$
>
> *and it is zero if and only if $\mu = 0$.*

The energy V can be finite or $+\infty$, and in the finite case there is a unique measure $\mu = \mu_\Sigma \in \mathcal{M}(\Sigma)$ for which the minimum defining energy V is reachable. Such a measure μ_Σ is called the *equilibrium distribution* or *equilibrium measure* of the compact set Σ. Fro such a function we have that its logarithm potential is always smaller or equal to the energy I_Σ. Define also the *logarithmic capacity*

$$cap(\Sigma) := \mathrm{e}^{-V}. \tag{4.20}$$

The capacity for an arbitrary Borel set E is defined by the supreme of the capacity of all the compact subsets $K \subseteq E$. Naturally, the capacity of a subset of a Borel set of capacity zero is said to be zero. The union of zero-capacity Borel sets can also be shown to be a zero-capacity Borel set. A property is said to hold *quasi-everywhere* (q.e.) on a set E if the set of non conforming points is of capacity zero. Then, we can say that

$$U^\mu_\Sigma(z) = V, \quad \text{q.e.} z \in \Sigma. \tag{4.21}$$

We should at this point reintroduce the external potential Q. The main difference now is that we can loose the compacity restriction on the sets Σ and impose that there is no positive mass around infinity. There is a natural way to introduce this idea. Let $\Sigma \subset \mathbb{C}$ be a closed set and $w : \Sigma \to [0,\infty)$. We call such a function a *weight function* on Σ.

Definition 10. *A weight function w is said to be **admissible** is it satisfies the following three conditions*

1. *w is upper semi-continuous;*
2. *$\Sigma_0 := \{z \in \Sigma | w(z) > 0\}$ has positive capacity;*
3. *if Σ is unbounded, then $|z|w(z) \to 0$ as $|z| \to \infty$, $z \in \Sigma$.*

We can define

$$w(z) := \mathrm{e}^{-nQ(z)}. \tag{4.22}$$

Then, for the weight to be admissible, $Q : \Sigma \to (\infty, \infty]$ must be lower semi-continuous, $Q(z) < \infty$ on a set of positive capacity and if Σ is unbounded, then

$$\lim_{|z|\to\infty, z\in\Sigma} \{Q(z) - \ln|z|\} = \infty. \tag{4.23}$$

Then, let $\mathcal{M}(\Sigma)$ be the collection of all positive unit Borel measures μ with support in Σ. Our definition for the weighted energy integral

$$I_w(\mu) = \int\int \ln|t-s|^{-1} \mathrm{d}\mu(t)\mathrm{d}\mu_x(s) + 2\int Q(s)\mathrm{d}\mu(s) \tag{4.24}$$

is recovered if both integrals exist and are finite. We can now move on to prove the existence of the *equilibrium measure* μ^Q for a given external field Q given by some polynomial

$$Q(x) = t_{2m}x^{2m} + t_{2m-1}x^{2m-1} + \cdots + t_0, \quad t_{2m} > 0, V(x) \geq 0. \tag{4.25}$$

Theorem 7. *(DEIFT,) Let Q be as in Equation 4.25. Then, there exists a unique probability measure $\mu = \mu^w$ such that*

$$E^w = \inf_{\mu\in\mathcal{M}^1} I^w(\mu) = I^w(\mu^w). \tag{4.26}$$

Moreover, μ^w has compact support.

4.2.1 Euler-Lagrange

Associates with the variational problem introduced on section 4.2

$$\begin{aligned} E^w &= \inf_{\mu\in\mathcal{M}^1} I^w(\mu) && (4.27)\\ &= \inf_{\mu\in\mathcal{M}^1} \left[\int\int \ln|t-s|^{-1}\mathrm{d}\mu(t)\mathrm{d}\mu(s) + 2\int Q(s)\mathrm{d}\mu(s)\right] && (4.28)\\ &= I^w(\mu^w) && (4.29)\end{aligned}$$

there are so-called *Euler-Lagrange* type variational equations. For some cases it is possible to solve these equations for μ^w.

Theorem 8. *(Variational Equations, (DEIFT,)) The equilibrium measure μ^w for the given variational problem satisfies the following conditions.*
There exists a real constant l such that

1. $$\int \left[2 \int \ln|x-y|^{-1} \mathrm{d}\mu^w(y) + 2Q(x) \right] \mathrm{d}\tilde{\mu}(x) \geq l \tag{4.30}$$

   for all $\tilde{\mu} \in \mathcal{M}^1(\mathbb{C})$ of compact support with $I^w(\tilde{\mu}) < \infty$.

2. $$2 \int \ln|x-y|^{-1} \mathrm{d}\mu^w(y) + 2Q(x) = l \tag{4.31}$$

   for μ^w-almost everywhere.

Conversely, if $\mu \in \mathcal{M}^1(\mathbb{C})$ has compact support, satisfies conditions (1) and (2) above, and $I^w(\mu) < \infty$, then $\mu = \mu^w$.

Proof. ($\Longrightarrow$) We recall that μ^w has compact support by Equation 7 and $I^w(\mu^w) = E^w < \infty$. Suppose $\mu^* \in \mathcal{M}^1(\mathbb{C})$ has compact support and $I^w(\mu^*) < \infty$. Then, defining some convex combination

$$\mu_t = t\mu^* + (1-t)\mu^w = \mu^w + t(\mu^* - \mu^w) \in \mathcal{M}^1(\mathbb{C}), \quad 0 \leq t \leq 1, \tag{4.32}$$

we should have

$$\begin{aligned} I^w(\mu_t) = \int\int \ln|t-s|^{-1} \mathrm{d}\mu^w(t)\mathrm{d}\mu^w(s) + 2\int Q(s)\mathrm{d}\mu^w(s) \\ + 2t \int\int \ln|t-s|^{-1} \mathrm{d}\mu^w(t)\mathrm{d}(\mu^* - \mu^w)(s) \\ + 2t \int Q(s)\mathrm{d}(\mu^* - \mu^w)(s) \\ + t^2 \int\int \ln|t-s|^{-1} \mathrm{d}(\mu^* - \mu^w)\mathrm{d}(\mu^* - \mu^w)(s). \end{aligned} \tag{4.33}$$

This is all well since μ^w, μ^* have compact support and $I^w(\mu^w), I^w(\mu^*) < \infty$. As $I^w(\mu_t) \geq I^w(\mu^w)$ for $0 \leq t \leq 1$ by assumption of minimality and Equation 1, is is necessary that

$$\int \left[2 \int \ln|x-y|^{-1} \mathrm{d}\mu^w(y) + 2Q(x) \right] \mathrm{d}(\mu^* - \mu^w)(x) \geq 0 \tag{4.34}$$

and therefore

$$\int \left[2 \int \ln|x-y|^{-1} \mathrm{d}\mu^w(y) + 2Q(x) \right] \mathrm{d}\mu^*(x) \geq l \tag{4.35}$$

where

$$\int \left[2 \int \ln|x-y|^{-1} \mathrm{d}\mu^w(y) + 2Q(x) \right] \mathrm{d}\mu^w(x) := l \tag{4.36}$$

proving the first assertion in (1). Now, let

$$B = \left\{ x \,\middle|\, 2 \int \ln|x-y|^{-1} \mathrm{d}\mu^w(y) + 2Q(x) < l \right\}. \tag{4.37}$$

Then, B must be a bounded set of $\mathbb{C}$. Suppose $\mu^*(B) > 0$ and set

$$\mu_B^* = \frac{\chi_B}{\mu^*(B)}\mu^* \tag{4.38}$$

where χ_B is the characteristic function of the set B. Then $\mu_B^* \in \mathcal{M}^1(\mathbb{C})$ has compact support and $I^w(\mu^*) < \infty$. inserting $\mathrm{d}\mu_B^*$ in Equation 4.35, we find

$$l \leq \int \left[2\int \ln|x-y|^{-1}\mathrm{d}\mu^w(y) + 2Q(x)\right]\mathrm{d}\frac{\chi_B}{\mu^*(B)}\mu^*(x) < l \tag{4.39}$$

a contradiction. Therefore,

$$\mu^*(B) = \mu^*\left(\left\{x \,\middle|\, 2\int \ln|x-y|^{-1}\mathrm{d}\mu^w(y) + 2Q(x) < l\right\}\right) = 0 \tag{4.40}$$

for all measures $\mu^* \in \mathcal{M}^1(\mathbb{C})$ with support compact and $I^w(\mu^*) < \infty$. in particular, it is true for $\mu^* = \mu^w$. Since,

$$0 = \int \left[2\int \ln|x-y|^{-1}\mathrm{d}\mu^w(y) + 2Q(x) - l\right]\mathrm{d}\mu^w(x) \tag{4.41}$$

$$= \int_{\mathbb{C}-B} \left[2\int \ln|x-y|^{-1}\mathrm{d}\mu^w(y) + 2Q(x) - l\right]\mathrm{d}\mu^w(x) \tag{4.42}$$

and (2) follows.

($\Longleftarrow$) If, now, $\mu \in \mathcal{M}^1$ satisfies (1) and (2), $I^w(\mu) < \infty$, and suppose μ has compact support. Then write $\mu^w = \mu + (\mu^w - \mu)$. We have, as before

$$I^w(\mu) \geq I^w(\mu^w) \tag{4.43}$$

$$\begin{aligned} &= I^w(\mu) + \int \left[2\int \ln|x-y|^{-1}\mathrm{d}\mu(y) + 2Q(x) - l\right]\mathrm{d}(\mu^w-\mu)(x) \\ &\qquad + \int\int \ln|t-s|^{-1}\mathrm{d}(\mu^w-\mu)(t)\mathrm{d}(\mu^w-\mu)(s). \end{aligned} \tag{4.44}$$

Using (2),

$$\int \left[2\int \ln|x-y|^{-1}\mathrm{d}\mu(y) + 2Q(x) - l\right]\mathrm{d}\mu^w(x) - l \geq l - l = 0 \tag{4.45}$$

but also the third term is strictly positive by Equation 1 unless it is identically null. If, however, $\mu^w - \mu \neq 0$, then we showed that

$$I^w(\mu) \geq I^w(\mu^w) > I^w(\mu), \tag{4.46}$$

a contradiction. Thus, if μ satisfies (1), (2), then it must be μ^w. □

If there is the knowledge that

$$\mathrm{d}\mu^w = \phi(x)\mathrm{d}x \tag{4.47}$$

for some continuous function ϕ of compact support, then

$$2\int \ln|x-y|^{-1}\mathrm{d}\mu(y) + 2Q(x) \tag{4.48}$$

is continuous and the previous theorem takes the following form

Theorem 9. *(Strong Variational Equations, (DEIFT,)) Suppose that* $\mathrm{d}\mu^w = \phi(x)\mathrm{d}x$ *for some continuous function* ϕ *of compact support. Then the conditions of Equation 7 can be replaced by*

1.
$$2\int \ln|x-y|^{-1}\mathrm{d}\mu^w(y) + 2Q(x) \geq l \tag{4.49}$$
for all $z \in \mathbb{C}$.

2.
$$2\int \ln|x-y|^{-1}\mathrm{d}\mu^w(y) + 2Q(x) = l \tag{4.50}$$
on $\{z : \phi(z) > 0\}$.

4.3 Recovering the Equilibrium Measure

We mentioned Equation 4.47 the possibility of knowing the density ϕ of the equilibrium measure μ^w in relation to the Lebesgue measure. We will know determine ϕ. As we did many times, assume that μ is finite and of compact support in the complex plane. For notation, let A denote the two-dimensional Lebesgue measure on $\mathbb{C}$ and Δ denote the Laplacian operator

$$\Delta F := \frac{\partial^2 F}{\partial x^2} + \frac{\partial^2 F}{\partial y^2}. \tag{4.51}$$

We have the theorem below.

Theorem 10. *(SAFF; TOTIK, 2024) If in a region* R *the potential* U^μ *has continuous second partial derivatives, then* μ *is absolutely continuous in* R *with respect to two-dimensional Lebesgue measure* A*, and on* R *we have the formula*

$$\mathrm{d}\mu = -\frac{1}{2\pi}\Delta U^\mu \mathrm{d}A \tag{4.52}$$

CHAPTER

5

ASYMPTOTICS

We have the sense in the previous chapters that we want to study some distributions when our matrices are large enough - be that for feasibility of computations or approximation of infinite dimension operators. We also have mentioned that our main interest here is in Invariant Ensembles, or specially in orthogonal polynomial ones. Because of this, studying large matrices is equivalent, in some sense, to studying the regime of large N for orthogonal polynomials. For some of those polynomials, a integral representation is available, allowing us to use asymptotic methods on integrals to solve for the large N behavior. This is the main discussion of this chapter.

5.1 Quadrature Technique

Before doing asymptotics on integrals we discuss a technique that will become important when approximating large summations: quadratures, i.e., approximating integrals by summations and vice-versa. One can see that asymptotics on integrals can then also perform a role on discrete summations. We first start briefly introducing a useful tool in the development of the formula we wish to use: the Bernoulli numbers B_n, polynomials $B_n(x)$ and periodic polynomials $\tilde{B}_n(x)$. The Bernoulli numbers B_n are a sequence of rational numbers that frequently appears in the Taylor series expansion of trigonometric-type functions. These number can also be seen as special values obtained by the Bernoulli polynomials $B_n(x)$. This is because these polynomials can be written explicitly as

$$B_k(x) = \sum_{n=0}^{k} \binom{k}{n} B_n x^{k-n}, \tag{5.1}$$

then, we must have that $B_k(0) = B_k$. More interestingly for us is the fact that these polynomials satisfy the differential equation

$$B_n'(x) = nB_{n-1}(x), \quad n = 1, 2, \cdots \tag{5.2}$$

Finally, the periodic Bernoulli polynomial is simply the Bernoulli polynomials evaluated at the fractional part of the argument, i.e., $\tilde{B}_n(x) = B_n(x - \lfloor x \rfloor)$. Now, let B_n and $B_n(x)$ denote the nth Bernoulli number and polynomial, respectively, and $\tilde{B}_n(x)$ the nth Bernoulli periodic function $B_n(x-\lfloor x \rfloor)$. Assume that a, m and n are integers such that $n > a$, $m > 0$, and $f^{(2m)}(x)$ is absolutely integrable over $[a, n]$. Then,

Theorem 11. *(Euler-Maclaurin Formula, (OLVER et al., 2010)) Following the context given above,*

$$\sum_{j=a}^{n} f(j) = \int_a^n f(x)\mathrm{d}x + \frac{1}{2}(f(a)+f(n)) + \sum_{s=1}^{m-1} \frac{B_{2s}}{(2s)!}\left(f^{(2s-1)}(n) - f^{(2s-1)}(a)\right)$$
$$+ \int_a^n \frac{B_{2m} - \tilde{B}_{2m}(x)}{(2m)!} f^{(2m)}(x)\mathrm{d}x. \quad (5.3)$$

This equation is called the *Euler-Maclaurin* formula. It can be extremely useful when estimating the asymptotic of the summation on the function f when its integral can be more easily evaluate. In essence, this is a quadrature technique; a numerical integration: the approximation of an integral $\int f(x)\mathrm{d}x$ of some function f by a discrete summation $\sum w_i f(x_i)$ over a set of points x_i with some weight function w_i. For example, in the classical Riemann sums, one approximates the integral $\int f(x)\mathrm{d}x$ over some interval $[a,b]$ with the limit of the summation $\sum f(x_i^*)\Delta x_i$, where $x_i^* \in [x_{i-1}, x_i]$ and $\Delta x_i = x_i - x_{i-1}$. Different methods can be generated by choosing x_i^* differently.

The trapezoidal rule, for example, is a Riemann sum where x_i^* is chosen to be the mean point $x_i^* = (x_{i-1}+x_i)/2$. In this way, we would estimate the area using trapezoids, as the name suggests. Then, one could write

$$\int_a^n f(x)\mathrm{d}x \approx \sum_{j=0}^{n} f(x_j^*) = \sum_{j=0}^{n} \frac{f(x_i + x_{i-1})}{2}, \quad (5.4)$$
$$= \sum_{j=1}^{n-1} f(x_j^*) + \frac{1}{2}(f(n)+f(0)) = \sum_{j=0}^{n-1} f(x_j^*) + \frac{1}{2}(f(n)-f(n)). \quad (5.5)$$

Of course, this has some error term. Suppose that for some (x_i, x_{i-1}) we have the same k-derivatives at both extreme points for every $k \geq 1$, i.e., the function is constant. Then, there is no error in the approximation. If however, the two extreme points differ only on the kth derivative, there is some curvature defined by this difference where the error comes from. Therefore, there is indication of some error function given in respect to higher order derivatives.

Proof. Following the computations in (ALPERT, 1999), take a function $f \in C^p(\mathbb{R})$ for $p \geq 1$. To derive the Euler-Maclaurin for the interval $[a,b]$, let $h = (b-a)/n$ and $c = a + jh$, then we can express

$$\int_a^b f(x)\mathrm{d}x = \sum_{j=0}^{n-1} \int_{a+jh}^{a+(j+1)h} f(x)\mathrm{d}x \quad (5.6)$$

where we can use the property (5.2) and integration by parts to rewrite each integral as

$$
\begin{aligned}
\int_c^{c+h} f(x)\mathrm{d}x &= h\int_0^1 B_0(1-x)f(c+xh)\mathrm{d}x && (5.7)\\
&= -h\frac{B_1(1-x)}{1!}f(c+xh)|_0^1 + h^2\int_0^1 \frac{B_1(1-x)}{1!}f'(c+xh)\mathrm{d}x, && (5.8)\\
&\vdots\\
&= -\sum_{r=0}^{p-1}\frac{h^{r+1}B_{r+1}(1-x)}{(r+1)!}f^{(r)}(c+xh)|_0^1 + h^{p+1}\int_0^1 \frac{h^{r+1}B_p(1-x)}{(p)!}f^{(p)}(c+xh)\mathrm{d}x, && (5.9)\\
&= h\frac{f(c)+f(c+h)}{2} - \sum_{r=0}^{p-1}\frac{h^{r+1}B_{r+1}(1-x)}{(r+1)!}\left[f^{(r)}(c+h)-f^{(r)}(c)\right]\\
&\qquad + h^{p+1}\int_0^1 \frac{B_p(1-x)}{(p)!}f^{(p)}(c+xh)\mathrm{d}x. && (5.10)
\end{aligned}
$$

where one can start to notice the trapezoidal rule creeping out on the first term. Notice that it was necessary to use that $-B_1(0) = B_1(1) = 1/2$ and $B_n(0) = B_n(1) = B_n$ for $n \neq 1$. Returning to the summation, we can rearrange terms as

$$
h\left[\frac{f(a)}{2} + f(a+h) + \cdots + f(b-h) + \frac{f(b)}{2}\right] = \frac{1}{2}(f(b)-f(a)) + \sum_{j=0}^{n-1} f(a+jh), \tag{5.11}
$$

$$
= \int_a^b f(x)\mathrm{d}x + \sum_{r=0}^{p-1}\frac{h^{r+1}B_{r+1}(1-x)}{(r+1)!}\left[f^{(r)}(b)-f^{(r)}(a)\right] - h^{p+1}\int_0^1 \frac{B_p(1-x)}{(p)!}f^{(p)}(c+xh)\mathrm{d}x. \tag{5.12}
$$

□

This is the Euler-Maclaurin formula, an extension of the trapezoidal rule including the higher order terms.

5.2 Asymptotic on Integrals

The techniques we will expand on here refers to a specific type of complex function $\phi(\lambda) : \mathbb{C} \to \mathbb{R}$ in the regimen where $\lambda \to \infty$. Those functions are of the form

$$
\phi(\lambda) := \int_\Gamma g(s)\,\mathrm{e}^{-\lambda f(s)}\,\mathrm{d}s, \tag{5.13}
$$

where Γ is a given contour in the plane and f, g are suitable functions. Suitable here means that we impose g and f to be continuous (or analytic); g is integrable; f have global minima with zero derivative (maybe in more than one point) and strictly positive second derivative at this point(s) of maximum. There is also a more technical regularity condition on f to ensure the approximations holds to still be made clear.

These conditions are not unnatural. Firstly, one can determine a large class of functions to which all of the requirements hold. Consider for example the following example.

Example 12. *(Airy Equation) Consider the function which satisfies the equation $\psi''(x) = x\psi(x)$. Taking the Fourrier transform we have*

$$-\omega^2\Phi(x) = \mathrm{i}\Phi'(x) \tag{5.14}$$

for which we can solve for $\Phi(x)$ and then take the inverse Fourrier Transform to get

$$\phi(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{i}/3\omega^3+\mathrm{i}x\omega}\,\mathrm{d}\omega. \tag{5.15}$$

The imaginary part of the argument is an odd function and therefore zero when integrated. The rest is expressed

$$\phi(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \cos\left(\frac{\omega^3}{3} + x\omega\right)\mathrm{d}\omega. \tag{5.16}$$

But how to know if this is the only solution? One can prove that for a more general definition of a Transform

$$\tilde{F}f(w) := \frac{1}{\sqrt{2\pi\mathrm{i}}}\int_\Gamma \mathrm{e}^{-\mathrm{i}\omega s} f(s)\mathrm{d}s \tag{5.17}$$

holds very similar properties of the Fourier Transformation and solves the equation for any contour that gives a convergent integral. We can actually choose independent contour such that each give us a different independent solution to the equation. More importantly, for this integral, the techniques applied here are of great use to understand its behavior as $\omega \to \infty$.

Moreover, one can somehow directly motive such conditions. Consider $\phi(\lambda) : \mathbb{R} \to \mathbb{R}$, f with a single global minima and $\Gamma \equiv \mathbb{R}$ - for simplicity; the argument's idea should hold given the appropriate considerations. If ϕ is as in Equation 5.13, and satisfying the conditions given for such function. Then, take x^* the single maximum critical point for f. We first note that if that is true, then, for any $|\epsilon| > 0$, the following limit should hold:

$$\lim_{\lambda\to\infty} \frac{\mathrm{e}^{-\lambda(f(x^*))}}{\mathrm{e}^{-\lambda(f(x^*-\epsilon))}} = \lim_{\lambda\to\infty} \mathrm{e}^{-\lambda(f(x^*)-f(x^*-\epsilon))} = \lim_{\lambda\to\infty} \mathrm{e}^{\lambda|c|} \to +\infty. \tag{5.18}$$

That is, the main contribution for the integral comes from a single point. The main idea then is to separate the integral in two regions: A first interval $(x^* - c/\sqrt{\lambda}, x^* + c/\sqrt{\lambda}), c > 0$, including this critical point and another interval that in theory should contribute only in error for the integral asymptotic. We start by approximating f around its global maximum by Taylor's theorem

$$f(x) = f(x^*) - \frac{1}{2}|f''(x^*)|(x-x^*)^2 + \epsilon_2(x-x^*) \tag{5.19}$$

and then perform a change of variables $\frac{x}{\sqrt{\lambda}} = s - x^*$, so that

$$\int_{x^*-\frac{c}{\sqrt{\lambda}}}^{x^*+\frac{c}{\sqrt{\lambda}}} g(s)\,\mathrm{e}^{-\lambda f(s)}\,\mathrm{d}(s) = \frac{\mathrm{e}^{-\lambda f(x^*)}}{\sqrt{\lambda}} \int_{-c}^{c} g\left(x^* + \frac{x}{\sqrt{\lambda}}\right) \mathrm{e}^{\frac{1}{2}|f''(x^*)|x^2 - \epsilon_2\left(\frac{x^3}{\sqrt{\lambda}}\right)}\,\mathrm{d}x. \tag{5.20}$$

Now, using the technical assumption of regularity on the function f, one can argue that the error term ϵ_2 goes to zero sufficiently fast, leaving a Gaussian Integral. More that that, one need to argue that the integral outside this small neighborhood of x^* is exponentially smaller then $\mathrm{e}^{\lambda\phi(x^*)}$. This should follow by an argument where the asymptotic for the integral

$$\int_{x^*+\frac{c}{\sqrt{\lambda}}}^{\infty} g(s)\,\mathrm{e}^{-\lambda f(s)}\,\mathrm{d}(s) \le \mathrm{e}^{-\lambda f(x^*+\frac{c}{\sqrt{\lambda}})}\,||g||_{L^1} \tag{5.21}$$

is exponentially less than the previous region.

With this, we only note a few details of interest. Firstly, more precise error estimates can be obtained if one uses higher order terms of the Taylor Series expansion of f. For maximizer x^* which have null second derivative (a double critical point), the Laplace Approximation can be used expanding to the third order terms of the Taylor Series $f(x) = f(x^*) = \frac{1}{6}f'''(x^*)(x-x^*)^3 + \epsilon_3(x-x^*)$ and making the change of variables $t/\sqrt[3]{n} = x - x^*$ instead. This argument and result could also be used on curves on the plane. This is motivated from the fact that every (smooth) curve is locally a straight line. This will lead us to the Method of *Steepest Descent*. These next subsections are in the spirit of proving a bit more formally the expansion given by the Laplace Method and introducing the mentioned Steepest Descent.

5.2.1 Watson's Lemma

We start this study in the real line by looking at the functions $F : \mathbb{R} \to \mathbb{R}$ of the form

$$F(\lambda) = \int_0^T f(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t \tag{5.22}$$

which can be identified as the Laplace transform of a function f supported on $[0,T]$. Note that this function fits well with the models we described previously, that is, it of the form such that one should expect the main contribution to come from the origin given some conditions on f. Let's formalize this result.

Theorem 12. *(Watson's Lemma) Take $f \in L^1(0,T)$ and suppose that for some $\delta > 0$ it is of the form*

$$f(t) = t^\alpha f_0(t), \quad 0 < t \le \delta, \tag{5.23}$$

where $\alpha \in (-1,\infty)$ and f_0 is continuously differentiable on $[0,\delta]$ with $f_0(0) \neq 0$. Then the function F in 5.22 satisfies

$$F(\lambda) = \frac{f_0(0)\Gamma(\alpha+1)}{\lambda^{\alpha+1}}(1 + O(\lambda^{-1})) \tag{5.24}$$

when $\lambda \to \infty$ and where Γ is the Gamma function.

Proof. A weak estimation of the integral could be given by

$$\int_0^T f(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t \le ||f||_{L^1} \tag{5.25}$$

where we only use the fact that $f \in L^1(0,T)$. However, we have more information on the behavior of the function in the section $0 < t < \delta$. To use this information we start by decomposing the function in two parts

$$F(\lambda) = \int_0^\delta f(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t + \int_\delta^T f(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t =: (1) + (2). \tag{5.26}$$

For (2), we can use the same weak estimate, leaving us with

$$F(\lambda) \le \int_0^\delta f(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t + ||f||_{L^1}\,\mathrm{e}^{-\delta\lambda}. \tag{5.27}$$

Now, we remind of the condition of continuous differentiability and the hypothesis on $f(t)$ to write (1) as

$$(1) = \int_0^\delta t^\alpha (f_0(0) + r(t))\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t = f_0(0)\int_0^\delta t^\alpha\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t + \int_0^\delta t^\alpha r(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t \tag{5.28}$$

$$=: f_0(0)\cdot(1.1) + (1.2). \tag{5.29}$$

For these two integrals we can try to simplify the problem by rewriting

$$(1.1) = \int_0^\infty t^\alpha\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t - \int_\delta^\infty t^\alpha\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} - \int_\delta^\infty t^\alpha\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t, \tag{5.30}$$

$$|(1.2)| \le ||f'||_{L^\infty}\int_0^\delta t^{\alpha+1}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t \tag{5.31}$$

$$= ||f'||_{L^\infty}\left(\int_0^\infty t^{\alpha+1}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t - \int_\delta^\infty t^{\alpha+1}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t\right) \tag{5.32}$$

$$= ||f'||_{L^\infty}\left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} - \int_\delta^\infty t^{\alpha+1}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t\right). \tag{5.33}$$

where $||f||_{L^\infty} = \inf\{c \ge 0 | \mu(\{|f| > c\}) = 0\}$ as usual. We are now only missing one ingredient to get the result, the order of

$$G(\lambda,\beta) := \int_\delta^\infty t^\beta\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t, \quad \lambda > 1, \quad \beta > -1. \tag{5.34}$$

Using Cauchy-Schwartz we can estimate

$$|G(\lambda,\beta)|^2 \le \left(\int_\delta^\infty \mathrm{e}^{-\lambda t}\,\mathrm{d}t\right)\left(\int_\delta^\infty t^{2\beta}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t\right) \tag{5.35}$$

$$= \left(\frac{e^{-\lambda\delta}}{\lambda}\right)\left(\int_\delta^\infty t^{2\beta}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t\right) \tag{5.36}$$

$$\overset{(u=\lambda t)}{=} \left(\frac{e^{-\lambda\delta}}{\lambda}\right)\left(\int_\delta^\infty \mathrm{e}^{-u}\,u^{2\beta}\frac{1}{\lambda^{2\beta+1}}\mathrm{d}u\right) \tag{5.37}$$

$$= \frac{\mathrm{e}^{-\lambda\delta}}{\lambda^{2\beta+2}}M_\beta \tag{5.38}$$

where M_β is some constant independent of λ. That leave us with

$$G(\lambda,\beta) = O\left(\frac{\mathrm{e}^{-\lambda\delta/2}}{\lambda^{\beta+1}}\right) = O\left(\mathrm{e}^{-\epsilon\lambda}\right) \tag{5.39}$$

where $\epsilon > 0$ depends on β and δ. Finally we can rewrite

$$(1.1) = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} - G(\lambda,\alpha) = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} + O\left(\mathrm{e}^{-\epsilon\lambda}\right) \tag{5.40}$$

$$|(1.2)| \le ||f'||_{L^\infty}\left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} - G(\lambda,\alpha+1)\right) = ||f'||_{L^\infty}\left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} + O\left(\mathrm{e}^{-\epsilon\lambda}\right)\right) = O\left(\lambda^{-\alpha-2}\right) \tag{5.41}$$

and then

$$F(\lambda) \le O\left(\lambda^{-\alpha-2}\right) + f_0(0)\left(\frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} + O\left(\mathrm{e}^{-\epsilon\lambda}\right)\right) + ||f||_{L^1}\,\mathrm{e}^{-\delta\lambda}. \tag{5.42}$$

By the dominance of the exponential error, we get the result

$$F(\lambda) = \frac{f_0(0)\Gamma(\alpha+1)}{\lambda^{\alpha+1}}(1 + O(\lambda^{-1})) \tag{5.43}$$

□

In the estimation above, the Gamma function and the factor of λ^{-1} is the result of integrating the linear exponent of $\mathrm{e}\,-\lambda t$. If we, however, change this exponent to be of the general type $-\lambda t^d$ we can still have a similar result on the behavior of the function.

Proposition 7. *For $T,S > 0$, define the function*

$$G(\lambda) := \int_{-S}^{T} g(t)\,\mathrm{e}^{-\lambda t^2}\,\mathrm{d}t \tag{5.44}$$

and assume $g(x)$ absolutely integrable in $[-S,T]$, of class C^1 in a fixed neighborhood of the origin. Then the function satisfies

$$G(\lambda) = \frac{g(0)\Gamma(1/2)}{\lambda^{1/2}} + O\left(\frac{1}{\lambda^{3/2}}\right) \tag{5.45}$$

when $\lambda \to \infty$ and where Γ is the Gamma function.

Proof. The proof is a reduction of the problem to Watson's Lemma. Assume, $T > S$ and write

$$G(\lambda) = \int_{-S}^{S} g(t)\,\mathrm{e}^{-\lambda t^2}\,\mathrm{d}t + \int_{S}^{T} g(t)\,\mathrm{e}^{-\lambda t^2}\,\mathrm{d}t \tag{5.46}$$

The second integral we know to be of order

$$\int_{S}^{T} g(t)\,\mathrm{e}^{-\lambda t^2}\,\mathrm{d}t \le ||g||_\infty\,\mathrm{e}^{-\lambda S^2}, \tag{5.47}$$

exponential on λ, so we drop it. For the first integral we rewrite

$$\int_{-S}^{S} g(t)\,\mathrm{e}^{-\lambda t^2}\,\mathrm{d}t = \int_{0}^{S} (g(t)+g(-t))\,\mathrm{e}^{-\lambda t^2}\,\mathrm{d}t \tag{5.48}$$

$$\overset{(u=t^2)}{=} \frac{1}{2}\int_{0}^{S^2} \left(\frac{g\left(\sqrt{u}\right)+g\left(-\sqrt{u}\right)}{\sqrt{u}}\right)\mathrm{e}^{-\lambda u}\,\mathrm{d}u \tag{5.49}$$

where now we can use Watson's Lemma. □

Example 13. *Take the function called Gamma given by the expression*

$$\Gamma(x) = \int_{0}^{\infty} \mathrm{e}^{-t}\,t^{x-1}\mathrm{d}t \tag{5.50}$$

We are interested in what happens to the function when we take the variable x to $+\infty$. To see what happens, let's fit it into Proposition 15. Rewrite equation 5.50 as

$$\int_{0}^{\infty} \mathrm{e}^{-t+x\log(t)-\log(t)}\,\mathrm{d}t. \tag{5.51}$$

This lead us to identify ϕ as $-\log(t)$, which would give us $\phi'(t) < 0$, that is, we would have it's minimum at $t = +\infty$. However $\mathrm{e}^{\log(t)} = t^1$ is not integrable near the origin. Something need to change about the identification. Let's introduce a variable to rescale the integral: $xs = t$, s.t.,

$$\Gamma(x) = x^x \int_{0}^{\infty} \mathrm{e}^{-\lambda(s-\log(s))}\,\mathrm{e}^{-s}\,\mathrm{d}s. \tag{5.52}$$

Where we have used $\lambda = x-1$. Thus we identify $\phi(s) = s-\log(s)$ and $g(s) = \mathrm{e}^{-s}$. in fact, now ϕ has a global minima at $s = 1$ and $\phi''(1) = 1 \neq 0$. Hence

$$\Gamma(x) = \frac{\sqrt{2\pi}x^x\,\mathrm{e}^{-x}}{(x-1)^{(-1/2)}}(1+O(x^{-1})) \tag{5.53}$$

$$= \sqrt{2\pi}x^{x-1/2}\,\mathrm{e}^{-x}(1+O(x^{-1})) \tag{5.54}$$

$$= (x-1/2)\log(x) - x + \frac{1}{2}\log(2\pi) \tag{5.55}$$

Which is called the Stirling formula.

5.2.2 Laplace Method

A seemly natural extension of the type of function we were dealing with are functions of the type

$$\Phi(\lambda) := \int_{a}^{b} g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t \tag{5.56}$$

for $a, b \in \mathbb{R}\cup\{-\infty,\infty\}$ and $a < b$. For now, ϕ is a real-valued function. To begin the study let's make some assumptions. The first one is that a is finite, that means that we can set $a = 0$ without loss of generality. The second assumption is that ϕ is a increasing function on $[0, b]$, so that $\phi(0) < \phi(t)$ for any t in the interval. With that we note that the coefficient $\mathrm{e}^{-\lambda\phi(t)}/\mathrm{e}^{-\lambda\phi(0)}$ decreases exponentially, that is, the integral is accumulated as the neighborhood of 0,

$$\Phi(\lambda) \approx \int_0^\delta g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t. \tag{5.57}$$

As we know $\phi(t)$ to be increasing, it's first non zero derivative must be of even order, so that when we expanded the function by Taylor,

$$\phi(t) \approx \phi(0) + \phi_{2k+1}t^{2k+1} \tag{5.58}$$

for some $k \geq 0$ and $|t| < \delta$. As ϕ_{2k+1}, the $(2k+1)$th-derivative, must be positive we also have that, as $\lambda \longrightarrow \infty$,

$$\Phi(\lambda) \approx \mathrm{e}^{-\lambda\phi(0)} \int_0^\delta g(t)\,\mathrm{e}^{-\lambda\phi_{2k+1}(t)\cdot t^{2k+1}}\,\mathrm{d}t \tag{5.59}$$

$$\overset{(u=t(\lambda\phi_{2k+1})^{1/(2k+1)})}{\approx} \mathrm{e}^{-\lambda\phi(0)} \frac{g(0)}{(\lambda\phi_{2k+1})^{\frac{1}{2k+1}}} \int_0^\infty \mathrm{e}^{-u^{2k+1}}\,\mathrm{d}u \tag{5.60}$$

$$\overset{(s=u^{2k+1})}{\approx} \mathrm{e}^{-\lambda\phi(0)} \frac{g(0)}{(2k+1)(\lambda\phi_{2k+1})^{\frac{1}{2k+1}}} \int_0^\infty \mathrm{e}^{-s}\,\mathrm{d}u \tag{5.61}$$

Where the integral can be evaluated as an Gamma function. More than anything we need to focus on an important assumption that was made: that ϕ has a global minima at one of the endpoints of integration. Is that were not the case, for example if $a < 0 < b$ and $\phi(t)$ had a global minima at $t = 0$ we would now have to consider the expansion

$$\phi(t) \approx \phi(0) + \phi_{2k}t^{2k} \tag{5.62}$$

which would lead us to the expression

$$\Phi(\lambda) \approx \mathrm{e}^{-\lambda\phi(0)} \frac{g(0)}{(\lambda\phi_{2k})^{\frac{1}{2k}}} \int_{-\infty}^\infty \mathrm{e}^{-u^{2k}}\,\mathrm{d}u \tag{5.63}$$

which differentiates of the previously result more importantly by the limits of integration. This is closely related to physical boundaries of the problem. Before enunciating the proper Laplace Method we state the Implicit Function Theorem, which will become important in a series of results that follows.

Theorem 13. *(Implicit Function Theorem) Assume that S is an open subset of $\mathbb{R}^{n+k}$ and that $F : S \longrightarrow \mathbb{R}^k$ is a function of class C^1. Assume also that (a,b) is a point in S such that*

$$F(a,b) = 0, \quad \textit{and} \quad \det(D_yF(a,b) \neq 0). \tag{5.64}$$

Then

1. *There exist $r_0, r_1 > 0$ such hat for every $x \in \mathbb{R}^n$ such that $|x-a| < r_0$, there exists a*

unique $y \in \mathbb{R}^k$ *such that* $|y-b| < r_1$

$$F(x,y) = 0. \tag{5.65}$$

That is, defines a function $y = f(x)$ *for* $x \in \mathbb{R}^n$ *near* a *with* $y = f(x)$ *close to* b.

2. *Moreover, the function* $f : B(r_0,a) \to B(r_1,b) \subset \mathbb{R}^k$ *from part* (1) *is of class* C^1, *and its derivatives may be determined by differentiating the identity*

$$F(x,f(x)) = 0 \tag{5.66}$$

and solving to find the partial derivatives of f

Now we turn to our interest.

Theorem 14. *(Laplace's Method, endpoint contributions) Let* Φ *be as in 5.56, with* a *finite and* $g \in L^1(a,b)$. *Assume that* ϕ *has a unique global minimum at* $t = a$. *In addition, suppose that for some* $\delta > 0$ *and some* $\alpha > -1$, *the function* g *takes the form*

$$g(t) = (t-a)^\alpha g_0(t) \tag{5.67}$$

for $|t-a| < \delta$ *and* g_0 *continuously differentiable in* $[a,a+\delta]$ *and* $g_0(a) \neq 0$, *whereas* ϕ *is twice continuously differentiable on* $[a,a+\delta]$ *with* $\phi'(a) > 0$. *Then the function* Φ *satisfies an estimate of the form*

$$\Phi(\lambda) = \frac{1}{\lambda^{\alpha+1}} \frac{g_0(a)\Gamma(\alpha+1)}{\phi'(a)^{\alpha+1}} \mathrm{e}^{-\lambda\phi(a)} \left(1 + O(\lambda^{-1})\right) \tag{5.68}$$

as $\lambda \to \infty$.

Proof. Start by assuming, without loss of generality, that $a = 0$ and dividing the integral in two intervals, one in the region where we have information about the behavior of g and another in the rest of the interval,

$$\Phi(\lambda) = \int_0^b g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t = \int_0^\delta g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t + \int_\delta^b g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t = (1) + (2). \tag{5.69}$$

Knowing that ϕ has a global minima at $t = 0$ we can assume, changing δ as needed, that $\phi(t) > \phi(\delta)$ for any $t > \delta$. That give us a simple limitation for (2):

$$(2) = \int_\delta^b g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t \leq \mathrm{e}^{-\lambda\phi(\delta)} \int_\delta^b g(t)\mathrm{d}t \leq \mathrm{e}^{-\lambda\phi(\delta)}\,||g||_{L^1[0,b]}. \tag{5.70}$$

We now get back to (1) and start by studying the difference give by $\phi(t) - \phi(0) = s$ in a similar way that we would think of a change of variable. Define the multivariate function

$$H(t,s) := \phi(t) - \phi(0) - s \tag{5.71}$$

such that $H(0,0) = 0$ and $\partial_t H(0,0) \neq 0$. This means we are in the condition to use the Theorem of the Implicit Function to say that there exists a unique bijective and continuously differentiable function $t(s) := t$ such that

$$\begin{cases} \phi(t(s)) - \phi(0) - s = 0, \\ t'(0) = \frac{\partial_s H(0,0)}{\partial_t H(0,0)} = \frac{1}{\phi'(0)}. \end{cases} \tag{5.72}$$

If we set $T = t^{-1}(\delta)$ we would then have

$$\int_0^\delta g(s)\,\mathrm{e}^{-\lambda\phi(s)}\,\mathrm{d}s = \int_0^T g(t(s))\,\mathrm{e}^{-\lambda\phi(t(s))}\,t'(s)\mathrm{d}s, \tag{5.73}$$

for now to apply the assumption on g. We write $g(t(s)) = t^\alpha(s)g_0(t(s))$ and because of its Taylor expansion,

$$t(s) = st'(0)(1+r(s)) = \frac{s}{\phi'(0)}(1+r(s)). \tag{5.74}$$

Joining both results,

$$\int_0^T g(t(s))\,\mathrm{e}^{-\lambda\phi(t(s))}\,t'(s)\mathrm{d}s = \mathrm{e}^{\lambda\phi(0)}\int_0^T g(t(s))\,\mathrm{e}^{-\lambda(\phi(t(s))-\phi(0))}\,t'(s)\mathrm{d}s \tag{5.75}$$

$$= \mathrm{e}^{\lambda\phi(0)}\int_0^T t^\alpha(s)g_0(t(s))\,\mathrm{e}^{-\lambda s}\,t'(s)\mathrm{d}s \tag{5.76}$$

$$= \frac{\mathrm{e}^{\lambda\phi(0)}}{\phi'(0)^\alpha}\int_0^T s^\alpha(1+r(s))^\alpha g_0(t(s))\,\mathrm{e}^{-\lambda s}\,t'(s)\mathrm{d}s \tag{5.77}$$

where, if we set $f(s) := s^\alpha(1+r(s))^\alpha g_0(t(s))$ we can apply Watson's Lemma. □

Theorem 15. *(Laplace's Method, interior contributions) Let Φ be as in 5.56, with a finite and $g \in L^1(a,b)$. Assume that ϕ has a unique global minimum at $t = c \in (a,b)$. In addition, suppose that for some $\delta > 0$ and some $\alpha > -1$, the function g takes the form*

$$g(t) = (t-a)^\alpha g_0(t) \tag{5.78}$$

for $|t-a| < \delta$ and g_0 continuously differentiable in $[c-\delta, c+\delta]$ and $g_0(a) \neq 0$, whereas ϕ is three times continuously differentiable on $[c-\delta, c+\delta]$ with $\phi'(a) > 0$. Then the function Φ satisfies an estimate of the form

$$\Phi(\lambda) = \mathrm{e}^{-\lambda\phi(c)}\left(\frac{\sqrt{2\pi}g(c)}{\sqrt{|\phi''(c)|}\lambda^{1/2}} + O(\lambda^{-3/2})\right) \tag{5.79}$$

as $\lambda \to \infty$.

Proof. Although the proof may be similar there is an important distinction in the solution. Suppose we wanted to try the same approach. We would first assume, w.l.g., that c = 0. Then, we could write

$$\Phi(\lambda) = \int_a^b g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t \tag{5.80}$$

$$= \int_a^{-\delta} g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t + \int_{-\delta}^{\delta} g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t + \int_{\delta}^{b} g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t \tag{5.81}$$

$$=: (1) + (2) + (3). \tag{5.82}$$

For (1) and (3) the result is as it was before

$$\begin{cases} (1) = \int_\delta^a g(-t)\,\mathrm{e}^{-\lambda\phi(-t)}\,\mathrm{d}t \leq \mathrm{e}^{\lambda\phi(-\delta)}\,||g||_{L^1[a,b]} \\ (3) = \int_\delta^b g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t \leq \mathrm{e}^{\lambda\phi(\delta)}\,||g||_{L^1[a,b]} \end{cases} \tag{5.83}$$

For (2) we would try to define a function

$$H(t,s) := \phi(t) - \phi(0) - s^2. \tag{5.84}$$

However we could not apply the Implicit Function Theorem here as its derivative $\partial_t H = \phi'$ vanishes at $t = 0$. A nice fact about this indetermination is that it doesn't come from the fact that there are no solutions to $H(t,s) = 0$ but from the fact that there are two. To solve this we will use a technique called *blow up*. First, we introduce a variable $v = v(t)$ such that $vs = t$. Now, we redefine our function of interested to be

$$L(s,v) := \frac{\phi(sv) - \phi(0)}{s^2} - 1. \tag{5.85}$$

for $s \in (-\delta,\delta) - \{0\}$. We would think this function is also singular but note that, using the Taylor Expansion of ϕ we can write

$$\phi(t) = \frac{\phi''(0)}{2}t^2 + t^3 R(t) \tag{5.86}$$

which leave us with

$$L(s,v) = \frac{\phi''(0)}{2}v^2 + sv^3 R(sv) - 1 \tag{5.87}$$

where we can indeed apply the Implicit Function Theorem. In fact, we know $L \in C^2$ and as we want to solve for $L(s,v) = 0$, we need that the v_0 we take to satisfy $L(0,v_0) = 0$, ie,

$$v_0 = \sqrt{\frac{2}{\phi''(0)}}. \tag{5.88}$$

Now, we also know that

$$\partial_v L(0,v_0) = \phi''(0)v_0 \neq 0. \tag{5.89}$$

So we get a bijective and continuously differentiable function $v = v(s)$ such that

$$\begin{cases} L(s,v(s)) = 0 \\ v(0) = v_0 \end{cases} \tag{5.90}$$

But, $L(s, v(s))$ means exactly that

$$\phi(t) - \phi(0) = s^2 \tag{5.91}$$

with $t = t(s) = sv(s)$. Getting back to our equation, we have that

$$\mathrm{e}^{\lambda\phi(0)} \int_{-\delta}^{\delta} g(t)\,\mathrm{e}^{-\lambda(\phi(t)-\phi(0))}\,\mathrm{d}t = \mathrm{e}^{\lambda\phi(0)} \int_{\alpha}^{\beta} g(sv(s))(v(s) + sv'(s))\,\mathrm{e}^{-\lambda s^2}\,\mathrm{d}t \tag{5.92}$$

where we can easily calculate $\alpha = -\sqrt{\phi(-\delta) - \phi(0)}$ and $\beta = \sqrt{\phi(\delta) - \phi(0)}$. We finish the proof with Proposition 7. □

Example 14. *(NICA; ORTMANN, 2025) For $x \in \mathbb{R}$ fixed, i.e. $x = o(1)$, we have the following asymptotic as $n \to \infty$:*

$$\mathrm{He}_n(x) = \sqrt{2}\left(\frac{n}{2}\right)^{n/2} \mathrm{e}^{x^2/4} \cos\left(\frac{n\pi}{2} - x\sqrt{n}\right)(1 + o(1)) \tag{5.93}$$

To see that, consider the integral representation of the Hermite Polynomial and re-write

$$\mathrm{He}_n(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (x + \mathrm{i}t)^n\,\mathrm{e}^{-t^2/2}\,\mathrm{d}t = \frac{2}{\sqrt{2\pi}} \Re\left\{\int_0^{\infty} (x + \mathrm{i}t)^n\,\mathrm{e}^{-t^2/2}\,\mathrm{d}t\right\} \tag{5.94}$$

$$= \frac{2}{\sqrt{2\pi}} \Re\left\{\int_{\Re(z)=x, Im(z)>0} \mathrm{e}^{n\ln(z) + \frac{1}{2}(z-x)^2}\,\frac{\mathrm{d}z}{\mathrm{i}}\right\} \tag{5.95}$$

$$= \frac{2}{\sqrt{2\pi}} \Re\left\{\int_{\Re(z)=0, Im(z)>0} \mathrm{e}^{n\ln(z) + \frac{1}{2}(z-x)^2}\,\frac{\mathrm{d}z}{\mathrm{i}}\right\} \tag{5.96}$$

where you start with a line integral in $\mathbb{C}$ and move the contour over to the positive imaginary axis - justified by the Cauchy's Theorem and the exponential decay of the integrand near infinity. Now, we focus on $n\ln(z) + \frac{1}{2}(z-x)^2$ which has maximum at $z^* = \mathrm{i}\sqrt{n} + o(\sqrt{n})$. The idea is to change variables by a factor of $\sqrt{z}$ so that the location of the maximum contribution stays $O(1)$ as n gets large. Specifically, we do

$$z = \mathrm{i}\sqrt{n}s, \quad \frac{\mathrm{d}z}{\mathrm{i}} = \sqrt{n}\mathrm{d}s, \quad \ln(z) = \ln(\sqrt{n}) + \ln(s) + \mathrm{i}\frac{\pi}{2} \tag{5.97}$$

which give us

$$\frac{1}{2}(z-x)^2 = -\frac{1}{2}ns^2 + \frac{1}{2}x^2 - \mathrm{i}\sqrt{n}sx \tag{5.98}$$

so that we remain with

$$\mathrm{He}_n(x) = \frac{2}{\sqrt{2\pi}} \Re\left\{\int_0^{\infty} \mathrm{e}^{n\ln(\sqrt{n})}\,\mathrm{e}^{n\ln(s)}\,\mathrm{e}^{n\mathrm{i}\frac{\pi}{2}}\,\mathrm{e}^{\frac{1}{2}x^2}\,\mathrm{e}^{-\mathrm{i}\sqrt{n}sx}\,\sqrt{n}\mathrm{d}s\right\} \tag{5.99}$$

$$= \frac{2}{\sqrt{2\pi}} \sqrt{n}\,\mathrm{e}^{n\ln(\sqrt{n})}\,\mathrm{e}^{\frac{1}{2}x^2} \int_0^{\infty} \mathrm{e}^{n\left(\ln(s) - \frac{1}{2}s^2\right)} \cos\left(n\frac{\pi}{2} - \sqrt{n}sx\right)\mathrm{d}s, \tag{5.100}$$

where we can apply Laplace's approximation discussed before. Noting that $\phi(s) = \ln(s) - \frac{1}{2}s^2$ has $\phi'(s) = \frac{1}{s} - s$ and $\phi''(s) = -\frac{1}{s^2} - 1$, with global maximum at $s^* = 1$ with $\phi(1) = \frac{1}{2}$ and $\phi''(1) = -2$, yielding

$$\mathrm{He}_n(x) = \frac{2}{\sqrt{2\pi}} \sqrt{n}\,\mathrm{e}^{n\ln(\sqrt{n})}\,\mathrm{e}^{\frac{1}{2}x^2} \left(\mathrm{e}^{n\phi(1)} \sqrt{\frac{2\pi}{n|\phi''(1)|}}\right) \cos\left(n\frac{\pi}{2} - \sqrt{n}x\right)(1 + o(1)), \tag{5.101}$$

which simplifies to the result.

5.2.3 Steepest Descent

Let's now finally explore integrals over the complex plane with complex valued functions. We retake the integral

$$\phi(\lambda) := \int_\Gamma g(s)\,\mathrm{e}^{-\lambda f(s)}\,\mathrm{d}s, \tag{5.102}$$

where Γ is a contour in the complex plane and f, g are given complex-valued, analytic functions. Suppose that Γ is parametrized by $\gamma : [a,b] \to \mathbb{C}$ and let us decompose the functions we have into their real and imaginary parts such as

$$f(z) = u(z) + \mathrm{i}v(z), \quad g(z) = p(z) + \mathrm{i}q(z), \quad \gamma(t) = \alpha(t) + \mathrm{i}\beta(t) \tag{5.103}$$

with all functions in the decomposition being real valued. Of course we now have

$$\mathrm{e}^{-\lambda f(z)} = \mathrm{e}^{-\lambda u(z)}\,\mathrm{e}^{-\mathrm{i}v(z)}, \tag{5.104}$$

and when λ grows large, we have the exponentially growing/decaying term $\mathrm{e}^{-\lambda u(z)}$ and a oscillating term coming from $\mathrm{e}^{\mathrm{i}v(z)}$ which is hard to control. The idea to get around this fact is to use paths in which this term is constant. If we have such a case, we would write

$$\begin{aligned}\Phi(\lambda) = \mathrm{e}^{-\mathrm{i}\lambda v_0} &\int_a^b (p(\gamma(t))\alpha'(t) - q(\gamma(t))\beta'(t))\,\mathrm{e}^{-\lambda u(\gamma(t))}\,\mathrm{d}t \\ &+ \mathrm{i}\,\mathrm{e}^{-\mathrm{i}\lambda v_0}\int_a^b (p(\gamma(t))\beta'(t) - q(\gamma(t))\alpha'(t))\,\mathrm{e}^{-\lambda u(\gamma(t))}\,\mathrm{d}t\end{aligned} \tag{5.105}$$

where each integral can now be solved by Laplace's method.

Theorem 16. *(Steepest Descent) Consider Φ as in 5.102 with a smooth contour Γ. Assume that the exponent $f = u + \mathrm{i}v$ is an analytic function on a neighborhood of Γ that satisfies*

$$v(z) \equiv v_0, \quad z \in \Gamma \tag{5.106}$$

In addition, suppose that u has a unique minimum at an interior point of Γ, say on $z_0 \in \Gamma$, with $f''(z_0) \neq 0$, and also that g is analytic on a neighborhood of Γ and satisfies

$$\int_\Gamma |g(z)||dz| < \infty. \tag{5.107}$$

Then Φ has a leading asymptotic as $\lambda \to +\infty$ given by

$$\Phi(\lambda) = \mathrm{e}^{-\lambda f(z_0)}\left(\frac{\sqrt{2\pi}\,g(z_0)\,\mathrm{e}^{\mathrm{i}\theta(z_0)}}{\lambda^{1/2}\sqrt{|f''(z_0)|}} + O(\lambda^{-3/2})\right), \quad \lambda \to +\infty, \tag{5.108}$$

where $\theta(z_0)$ is the angle of the oriented tangent of Γ at z_0.

Proof. Fix a parametrization $\gamma : [a,b] \to \mathbb{C}$ of Γ with $\gamma(c) = z_0, a < c < b$, and decompose Φ as before in 5.105. By the assumptions on the functions f, g we can apply Laplace's method on both equations, and yields, as $\lambda \to +\infty$,

$$\int_a^b (p(\gamma(t))\alpha'(t) - q(\gamma(t))\beta'(t))\, \mathrm{e}^{-\lambda u(\gamma(t))}\, \mathrm{d}t = \sqrt{2\pi}\, \mathrm{e}^{-\lambda u(z_0)} \left(\frac{p(z_0)\alpha'(c) - q(z_0)\beta'(c)}{\sqrt{|(u\circ\gamma)''(c)|}\lambda^{1/2}} + O\left(\lambda^{3/2}\right) \right) \tag{5.109}$$

and

$$\int_a^b (p(\gamma(t))\beta'(t) - q(\gamma(t))\alpha'(t))\, \mathrm{e}^{-\lambda u(\gamma(t))}\, \mathrm{d}t = \sqrt{2\pi}\, \mathrm{e}^{-\lambda u(z_0)} \left(\frac{p(z_0)\beta'(c) - q(z_0)\alpha'(c)}{\sqrt{|(u\circ\gamma)''(c)|}\lambda^{1/2}} + O\left(\lambda^{3/2}\right) \right) \tag{5.110}$$

Using that $v_0 + u(z_0) = f(z_0)$ and also that,

$$(p(z_0)\alpha'(c) - q(z_0)\beta'(c)) + \mathrm{i}(p(z_0)\beta'(c) - q(z_0)\alpha'(c)) = g(z_0)\gamma'(c). \tag{5.111}$$

Combining the estimative for both parts we get

$$\Phi(\lambda) = \sqrt{2\pi}\, \mathrm{e}^{-\lambda f(z_0)} \left(\frac{g(z_0)\gamma'(c)}{\sqrt{|(u\circ\gamma)''(c)|}\lambda^{1/2}} + O\left(\lambda^{3/2}\right) \right) \tag{5.112}$$

and it suffices now to get an expression for the second derivative in the denominator. Knowing the imaginary part is constant we can write

$$\frac{\mathrm{d}}{\mathrm{d}t} u(\gamma(t)) = \frac{\mathrm{d}}{\mathrm{d}t} f(\gamma(t)) = \gamma'(t) f'(\gamma(t)) \tag{5.113}$$

and

$$\frac{\mathrm{d}^2}{\mathrm{d}t} u(\gamma(t)) = \frac{\mathrm{d}^2}{\mathrm{d}t} f(\gamma(t)) = (\gamma'(t))^2 f''(\gamma(t)) + \gamma''(t) f'(\gamma(t)). \tag{5.114}$$

Hence,

$$\frac{\gamma'(c)}{\sqrt{|(u\circ\gamma)''(c)|}} = \frac{1}{\sqrt{|f''(z_0)|}} \frac{\gamma'(c)}{|\gamma'(c)|} \tag{5.115}$$

which concludes the proof. □

We now need do discuss why this assumption of constant value of $Im(f(z))$ is not an assumption as strong as one might assume at first. Suppose we want to integrate in a random contour Γ of the plane. Of course, our assumption that the functions were analytic permit us to deform the contour keeping the value of the integral. We are going to use this to find an contour $\tilde{\Gamma}$ such that sections of the contour are such that $Im(f(z))$ is constant. We already know that being analytic means satisfying the *Cauchy-Riemann* equations,

$$u_x(z) = v_y(z), \quad \text{and} \quad u_y(z) = -v_x(z). \tag{5.116}$$

An immediate result from these equation is that the gradients of u and v must be normal. Indeed, if $\langle \cdot, \cdot \rangle$ is the inner product on $\mathbb{R}^2$,

$$\langle \nabla u(z), \nabla v(z) \rangle = u_x(z)v_x(z) + u_y(z)v_y(z) = u_x(z)v_x(z) - v_x(z)u_x(z) = 0.$$

As we're following the curves where I$m(f(z))$ is constant, we're also following the path of Steepest Descent of the real part of the integral. As we also are looking to use Laplace's method on the integrals, we may want to have our paths going through the critical points of the function. This fully motivates the method described.

Example 15. *There are three asymptotics regimes for* $\mathrm{He}_n(x)$ *when* $x \to \infty$ *depending on the value of* $\lim_{N\to\infty} x/2\sqrt{N}$ *for* $N = n+1$, *each of which has qualitatively different behavior*

- *Small x values* $-2\sqrt{N} < x < 2\sqrt{N}$*: in this regime the Hermite Polynomials will be oscillating, and a have a similar qualitative behavior as in Example 14, with modification to phase and amplitude. These polynomials will have some connections to the arcsine law and semi-circle law.*
- *Large x values* $|x| > 2\sqrt{N}$*: In this regime the polynomial doesn't have more oscillations and is monotone increasing. As x grows larger, this approximation behaves* $\approx x^N \,\mathrm{e}^{-N^2/2x^2}$.
- *Edge x values* $x = \pm 2\sqrt{N} + O(N^{-\frac{1}{6}})$*: in this regime the polynomials will be approximated by the Airy Function. This interpolates between the oscillatory small x regime and the monotone large x regimes.*

We write the integral representation given by:

$$\begin{aligned}\mathrm{He}_n(x) &:= \frac{n!}{2\pi \mathrm{i}} \int_\Gamma z^{-n-1}\,\mathrm{e}^{xz-\frac{z^2}{2}}\,\mathrm{d}z \\ &= \frac{n!}{2\pi \mathrm{i}} \int_\Gamma \mathrm{e}^{xz-\frac{z^2}{2}-(n+1)\ln(z)}\,\mathrm{d}z\end{aligned} \tag{5.117}$$

where Γ *is a contour that goes around the origin coming and going from minus infinity such as* $\Gamma = \{\lambda - \mathrm{i}\delta : \lambda < 0\} \cup \{\delta\,\mathrm{e}^{\mathrm{i}\theta} : -\pi/2 < \theta < \pi/2\} \cup \{\lambda + \mathrm{i}\delta : \lambda < 0\}$. *This is equivalent to using* $\Gamma = \{\mathrm{e}^{\mathrm{i}\theta} : 0 \le \theta < 2\pi\}$. *To determine* He_n, *set* $N = n+1$ *and consider the scaling* $x \mapsto N^a\alpha$ *and* $z = N^b\beta$

$$\begin{aligned}\mathrm{He}_n(N^a\alpha) &= \frac{(N-1)!}{2\pi\mathrm{i}} \int_\Gamma \mathrm{e}^{(N^a\alpha)(N^b\beta)-\frac{1}{2}(N^b\beta)^2-N\ln(N^b\beta)}\,\mathrm{d}\beta \\ &\overset{(a=b=1/2)}{=} \frac{(N-1)!N^{\frac{1}{2}(1-N)}}{2\pi\mathrm{i}} \int_\Gamma \mathrm{e}^{-N(-\alpha\beta+\frac{1}{2}\beta^2+\ln(\beta))}\,\mathrm{d}\beta\end{aligned} \tag{5.118}$$

So that we have

$$\mathrm{He}_n(\sqrt{N}\alpha) = C_n^0 \int_\Gamma \mathrm{e}^{-N\phi_\alpha(\beta)}\,\mathrm{d}\beta \tag{5.119}$$

with

$$\phi_\alpha(\beta) = \frac{1}{2}\beta^2 - \alpha\beta + \ln(\beta). \tag{5.120}$$

As we want to calculate the asymptotic by steepest descent, we need to determine the critical points of such a function and then the steepest descent paths passing through this points. We can easily determine the critical points as $\beta_c^{\pm} = \frac{\alpha \pm \sqrt{\alpha^2-4}}{2}$ *- solutions to the quadratic formula given by solving the critical point of the function (the zeros of its derivatives). Now, the behavior of the critical points will depend on the interval of the real parameter* α*.*

$$\beta_c^{\pm} = \frac{\alpha \pm \sqrt{\alpha^2 - 4}}{2} = \begin{cases} \mathrm{e}^{\pm\gamma}, & \alpha = 2\cosh(\gamma) > 2, \quad \gamma > 0; \\ \mathrm{e}^{\pm \mathrm{i}\gamma}, & \alpha = 2\cos(\gamma) \in [0,2), \quad \gamma \in (0, \pi/2]; \\ 1, & \alpha = 2. \end{cases} \tag{5.121}$$

each of the cases refers to one of the regimes we described before.

CHAPTER

6

PLANAR ENSEMBLES

We are going to discuss in this chapter the ideas developed in the paper (BYUN; KANG; SEO, 2023), where asymptotics were develop for a large class of random matrix ensembles using relatively straightforward asymptotic techniques taking advantage of the orthogonal polynomials present in these models. In the end, we hope to clarify the origin for the novel limit-behavior expression for the so-called *Partition Function* $\mathcal{Z}_N^\beta$. Namely, we get an expression of the type:

$$\ln(\mathcal{Z}_{N,\beta=2}^Q) = -I^Q N^2 + \frac{1}{2} N \ln N + E^Q N + G \ln N + F^Q + o(1). \tag{6.1}$$

These functions perform the role of the normalization for the probability measure relative to the Coulomb Gas being worked on - a system of interacting particles confined by some potential. In this instance, we deal with complex plane gases, parameterized by some variable $\beta = 2$. This relates in RMT, to the famous *random normal matrix* and *planar symplectic ensembles*, depending on the symmetry imposed. If one takes, for example, the quadratic potential keeping the $2-\beta$ regime, one gets ensembles correspondent to the *complex* and *planar* Ginibre ensembles, falling back into the classical models. For the results, another restriction is made, the confining potentials must be radially symmetric. For a large class of potentials Q, the first term I^Q was known for a long time, being given by the energy of the associated equilibrium measure. The coefficient $1/2$ of the order $N \ln N$, as well as the entropy term E^Q, were known from recent work of Leblé and Serfaty (LEBLÉ; SERFATY, 2017), also for rather large classes of Q. The remaining terms in this expansion are new. The term G appears to depend only on whether the limiting spectrum is an annulus or a disc which, surprisingly, seems to not have been observed in the previous literature.

Although some recent work may indicate the direction to expand some of these results to larger classes of potentials (HEDENMALM; WENNMAN, 2020), this is, so far, still a necessary condition for the techniques presented henceforth. Once the model is established, the techniques are involved, but should be reasonable: We express the partition function in terms of a summation of some orthogonal polynomials, for which we have a integral representation.

Then, the integral can be estimated using the Laplace technique - separating the integral in a central main-contributing region and a bounded error-contributing complement region. Finally, the limit on the summation can be estimated with some quadrature techniques, namely the Euler-Maclaurin formula, giving us the final results after computations.

6.1 Solving for the Free Energy

Check Introduction to Random Matrices theory and Practice Chapter 4 for more details.

6.2 Working Model

In this section we properly define the model and specifics of the working model to which we will apply the asymptotics techniques. We have discussed in **??** how eigenvalue statistics can be interpret as some appropriate Coulomb gas. On the complex plane one could have for example the Coulomb gas

$$\mathrm{dp}_N^{(\beta)}(z_1,\cdots,z_N) := \frac{1}{\mathcal{Z}_N^{(\beta)}}\prod_{j>k=1}^{N}|z_j-z_k|^\beta\prod_{j=1}^{N}\mathrm{e}^{-\frac{\beta N}{2}Q(z_j)}\,\mathrm{d}A(z_j) \tag{6.2}$$

where $\mathrm{d}A(z) := \mathrm{d}^2z/\pi$ is the area measure. This measure would imply the normalization constant - or in our context, the *Partition Function*,

$$\mathcal{Z}_N^{(\beta)} := \int_{\mathbb{C}^N}\prod_{j>k=1}^{N}|z_j-z_k|^\beta\prod_{j=1}^{N}\mathrm{e}^{-\frac{\beta N}{2}Q(z_j)}\,\mathrm{d}A(z_j). \tag{6.3}$$

Definition 11. $\mathrm{p}_N^{(\beta)}$ *defined as above is the expression, if* $\beta = 2$*, for the measure of the* ***random normal matrix*** *ensemble. If, for example, one sets* $Q(z) = |z|^2$ *this would be the so-called* ***complex Ginibre ensemble****.*

We could also define the measure for the configurational canonical Coulomb gas ensemble in the upper-half plane - which has an additional complex conjugation symmetry and is governed by the law

$$\mathrm{d}\tilde{\mathrm{p}}_N^{(\beta)}(z_1,\cdots,z_N) := \frac{1}{\mathcal{Z}_N^{(\tilde{b}eta)}}\prod_{j>k=1}^{N}|z_j-z_k|^\beta|z_j-\bar{z}_k|^\beta\prod_{j=1}^{N}|z_j-\bar{z}_j|^\beta\,\mathrm{e}^{-\frac{\beta N}{2}Q(z_j)}\,\mathrm{d}A(z_j) \tag{6.4}$$

and would imply the normalization constant

$$\mathcal{Z}_N^{(\tilde{b}eta)} := \int_{\mathbb{C}^N}\prod_{j>k=1}^{N}|z_j-z_k|^\beta|z_j-\bar{z}_k|^\beta\prod_{j=1}^{N}|z_j-\bar{z}_j|^\beta\,\mathrm{e}^{-\frac{\beta N}{2}Q(z_j)}\,\mathrm{d}A(z_j) \tag{6.5}$$

Definition 12. $\tilde{\mathrm{p}}_N^{(\beta)}$ *defined as above is the expression, if* $\beta = 2$*, for the measure of the* ***planar symplectic ensemble****. For* $Q(z) = |z|^2$ *this would be the so-called* ***symplectic Ginibre ensemble****.*

Virtually all results we saw for the random normal matrix ensemble are valid in the context of the planar symplectic ensemble with only a few adaptions. However, in the name of history telling and simplicity we will omit this measure: Adding this case to our chapter would cloud the point. For now on, we will only consider $\beta = 2$, this has some algebraic advantages that become clear by the Determinantal Point Processes (see Chapter 3 for details) - because of this, we omit the β in the variables ($\mathcal{Z}_N^{(\beta)} := \mathcal{Z}_N$).

For the models of random matrices we are interested in, we must consider $\{P_n(z) = \sum_{j=0}^n a_j z^j\}_n^{(w)}$ a family of orthogonal polynomials in respect to w where P_m has order m. We're dealing with weight functions of the type

$$w(z) = \mathrm{e}^{-NQ(z)}, \tag{6.6}$$

where $Q(z)$ is some complex potential that satisfies:

1. We have $Q(z) = q(|z|)$ for some $q : \mathbb{R} \mapsto \mathbb{R}$, i.e., Q is a radially symmetric potential;
2. The potential Q grows sufficiently fast, that is,
$$\liminf_{|z|\to\infty} \frac{Q(z)}{2\ln(|z)} > 1; \tag{6.7}$$
3. Furthermore, assume Q to be is smooth, subharmonic in $\mathbb{C}$, and strictly subharmonic in a neighborhood of the droplet.

There is still a question of what are these polynomials exactly.

Lemma 2. *The set* $\{z^j\}_j^{(w)}$ *is the proper orthogonal family in relation to the weight function* $w(z) = \mathrm{e}^{-NQ(z)}$ *given.*

Proof. We know by the assumption of orthogonality that

$$\langle P_n, P_m \rangle_w = \begin{cases} 0 & \text{if } m = n, \\ 1 & \text{if } m \neq n. \end{cases} \tag{6.8}$$

Let us take some arbitrary base of the polynomial space $\{z^j\}_j^{(w)}$. We can apply the Gram Schmidt scheme to get the proper orthogonality in respect to the weight w. If $P_n = x^n$, then the orthogonal element of the basis $\tilde{P}_n$ should be

$$\tilde{P}_n = P_n - \sum_{j=0}^{n-1} \frac{\langle P_n, P_j \rangle_w}{\langle P_j, P_j, \rangle_w} P_j, \tag{6.9}$$

but notice that

$$\langle P_n, P_j\rangle_w = \int_{\mathbb{C}} z^{n-j}\, \mathrm{e}^{-NQ(z)}\, \mathrm{d}A(z), \tag{6.10}$$

$$= \int_0^\infty \int_0^{2\pi} r^{n-j}\, \mathrm{e}^{\mathrm{i}\theta(n-j)}\, \mathrm{e}^{-Nq(r)}\, r\mathrm{d}r\mathrm{d}\theta, \tag{6.11}$$

$$= \int_0^{2\pi} \mathrm{e}^{\mathrm{i}\theta(n-j)}\, \mathrm{d}\theta \int_0^\infty r^{n-j+1}\, \mathrm{e}^{-Nq(r)}\, \mathrm{d}r, \tag{6.12}$$

$$= 0, \quad \text{since } n \neq j. \tag{6.13}$$

Therefore, we have that $\{z^j\}_j^{(w)}$ is the proper orthogonal family. More than that, if we impose the normality,

$$\int_{\mathbb{C}} a_n^2 z^{n-n}\, \mathrm{e}^{-NQ(z)}\, \mathrm{d}A(z) = \int_{\mathbb{C}} a_n^2\, \mathrm{e}^{-NQ(z)}\, \mathrm{d}A(z) = 1 \tag{6.14}$$

$$\implies a_n = \sqrt{\frac{1}{\int_{\mathbb{C}} w(z)\mathrm{d}A(z)}}. \tag{6.15}$$

□

6.3 Asymptotics on Droplets

We will use some potential $Q(z) : \mathbb{C} \to \mathbb{R}$ which satisfies some nice growth and smoothness conditions, as specified in section 6.2. Because of the radial symmetry, one can take some function $q : \mathbb{R} \mapsto \mathbb{R}$ such that $q(|z|) = Q(z)$; then we can say that the previous conditions on the potential Q implies that $rq'(r)$ is increasing in $\mathbb{C}$ and strictly increasing in the droplet. Consider the result:

Lemma 3. *When dealing with a p.d.f. of invariant measure, one can write*

$$\mathcal{Z}_{n,2} = n! \prod_{k=0}^{n-1} r_k^{-1}, \tag{6.16}$$

where r_k is the normalizing constant as in $\langle q_j^{(2)}, q_k^{(2)}\rangle^{(2)} = r_j^{(2)} \chi_{j=k}$.

Proof. We start stating the following equality which follows from 4 that

$$\det\left[K_n^{(2)}(\lambda_i,\lambda_j)\right]_{i,j=1}^n = p^{(w)}(\lambda_1,\lambda_2,\ldots,\lambda_n;\beta), \tag{6.17}$$

$$\prod_{k=1}^n r_k^{-1} \det\left[\sqrt{w(x)w(y)}\sum_{k=0}^{N-1} q_k^{(2)}(x) q_k^{(2)}(y)\right]_{i,j=1}^n = \prod_{1<i\leq j<n} (\lambda_j-\lambda_i)^2 \prod_{k=0}^n w(\lambda_k) \tag{6.18}$$

therefore

$$n!\det\left[\tilde{K}_n^{(2)}(\lambda_i,\lambda_j)\right]_{i,j=1}^n = n!\prod_{k=1}^n r_k \prod_{1<i\leq j<n} (\lambda_j-\lambda_i)^2 \prod_{k=0}^n w(\lambda_k) \tag{6.19}$$

where

$$\mathcal{Z}_{n,2} = n! \prod_{k=1}^{n} r_k \tag{6.20}$$

□

Thus, one can write the free energy

$$\ln \mathcal{Z}_N = \ln N! - 2\sum_{j=0}^{N-1} \ln h_j, \qquad h_k := \int_{\mathbb{C}} |z|^{2j}\, \mathrm{e}^{-NQ(z)}\, \mathrm{d}A(z). \tag{6.21}$$

where h_k is the norming constant for the monomial family of orthogonal polynomials in respect to w. One could also write, for the planar symplectic ensemble

$$\ln \tilde{\mathcal{Z}}_N = \ln N! - 2\sum_{j=0}^{N-1} \ln 2\tilde{h}_{2j+1} \qquad \tilde{h}_k := \int_{\mathbb{C}} |z|^{2j}\, \mathrm{e}^{-2NQ(z)}\, \mathrm{d}A(z). \tag{6.22}$$

The expressions derived for the normal ensemble would be easily adapted for this model and it is done in (BYUN; KANG; SEO, 2023), however we will not follow that computation in name of clarity and storytelling.

In order to derive the asymptotic formulas for the free energy we need to derive the asymptotics for the orthogonal polynomials and its norming constants. The first step in the asymptotics for the normal matrices is to prepare for the Laplace method. Our goal is to identify a region that gives the main contribution, in order, for value of the integral - and consequently to identify a error-contributing region of the integration region. Let's reformulate our integral to identify the integral as usual in the Laplace method, such as in Chapter 5. We know $Q(z) = q(|z|)$ a radially symmetric potential such that, defining $z := r\,\mathrm{e}^{\mathrm{i}\theta}$, that is, $|z| = r$, we would have

$$h_j = \int_{\Theta}\int_{\mathbb{R}_+} r^{2j}\, \mathrm{e}^{-Nq(r)}\, \frac{r}{\pi} \mathrm{d}r\mathrm{d}\theta \tag{6.23}$$

$$= \frac{2\pi}{\pi}\int_{\mathbb{R}_+} r^{2j+1}\, \mathrm{e}^{-Nq(r)}\, \mathrm{d}r \tag{6.24}$$

$$= \int_{\mathbb{R}_+} 2r\, \mathrm{e}^{-N\left(q(r) - \frac{2j}{N}\ln(r)\right)}\, \mathrm{d}r \tag{6.25}$$

We first redefine our potential such that the integrand has a expression better suited to apply the Laplace method. Using $V_{\tau(j)} = q(r) - 2\tau(j)\ln(r)$, where $\tau(j) := j/N \in [0,1)$, one can rewrite the integral as

$$h_j = \int_{\mathbb{R}_+} 2r\, \mathrm{e}^{-NV_{\tau(j)}(r)}\, \mathrm{d}r \tag{6.26}$$

Where we have set the stage for the Laplace technique. Note that now τ will perform the function of the previous index j and relates to which coefficient we're calculating. Taking a look at the first derivative of the newly defined potential we find for the critical point

$$V_\tau'(r) = q'(r) - \frac{2\tau}{r} \qquad \Longrightarrow \qquad r_c q'(r_c) = 2\tau. \tag{6.27}$$

This is: We can find a critical point of the potential r_c, parameterized by the degree of the polynomial we are calculating, or more concretely, by τ. Of course, because we know the range of τ, it is worth defining the limital points

$$\begin{cases} r_0, & \text{the largest solution of } r_0 q'(r_0) = 0 \\ r_1, & \text{the smallest solution of } r_1 q'(r_1) = 2 \end{cases} \tag{6.28}$$

Our main goal here is to evaluate the integral using the surroundings of the critical point. If this point is unique, then we need no more than to calculate the integral in a single small interval around $r_c(\tau)$. This is, of course, true: We have mentioned that the properties on the potential directly implies that $rq'(r)$ is a strictly increasing function on $[r_0, r_1]$, then the solution must be unique. Now, as outside the droplet the growth is not strict, the definitions of r_0 ad r_1 had to be as mentioned before: Respectively the largest and smallest solution to the proper instance of $rq'(r) = 2\tau$. An important distinction must be made between the cases when $r_0 = 0$ and $r_0 > 0$, corresponding to subsection 6.3.2 and subsection 6.3.1, respectively. We begin noticing that $r_\tau | r_\tau q'(r_\tau) = 2\tau$ is an increasing function on τ and that $r_0 \leq r_\tau \leq r_1 \quad \forall \tau \in [0, 2]$. The following proposition should be important when computing the asymptotics.

Proposition 8. *The Gaussian integrals below can be evaluated as follows.*

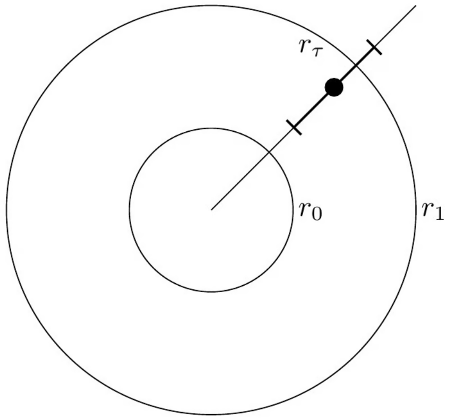
$$\int_{\mathbb{R}} \mathrm{e}^{-2ar^2} \mathrm{d}r = \sqrt{\frac{\pi}{2}} \frac{1}{a^{1/2}} \implies \int_{\mathbb{R}} \mathrm{e}^{-2\left(\frac{V^{(2)}_{\tau(j)}(r_{\tau(j)})}{4}\right) y^2} = \sqrt{\frac{\pi}{2}} \left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})} \right)^{\frac{1}{2}}, \tag{6.29}$$

$$\int_{\mathbb{R}} \mathrm{e}^{-2ar^2} r^4 \mathrm{d}r = \sqrt{\frac{\pi}{2}} \frac{3}{16} \frac{1}{a^{5/2}} \implies \int_{\mathbb{R}} \mathrm{e}^{-2\left(\frac{V^{(2)}_{\tau(j)}(r_{\tau(j)})}{4}\right) y^2} y^4 = \sqrt{\frac{\pi}{2}} \frac{3}{16} \left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})} \right)^{\frac{5}{2}}, \tag{6.30}$$

$$\int_{\mathbb{R}} \mathrm{e}^{-2ar^2} r^6 \mathrm{d}r = \sqrt{\frac{\pi}{2}} \frac{15}{64} \frac{1}{a^{7/2}} \implies \int_{\mathbb{R}} \mathrm{e}^{-2\left(\frac{V^{(2)}_{\tau(j)}(r_{\tau(j)})}{4}\right) y^2} y^6 = \sqrt{\frac{\pi}{2}} \frac{15}{64} \left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})} \right)^{\frac{7}{2}}. \tag{6.31}$$

6.3.1 Asymptotics - Annulus

We start by considering, in this regime, $r_0 > 0$. We know the measure in Equation 6.2 to be the law for the distribution of the particles in this problem. When we say that $r_0 > 0$, we are saying that the region of space $[r_0, r_1]$ where this measure is non-zero for $N \longrightarrow \infty$ has the annulus shape. If we think of the equivalent Coulomb gas, this means the particles will concentrate in this shape when we perform an asymptotic on N. This is the support for the limit measure.



Theorem 17. *For the annulus case, where $r_0 > 0$, we should have the following asymptotic for the free energy*

$$\ln \mathcal{Z}_N = -2N^2 I_Q[\mu_Q] + N\left(3\ln(2\pi) - E_Q[\mu_Q] - 1\right) + \frac{1}{2}\ln(N) + 2\int_S \mathfrak{B}_1 \mathrm{d}\mu_Q + O(N^{-1}) \tag{6.32}$$

where we have set

$$\mathfrak{B}_1(r_{\tau(j)}) = \frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{V^{(2)}_{\tau(j)}(r_{\tau(j)})^2}\frac{1}{4} - \frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})^2}{V^{(2)}_{\tau(j)}(r_{\tau(j)})^{\frac{5}{2}}}\frac{5}{12} + \frac{1}{V^{(2)}_{\tau(j)}(r_{\tau(j)})}\frac{2}{3r_{\tau(j)}}, \tag{6.33}$$

$$I_Q[\mu_Q] = \left[\frac{1}{4}\int_{r_0}^{r_1} q(s)sV^{(2)}_{\tau(j)}(s)\mathrm{d}s - \ln(r_1) + \frac{1}{2}q(r_1)\right], \tag{6.34}$$

and

$$E_Q[\mu_Q] = \frac{1}{2}\int_{r_0}^{r_1} \ln(V^{(2)}(s))(sq'(s))'\mathrm{d}s. \tag{6.35}$$

We start by studying the asymptotic for h_j and then for the summation $\sum \ln(h_j)$. This theorem then should follow from the following lemmas concerning this two asymptotics.

6.3.1.1 Asymptotic for the orthogonal polynomial

Theorem 18. *We have the following asymptotic for h_j,*

$$h_j = \frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}}\left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})}\right)^{\frac{1}{2}}\left[1+\frac{1}{N}\mathfrak{B}_1(r_{\tau(j)})+O(N^{-2})\right]. \tag{6.36}$$

where

$$\mathfrak{B}_1(r_{\tau(j)}) = \frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2}\frac{1}{4} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^{\frac{5}{2}}}\frac{5}{12} + \frac{1}{V_{\tau(j)}^{(2)}(r_{\tau(j)})}\frac{2}{3r_{\tau(j)}} \tag{6.37}$$

The proof for such theorem passes trough Lemmas 4 and 5. We begin as is usual when applying asymptotic techniques in integrals; separating a central part where the integral value is concentrated and another part where we can control the order of contribution. This is of course possible because the integrand is proportional to some exponential, with negative exponent, with a unique critical point, around which, the value of the expression concentrates. Defined $\delta_N = \ln(N)/\sqrt{N}$ - this will be our the cut-size for the interest region and will be justified during the calculations - we have

$$h_j = \int_{|r-r_j|>\delta_N} 2r\,\mathrm{e}^{-NV_{\tau(j)}(r)}\,\mathrm{d}r + \int_{|r-r_j|<\delta_N} 2r\,\mathrm{e}^{-NV_{\tau(j)}(r)}\,\mathrm{d}r = (1)+(2). \tag{6.38}$$

As mentioned, the first integral must be somehow controlled and the second estimated as the main value for the result. We begin by showing that the first term has an error contribution.

Lemma 4. *If we choose M to be such that*

$$\int_{|r-r_j|>\delta_N}\left(\mathrm{e}^{-q(r)}r^2\right)^M r\mathrm{d}r < \infty, \tag{6.39}$$

then

$$(1) = \int_{|r-r_j|>\delta_N} 2r\,\mathrm{e}^{-NV_{\tau(j)}(r)}\,\mathrm{d}r = \mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}\,\epsilon_N \tag{6.40}$$

with $\epsilon_N = O(\mathrm{e}^{-c(\ln(N))^2})$.

Proof. Let's first show that we can control the first part of the integral, for that, rewrite

$$(1) = \mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}\int_{|r-r_j|>\delta_N} 2r\,\mathrm{e}^{-N\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r, \tag{6.41}$$

$$= \mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}\int_{|r-r_j|>\delta_N} 2r\,\mathrm{e}^{-N\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{e}^{-(M-M)\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r, \tag{6.42}$$

$$= \mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}\int_{|r-r_j|>\delta_N} 2r\,\mathrm{e}^{-M\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{e}^{-(N-M)\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r. \tag{6.43}$$

Now, we need some way to estimate the difference $(V_{\tau(j)}(r) - V_{\tau(j)}(r_{\tau(j)})$. If we are able to say that this difference is at least $O(N^{-1})$, we could control the growth of the integral choosing M to be large enough such that

$$\int_{|r-r_j|>\delta_N} \left(\mathrm{e}^{-q(r)}\, r^{2\tau(j)}\right)^M r\mathrm{d}r \tag{6.44}$$

converges. Using the definition of the function V and the mean value theorem, we start by evaluating the difference $\Delta(j) = V_{\tau(j+1)}(r_{\tau(j+1)}) - V_{\tau(j)}(r_{\tau(j)})$ in the following manner

$$\Delta(j) = q(r_{\tau(j+1)}) - q(r_{\tau(j)}) - 2\left(\tau(j+1)\ln(r_{\tau(j+1)}) - \tau(j)\ln(r_{\tau(j)})\right), \tag{6.45}$$

$$= q'(c_1)(r_{\tau(j+1)} - r_{\tau(j)}) - \frac{2}{N}\left(\ln(r_{\tau(j+1)}) + j\ln\left(\frac{r_{\tau(j+1)}}{r_{\tau(j)}}\right)\right), \tag{6.46}$$

$$= q'(c_1)r'(c_2)(\tau(j+1) - \tau(j)) - \frac{2}{N}\left(\ln(r_{\tau(j+1)}) + jO(N^{-1})\right), \tag{6.47}$$

$$= q'(c_1)r'(c_2)O(N{-}1) - O(N^{-1}), \tag{6.48}$$

$$= O(N^{-1}). \tag{6.49}$$

The last equality holds as both functions are continuous and c_1, c_2 are values that are limited; in a compact disk on the plane. Then, we can add zero and use the estimation

$$V_{\tau(j+1)}(r_{\tau(j+1)}) - V_{\tau(j+1)}(r) + V_{\tau(j+1)}(r) - V_{\tau(j)}(r_{\tau(j)}) = O(N^{-1}). \tag{6.50}$$

to derive the result we are after. It is enough to estimate $V_{\tau(j+1)}(r) - V_{\tau(j)}(r_{\tau(j)})$. For that, because $rq'(r)$ is increasing, for all $r > r_{\tau(j)} + \delta_N$ we can say that

$$V_{\tau(j+1)}(r) \geq V_{\tau(j+1)}(r_{\tau(j)} + \delta_N) \tag{6.51}$$

$$= V_{\tau(j+1)}(r_{\tau(j+1)}) + \frac{1}{2}V''_{\tau(j+1)}(r_{\tau(j+1)})(r_{\tau(j)} + \delta_N - r_{\tau(j+1)})^2 + O(\delta_N^3), \tag{6.52}$$

and therefore can write

$$V_{\tau(j+1)}(r) - V_{\tau(j)}(r_{\tau(j)}) = \frac{1}{2}V''_{\tau(j+1)}(r_{\tau(j+1)})(r_{\tau(j)} + \delta_N - r_{\tau(j+1)})^2 + O(\delta_N^3) \tag{6.53}$$

$$= \delta^2 c + O(N^{-1}) + O(\delta_N^3) \tag{6.54}$$

$$\geq \delta^2 c \tag{6.55}$$

Finally, using this one can say

$$V_{\tau(j+1)}(r) - V_{\tau(j+1)}(r_{\tau(j+1)}) \geq c\delta_N^2. \tag{6.56}$$

Getting this inequality back to the integral expression end up with

$$(1) = \mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})} \int_{|r-r_j|>\delta_N} 2r\,\mathrm{e}^{-M\left(V_{\tau(j)}(r) - V_{\tau(j)}(r_{\tau(j)})\right)}\, \mathrm{e}^{-(N-M)\left(V_{\tau(j)}(r) - V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r \tag{6.57}$$

$$= \mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}\, \mathrm{e}^{-c\delta_N^2(N-M)} \int_{|r-r_j|>\delta_N} 2r\,\mathrm{e}^{-M\left(V_{\tau(j)}(r) - V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r \tag{6.58}$$

where you can check that

$$\int_{|r-r_j|>\delta_N} 2r\,\mathrm{e}^{-M\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r \tag{6.59}$$

$$= \int_{|r-r_j|>\delta_N} 2r\,\mathrm{e}^{-M\left((q(r)-2\tau(j)\ln(r))-(q(r_{\tau(j)})-2\tau(j)\ln(r_{\tau(j)}))\right)}\,\mathrm{d}r, \tag{6.60}$$

$$= 2\left(\mathrm{e}^{-q(r_{\tau(j)})}\,r_{\tau(j)}^{-2\tau(j)}\right)^M \int_{|r-r_j|>\delta_N} \left(\mathrm{e}^{-q(r)}\,r^{2\tau(j)}\right)^M r\mathrm{d}r. \tag{6.61}$$

By assumption, we chose M to be a big enough number such that the integral above converges. If that is true,

$$(1) = \int_{|r-r_j|>\delta_N} 2r\,\mathrm{e}^{-NV_{\tau(j)}(r)}\,\mathrm{d}r = \mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}\,\epsilon_N, \tag{6.62}$$

with $\epsilon_N = O(\mathrm{e}^{-c(\ln(N))^2})$. □

We now move on to the value-contributing part of the integral. The other order term should be absolved in this one if Theorem 18 is to be right.

Lemma 5. *For the second part of the integral we have*

$$(2) = \frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}}\left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})}\right)^{\frac{1}{2}}\left[1+\frac{1}{N}\mathfrak{B}_1(r_{\tau(j)})+O(N^{-2})\right] \tag{6.63}$$

where

$$\mathfrak{B}_1(r_{\tau(j)}) = \frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2}\frac{1}{4} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^{\frac{5}{2}}}\frac{5}{12} + \frac{1}{V_{\tau(j)}^{(2)}(r_{\tau(j)})}\frac{2}{3r_{\tau(j)}}. \tag{6.64}$$

Proof. Now we must deal with the second part of the integral. As we're dealing with a symmetric (in relation to the critical point) integral, we might as well set a centralized variable $x = r - r_\tau$, with this, the integral becomes

$$(2) = \int_{-\delta_N}^{\delta_N} 2(x+r_{\tau(j)})\,\mathrm{e}^{-NV_{\tau(j)}(x+r_{\tau(j)})}\,\mathrm{d}x \tag{6.65}$$

$$= \mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}\int_{-\delta_N}^{\delta_N} 2(x+r_{\tau(j)})\,\mathrm{e}^{-N\left[V_{\tau(j)}(x+r_{\tau(j)})-V_{\tau(j)}(r_{\tau(j)})\right]}\,\mathrm{d}x. \tag{6.66}$$

There is also a question of scale. Our interval grows with $\sqrt{N}/\ln(N)$. By re-scaling the reference variable using $y = \sqrt{N}x$ we get

$$(2) = \frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}}\int_{-\ln(N)}^{\ln(N)} 2\left(\frac{y}{\sqrt{N}}+r_{\tau(j)}\right)\mathrm{e}^{-N\left[V_{\tau(j)}(\frac{y}{\sqrt{N}}+r_{\tau(j)})-V_{\tau(j)}(r_{\tau(j)})\right]}\,\mathrm{d}y. \tag{6.67}$$

Now, the idea here is quite direct: to use the Taylor expansion for $V_{\tau(j)}\left(\frac{y}{\sqrt{N}}+r_{\tau(j)}\right)$ near the critical point $r_{\tau(j)}$. That is,

$$(2)=\frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}}\int_{-\ln(N)}^{\ln(N)}2\left(\frac{y}{\sqrt{N}}+r_{\tau(j)}\right)\mathrm{e}^{-N\left(\frac{y^2}{2N}V^{(2)}_{\tau(j)}(r_{\tau(j)})+\frac{y^3}{3!N^{\frac{3}{2}}}V^{(3)}_{\tau(j)}(r_{\tau(j)})+\frac{y^4}{4!N^2}V^{(4)}_{\tau(j)}(r_{\tau(j)})+O\left(\left(\frac{y}{\sqrt{N}}\right)^5\right)\right)}\mathrm{d}y, \tag{6.68}$$

$$=\frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{(2r_{\tau(j)})^{-1}\sqrt{N}}\int_{-\ln(N)}^{\ln(N)}\left(\frac{y}{r_{\tau(j)}\sqrt{N}}+1\right)\mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})-\frac{y^3}{3!\sqrt{N}}V^{(3)}_{\tau(j)}(r_{\tau(j)})-\frac{y^4}{4!N}V^{(4)}_{\tau(j)}(r_{\tau(j)})+O\left(\frac{y^5}{N^{\frac{3}{2}}}\right)}\mathrm{d}y, \tag{6.69}$$

$$=\frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{(2r_{\tau(j)})^{-1}\sqrt{N}}\int_{-\ln(N)}^{\ln(N)}\left(\frac{y}{r_{\tau(j)}\sqrt{N}}+1\right)\mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})}\left(\mathrm{e}^{-\frac{y^3}{3!\sqrt{N}}V^{(3)}_{\tau(j)}(r_{\tau(j)})-\frac{y^4}{4!N}V^{(4)}_{\tau(j)}(r_{\tau(j)})+O\left(\frac{y^5}{N^{\frac{3}{2}}}\right)}\right)\mathrm{d}y. \tag{6.70}$$

Now we take the Taylor Series for

$$\mathrm{e}^{-\frac{y^3}{3!\sqrt{N}}V^{(3)}_{\tau(j)}(r_{\tau(j)})-\frac{y^4}{4!N}V^{(4)}_{\tau(j)}(r_{\tau(j)})+O\left(\frac{y^5}{N^{\frac{3}{2}}}\right)}=1-\left[\frac{y^3}{3!\sqrt{N}}V^{(3)}_{\tau(j)}(r_{\tau(j)})+\frac{y^4}{4!N}V^{(4)}_{\tau(j)}(r_{\tau(j)})+O\left(\frac{y^5}{N^{\frac{3}{2}}}\right)\right]$$
$$-\frac{1}{2}\left[\frac{y^6}{N}\left(\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})}{6}\right)^2+\frac{y^8}{N^2}\left(\frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{24}\right)^2+\frac{y^7}{N^3}\left(\frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{6}\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})}{24}\right)+O\left(\frac{y^8}{N^2}\right)\right]+O\left(\frac{y^a}{N^b}\right). \tag{6.71}$$

To get the asymptotic expansion (since odd terms vanish) we reorganize the terms in the integral again to obtain

$$(2)=2r_{\tau(j)}\frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}}\int_{-\ln(N)}^{\ln(N)}\mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})}\left[1-\frac{y^4}{3!r_{\tau(j)}N}V^{(3)}_{\tau(j)}(r_{\tau(j)})\right.$$
$$\left.-\frac{y^4}{4!N}V^{(4)}_{\tau(j)}(r_{\tau(j)})-\frac{y^6}{2\cdot(3!)^2N}V^{(3)}_{\tau(j)}(r_{\tau(j)})^2\right]\mathrm{d}y, \tag{6.72}$$

then, distributing the integral,

$$(2)=2r_{\tau(j)}\frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}}\left[\left(-\frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{24N}-\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})}{6r_{\tau(j)}N}\right)\int_{-\ln(N)}^{\ln(N)}\mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})}y^4\right.$$
$$\left.-\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})^2}{72N}\int_{-\ln(N)}^{\ln(N)}\mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})}y^6+\int_{-\ln(N)}^{\ln(N)}\mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})}\right]. \tag{6.73}$$

At this point, the result can be directly evaluated using the Gaussian integrals in Proposition 8.

So that we can finally substitute the values to get

$$(2) = \sqrt{2\pi} r_{\tau(j)} \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \left[\left(-\frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{24N} - \frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})}{6r_{\tau(j)}N} \right) \left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \right.$$
$$\left. - \frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})^2}{72N} \frac{3}{16} \left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})} \right)^{\frac{5}{2}} + \frac{15}{64} \left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})} \right)^{\frac{7}{2}} \right], \quad (6.74)$$

and, rearranging the terms we should get our result

$$(2) = \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \left(\frac{8\pi r^2_{\tau(j)}}{V^{(2)}_{\tau(j)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left[1 + \frac{1}{N} B_1(r_{\tau(j)}) + O(N^{-2}) \right] \quad (6.75)$$

where we defined $\mathfrak{B}_1(r_{\tau(j)})$ to be given by

$$\mathfrak{B}_1(r_{\tau(j)}) = \frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{V^{(2)}_{\tau(j)}(r_{\tau(j)})^2} \frac{1}{4} - \frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})^2}{V^{(2)}_{\tau(j)}(r_{\tau(j)})^{\frac{5}{2}}} \frac{5}{12} + \frac{1}{V^{(2)}_{\tau(j)}(r_{\tau(j)})} \frac{2}{3r_{\tau(j)}} \quad (6.76)$$

Notice that the error term of Lemma 4 is absorbed into the error order of Equation 6.75. Therefore we have the asymptotic expression for h_j given by (6.36) and the proof for Equation 18 follows directly. □

6.3.1.2 Asymptotics on the Summation

So now we need to return to (6.21) and calculate each term for $\sum_{j=0}^{N-1} \ln h_j$ using th expansion in terms of Theorem 18

$$\sum_{j=0}^{N-1} -NV_{\tau(j)}(r_{\tau(j)}) + \frac{1}{2} \left(\ln(8\pi r^2_{\tau(j)}) - \ln N - \ln V^{(2)}_{\tau(j)}(r_{\tau(j)}) \right) + \frac{1}{N} \mathfrak{B}_1(r_{\tau(j)}) + O(N^{-2}). \quad (6.77)$$

That is, we need to prove the following theorem.

Theorem 19.

$$\sum_{j=0}^{N-1} \ln h_j = -N^2 I_Q[\mu_Q] - \frac{N}{2} \ln N + N \left(\frac{3\ln(2\pi)}{2} - \frac{1}{2} E_Q[\mu_Q] \right)$$
$$- \frac{1}{2} \ln \left(\frac{r_1}{r_0} \right) + \frac{1}{4} \ln \left(\frac{V^{(2)}(r_1)}{V^{(2)}(r_0)} \right) + \int_S \mathfrak{B}_1 \, d\mu_Q + O(N^{-1}) \quad (6.78)$$

where we have set

$$\mathfrak{B}_1(r_{\tau(j)}) = \frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{V^{(2)}_{\tau(j)}(r_{\tau(j)})^2} \frac{1}{4} - \frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})^2}{V^{(2)}_{\tau(j)}(r_{\tau(j)})^{\frac{5}{2}}} \frac{5}{12} + \frac{1}{V^{(2)}_{\tau(j)}(r_{\tau(j)})} \frac{2}{3r_{\tau(j)}}, \quad (6.79)$$

$$I_Q[\mu_Q] = \left[\frac{1}{4}\int_{r_0}^{r_1} q(s)sV^{(2)}_{\tau(j)}(s)\mathrm{d}s - \ln(r_1) + \frac{1}{2}q(r_1)\right], \tag{6.80}$$

and

$$E_Q[\mu_Q] = \frac{1}{2}\int_{r_0}^{r_1} \ln(V^{(2)}(s))(sq'(s))'\mathrm{d}s. \tag{6.81}$$

The proof comes from gathering the terms in Equation 6.77 with the expressions in Equation 6.82, Equation 6.231 and Equation 6.125. To get there, for each term depending on j we need an summation expression. We go term by term. Lemma 6 refers to the asymptotic of

$$\sum_{j=0}^{N-1} -NV_{\tau(j)}(r_{\tau(j)}),$$

Lemma 7 refers to the asymptotics of both terms

$$\sum_{j=0}^{N-1}\left(\ln(2\pi r^2_{\tau(j)}) - \ln V^{(2)}_{\tau(j)}(r_{\tau(j)})\right).$$

Then, the final expression becomes a question of gathering terms.

Lemma 6.

$$-N\sum_{j=0}^{N-1} V_{\tau(j)}(r_{\tau(j)}) = -N^2 I^Q[\mu_Q] - 2NU_{\mu_Q}(0) - \frac{1}{6}\ln\left(\frac{r_0}{r_1}\right) + O(N^{-2}) \tag{6.82}$$

where

$$I^Q[\mu_Q] := \left[\frac{1}{4}\int_{r_0}^{r_1} q(s)sV^{(2)}_{\tau(j)}(s)\mathrm{d}s - \ln(r_1) + \frac{1}{2}q(r_1)\right] \tag{6.83}$$

and

$$2U_{\mu_Q}(0) := (r_1) - q(r_0) - 2\ln(r_1). \tag{6.84}$$

Proof. Using the Euler-Maclaurin formula

$$\sum_{j=0}^{N-1} V_{\tau(j)}(r_{\tau(j)}) = \int_0^N V_{\tau(j)}(r_{\tau(j)})\mathrm{d}t - \frac{1}{2}\left(V_{\tau(j)}(r_{\tau(j)}) - V_{\tau(0)}(r_{\tau(0)})\right) + \frac{1}{12}\left[\partial_t V_{\tau(j)}(r_{\tau(j)})|_{t=N} - \partial_t V_{\tau(j)}(r_{\tau(j)})|_{t=0}\right] + O(N^{-3}). \tag{6.85}$$

The first term in (6.200) is solved doing a integral by parts. We remember the expression for the potential $V_{\tau(j)}(r_{\tau(j)}) = q(r_{\tau(j)}) - 2\tau(j)\ln r$, with $\tau(j) = j/N$. Therefore the differential can be written as

$$\mathrm{d}(r_{\tau(j)}q'(r_{\tau(j)})) = 2\mathrm{d}\tau(j) \tag{6.86}$$

$$\mathrm{d}rq'(r) + \mathrm{d}rrq''(r) = 2\mathrm{d}\tau(j) \tag{6.87}$$

$$\mathrm{d}r(q'(r) + rq''(r)) = 2\mathrm{d}\tau(j). \tag{6.88}$$

This implies that the derivative for r is

$$\frac{\mathrm{d}r}{\mathrm{d}\tau} = \frac{2}{r_{\tau(j)} V^{(2)}_{\tau(j)}(r_{\tau(j)})} = \frac{2}{(rq'(r))'}. \tag{6.89}$$

Note that in the critical point $r_{\tau(j)}$, we should have $r_{\tau(j)} q'(r_{\tau(j)}) = 2\tau$. We then are able to rewrite, setting $s = r_{\tau(j)}$,

$$\begin{aligned}
\int_0^N V_{\tau(t)}(r_{\tau(t)})\mathrm{d}t &= \int_0^N (q(r_{\tau(t)}) - 2\tau(t)\ln(r_{\tau(t)}))\mathrm{d}t && (6.90)\\
&= \frac{1}{2}\int_{r_0}^{r_1} (q(s) - sq'(s)\ln(s))sV^{(2)}_{\tau(j)}(s)N\mathrm{d}s && (6.91)\\
&= N\left[\frac{1}{2}\int_{r_0}^{r_1} q(s)sV^{(2)}_{\tau(j)}(s)\mathrm{d}s - \frac{1}{2}\int_{r_0}^{r_1} s^2q'(s)\ln(s)V^{(2)}_{\tau(j)}(s)\mathrm{d}s\right] && (6.92)
\end{aligned}$$

and finally using that $r_1 q'(r_1) = 2$ and $r_0 q'(r_0) = 0$ we can apply the integration by parts on the second term and get

$$\begin{aligned}
\frac{1}{2}\int_{r_0}^{r_1} s^2q'(s)\ln(s)V^{(2)}_{\tau(j)}(s)\mathrm{d}s &= \frac{1}{2}\int_{r_0}^{r_1} s^2q'(s)\ln(s)\frac{1}{s}(sq'(s))'\mathrm{d}s && (6.93)\\
&= \frac{1}{2}\int_{r_0}^{r_1} sq'(s)\ln(s)(sq'(s))'\mathrm{d}s && (6.94)\\
&= \frac{1}{4}\int_{r_0}^{r_1} ((sq'(s))^2)'\ln(s)\mathrm{d}s && (6.95)\\
&= \frac{1}{4}(sq'(s))^2\ln(s)|_{r_0}^{r_1} + \frac{1}{4}\int_{r_0}^{r_1} (sq'(s))^2\frac{1}{s}\mathrm{d}s && (6.96)\\
&= \ln(r_1) + \frac{1}{4}\int_{r_0}^{r_1} sq'(s)^2\mathrm{d}s && (6.97)\\
&= \ln(r_1) + \frac{1}{2}q(r_1) - \frac{1}{4}\int_{r_0}^{r_1} q(s)(sq'(s))'\mathrm{d}s && (6.98)
\end{aligned}$$

Leaving us with

$$\begin{aligned}
\int_0^N V_{\tau(j)}(r_{\tau(j)})\mathrm{d}t &= N\left[\frac{1}{2}\int_{r_0}^{r_1} q(s)sV^{(2)}_{\tau(j)}(s)\mathrm{d}s - \ln(r_1) + \frac{1}{2}q(r_1) - \frac{1}{4}\int_{r_0}^{r_1} q(s)(sq'(s))'\mathrm{d}s\right] && (6.99)\\
&= N\left[\frac{1}{4}\int_{r_0}^{r_1} q(s)sV^{(2)}_{\tau(j)}(s)\mathrm{d}s - \ln(r_1) + \frac{1}{2}q(r_1)\right] && (6.100)\\
&:= NI_Q[\mu_Q] && (6.101)
\end{aligned}$$

We also need an expression for $V_{\tau(j)}(r_{\tau(j)}) - V_{\tau(0)}(r_{\tau(0)})$. This is direct noting that

$$V_{\tau(j)}(r_{\tau(j)}) - V_{\tau(0)}(r_{\tau(0)}) = V_1(r_1) - V_0(r_0) = q(r_1) - q(r_0) - 2\ln(r_1) =: 2U_{\mu_Q}(0) \tag{6.102}$$

And finally we can the remaining term , we use the Leibniz rule and obtain

$$\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=N} = \partial_t \left(q(r_{\tau(t)}) - 2\tau(t)\ln(r_{\tau(t)})\right)_{t=N} \tag{6.103}$$
$$= \partial_\tau \left(q(r_{\tau(t)}) - 2\tau(t)\ln(r_{\tau(t)})\right)_{\tau=1} \partial_t \left(\tau(t)\right)_{t=N} \tag{6.104}$$
$$= \frac{1}{N}\left(\partial_\tau(q(r_\tau)) - 2\left(\ln(r_\tau) + \tau\partial_\tau\left(\ln(r_\tau)\right)_{\tau=1}\right)\right) \tag{6.105}$$
$$= \frac{1}{N}\left[\partial_t(q(r_{\tau(t)}) - 2\ln(r_{\tau(t)}))_{\tau=1} - 2\ln(r_1)\right] \tag{6.106}$$

and

$$\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=0} = \frac{1}{N}\left[\partial_t(q(r_{\tau(t)}) - 2\ln(r_{\tau(t)}))_{\tau=0} - 2\ln(r_0)\right]. \tag{6.107}$$

And since we can calculate, from $r_1 q'(r_1) = 2$ and $r_0 q'(r_0) = 0$

$$\partial_\tau q(r_\tau)|_{\tau=0} = \left(\frac{2q'(r)}{(rq'(r))'}\right)_{r_0} = 0, \tag{6.108}$$

$$\partial_\tau q(r_\tau)|_{\tau=1} = \left(\frac{2q'(r)}{(rq'(r))'}\right)_{r_1} = \frac{4}{r_1(q'(r_1) + r_1 q''(r_1))}, \tag{6.109}$$

$$\partial_\tau(2\ln(r_\tau))|_{\tau=1} = \left(\frac{4}{r(rq'(r))'}\right)_{r_1} = \frac{4}{r_1(q'(r_1) + r_1 q''(r_1))}, \tag{6.110}$$

we can also express the difference as

$$\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=N} - \partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=0} = \frac{1}{12N}\left[2(\ln(r_0) - \ln(r_1))\right] = \frac{1}{6N}\ln\left(\frac{r_0}{r_1}\right). \tag{6.111}$$

And the proof for the summation $\sum_{j=0}^{N-1} V_{\tau(j)}(r_{\tau(j)})$ is complete if one gathers the terms (6.101), (6.111) and (6.102),

□

We move on to the next term from where we need to determine both terms in

$$\sum_{j=0}^{N-1}\left(\ln(2\pi r_{\tau(j)}^2) - \ln V_{\tau(j)}^{(2)}(r_{\tau(j)})\right). \tag{6.112}$$

Lemma 7.

$$\sum_{j=0}^{N-1} \ln V_{\tau(j)}^{(2)}(r_{\tau(j)}) = \frac{N}{2}\int_{r_0}^{r_1} \ln(V^{(2)}(s))(sq'(s))' \mathrm{d}s - \frac{1}{2}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) + O(N^{-1}) \tag{6.113}$$

and

$$\sum_{j=0}^{N-1} \ln(8\pi r_{\tau(j)}^2) = N\ln(8\pi) + 2\left(\frac{N}{2}\int_{r_0}^{r_1} \ln(s)(sq'(s))' \mathrm{d}s - \frac{1}{2}\ln\left(\frac{r_1}{r_0}\right) + O(N^{-1})\right) \tag{6.114}$$

Proof. Lets start now with the second summation, to which we apply Euler-Maclaurin again

$$\sum_{j=0}^{N-1} \ln V^{(2)}_{\tau(j)}(r_{\tau(j)}) = \int_0^N \ln V^{(2)}_{\tau(t)}(r_{\tau(t)})\mathrm{d}t - \frac{1}{2}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) + \\ \frac{1}{12}\left[\partial_t \ln V^{(2)}_{\tau(t)}(r_{\tau(t)})|_{\tau=1} - \partial_t \ln V^{(2)}_{\tau(t)}(r_{\tau(t)})|_{\tau=0}\right] + O(N^{-3}) \quad (6.115)$$

We first tackle the integral on the right hand side, changing variables as $s = r_{\tau(t)}$ we would have

$$\int_0^N \ln V^{(2)}_{\tau(t)}(r_{\tau(t)})\mathrm{d}t = \frac{N}{2}\int_{r_0}^{r_1} \ln(V^{(2)}(s))(sq'(s))'\mathrm{d}s \quad (6.116)$$

And, noticing that

$$\partial_t \ln V^{(2)}_{\tau(t)}(r_{\tau(t)}) = \partial_V\left(\ln V^{(2)}_{\tau(t)}(r_{\tau(t)})\right)\partial_r\left(V^{(2)}_{\tau(t)}(r_{\tau(t)})\right)\partial_\tau\left(r_{\tau(t)}\right) = \frac{\left(\partial_r V^{(2)}\right)(r_{\tau(t)})}{r_{\tau(t)}\left(V^{(2)}(r_{\tau(t)})\right)^2} \quad (6.117)$$

We have for the third term on the right hand side,

$$\partial_t \ln V^{(2)}_{\tau(t)}(r_{\tau(t)})|_{\tau=1} - \partial_t \ln V^{(2)}_{\tau(t)}(r_{\tau(t)})|_{\tau=0} = \frac{1}{2N}\left(\frac{\left(\partial_r V^{(2)}\right)(r_1)}{r_1\left(V^{(2)}(r_1)\right)^2} - \frac{\left(\partial_r V^{(2)}\right)(r_0)}{r_0\left(V^{(2)}(r_0)\right)^2}\right) \quad (6.118)$$

$$= O(N^{-1}) \quad (6.119)$$

then,

$$\sum_{j=0}^{N-1} \ln V^{(2)}_{\tau(j)}(r_{\tau(j)}) = \frac{N}{2}\int_{r_0}^{r_1} \ln(V^{(2)}(s))(sq'(s))'\mathrm{d}s - \frac{1}{2}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) + O(N^{-1}) \quad (6.120)$$

Similarly, for the first summation, we apply Euler-Maclaurin again

$$\sum_{j=0}^{N-1} \ln(8\pi r^2_{\tau(j)}) = \ln(8\pi) + 2\left(\int_0^N \ln(r_{\tau(t)})\mathrm{d}t - \frac{1}{2}\ln\left(\frac{r_1}{r_0}\right)\right. \\ \left. + \frac{1}{12}\left[\partial_t \ln r_{\tau(t)}|_{\tau=1} - \partial_t \ln r_{\tau(t)}|_{\tau=0}\right] + O(N^{-3})\right) \quad (6.121)$$

We first tackle the integral on the right hand side, changing variables as $s = r_{\tau(t)}$ we would have

$$\int_0^N \ln r_{\tau(t)}\mathrm{d}t = \frac{N}{2}\int_{r_0}^{r_1} \ln(s)(sq'(s))'\mathrm{d}s \quad (6.122)$$

And, noticing that

$$\partial_t \ln r_{\tau(t)} = \partial_r\left(\ln(r_{\tau(t)})\right)\partial_\tau\left(r_{\tau(t)}\right) = \frac{2}{V^{(2)}(r_{\tau(t)})\left(r_{\tau(t)}\right)^2} \quad (6.123)$$

We have for the third term on the right hand side,

$$\partial_t \ln r_{\tau(t)}|_{\tau=1} - \partial_t \ln r_{\tau(t)}|_{\tau=0} = \frac{1}{2N}\left(\frac{2}{V^{(2)}(r_1)\left(r_1\right)^2} - \frac{2}{V^{(2)}(r_0)\left(r_0\right)^2}\right) = O(N^{-1}) \quad (6.124)$$

then

$$\sum_{j=0}^{N-1} \ln(8\pi r^2_{\tau(j)}) = N\ln(8\pi) + 2\left(\frac{N}{2}\int_{r_0}^{r_1} \ln(s)(sq'(s))'\mathrm{d}s - \frac{1}{2}\ln\left(\frac{r_1}{r_0}\right) + O(N^{-1})\right) \quad (6.125)$$

□

6.3.1.3 Free Energy Annulus

We can finally prove Theorem 17.

Proof. (Theorem 17) From

$$\sum_{j=0}^{N-1} \ln(h_j) = -N^2 I_Q[\mu_Q] - \frac{N}{2}\ln N + N\left(\frac{3\ln(2\pi)}{2} - \frac{1}{2}E_Q[\mu_Q]\right) - \frac{1}{2}\ln\left(\frac{r_1}{r_0}\right) + \frac{1}{4}\ln\left(\frac{V^{(2)}(r_1)}{V^{(2)}(r_0)}\right) + \int_S \mathfrak{B}_1 \mathrm{d}\mu_Q + O(N^{-1}) \quad (6.126)$$

We can get the asymptotic expansion for the Free energy using that

$$\ln(N!) = N\ln(N) - N + \frac{1}{2}\ln(N) + \frac{1}{2}\ln(2\pi) + O(N^{-1}) \quad (6.127)$$

that is,

$$\ln \mathcal{Z}_N = \ln N! - 2\sum_{j=0}^{N-1} \ln h_j \quad (6.128)$$

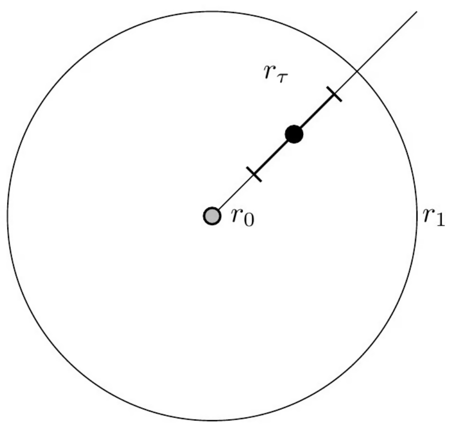
$$= N\ln(N) - N + \frac{1}{2}\ln(N) + \frac{1}{2}\ln(2\pi) + 2\Bigg(-N^2 I_Q[\mu_Q] - \frac{N}{2}\ln N + N\left(\frac{3\ln(2\pi)}{2} - \frac{1}{2}E_Q[\mu_Q]\right) - \frac{1}{2}\ln\left(\frac{r_1}{r_0}\right) + \frac{1}{4}\ln\left(\frac{V^{(2)}(r_1)}{V^{(2)}(r_0)}\right) + \int_S \mathfrak{B}_1 \mathrm{d}\mu_Q\Bigg) + O(N^{-1}) \quad (6.129)$$

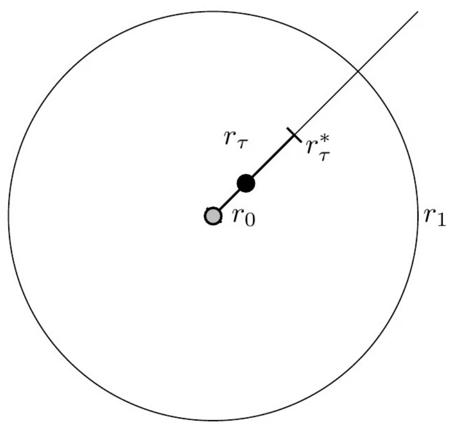
$$= -2N^2 I_Q[\mu_Q] + N\left(3\ln(2\pi) - E_Q[\mu_Q] - 1\right) + \frac{1}{2}\ln(N) + 2\int_S \mathfrak{B}_1 \mathrm{d}\mu_Q + O(N^{-1}) \quad (6.130)$$

□

6.3.2 Asymptotics - Disk

In this case, we consider $r_0 = 0$. As before, the measure in Equation 4.1 to be the law for distribution of the particles in this model. When we say $r_0 = 0$, we say that the support for this configuration space of particles is a Disk in the complex plane. The expression we get is different from taking the limit in the case $r_0 > 0$.





Theorem 20. *For the annulus case, where $r_0 = 0$, we should have the following asymptotic for the free energy*

$$\ln \mathcal{Z}_N = \tag{6.131}$$

where we have set

$$\mathfrak{B}_1(r_{\tau(j)}) = \frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{V^{(2)}_{\tau(j)}(r_{\tau(j)})^2}\frac{1}{4} - \frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})^2}{V^{(2)}_{\tau(j)}(r_{\tau(j)})^{\frac{5}{2}}}\frac{5}{12} + \frac{1}{V^{(2)}_{\tau(j)}(r_{\tau(j)})}\frac{2}{3r_{\tau(j)}}, \tag{6.132}$$

$$I_Q[\mu_Q] = \left[\frac{1}{4}\int_{r_0}^{r_1} q(s)sV^{(2)}_{\tau(j)}(s)\mathrm{d}s - \ln(r_1) + \frac{1}{2}q(r_1)\right] \tag{6.133}$$

and

$$E_Q[\mu_Q] = \frac{1}{2}\int_{r_0}^{r_1} \ln(V^{(2)}(s))(sq'(s))'\mathrm{d}s. \tag{6.134}$$

Again, we start by studying the asymptotic for h_j and then for the summation $\sum \ln(h_j)$. This theorem then should follow from the following lemmas concerning this two asymptotics.

6.3.2.1 Asymptotic for the orthogonal polynomial

In the asymptotic expansions of h_j, one should distinguish the following two cases depending on a small constant $1/5 > \epsilon > 0$.

- Case 1: $r_{j/N} >> N^{-\epsilon}$. For the annular droplet case where $r_0 > 0$, this case covers all $j = 0, 1, \cdots, N-1$. On the other hand, for the disc droplet case where $r_0 = 0$, this case covers only $j = m_N, m_N+1, \cdots, N-1$ for $m_N = N^{\epsilon}$ with some $\epsilon > 0$;
- Case 2: $r_{j/N} << N^{-\epsilon}$. This covers the remaining disc droplet case with $j = 0, 1, \cdots, m_N - 1$. Notably, the asymptotic expansion involves gamma functions in this case.

We separate this in two proofs, the first one for $j = m_N, m_N+1, \cdots, N-1$ is very similar to our previous proof so we begin by this case.

Theorem 21. *For $j \geq m_N$,*

$$h_j = \frac{\mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}}\left(\frac{8\pi r^2_{\tau(j)}}{V^{(2)}(r_{\tau(j)})}\right)^{\frac{1}{2}}\left(1 + \frac{1}{N}\mathfrak{B}_1(r_{\tau(j)}) + \epsilon_{N,1}\right) \tag{6.135}$$

where

$$\mathfrak{B}_1(r_{\tau(j)}) = \frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{V^{(2)}_{\tau(j)}(r_{\tau(j)})^2}\frac{1}{4} - \frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})^2}{V^{(2)}_{\tau(j)}(r_{\tau(j)})^{\frac{5}{2}}}\frac{5}{12} + \frac{1}{V^{(2)}_{\tau(j)}(r_{\tau(j)})}\frac{2}{3r_{\tau(j)}}, \tag{6.136}$$

and

$$\epsilon_{N,1} = O(j^{-\frac{3}{2}}(\ln N)^{\alpha}) \tag{6.137}$$

for some $\alpha > 0$.

To actually expand on the details, we need some useful facts. We will make use in this case of the proposition

Proposition 9. *Due to strict subharmonicity of Q in a neighborhood of the droplet, r_τ, defined as before, satisfies*

$$r_\tau = \left(\frac{2\tau}{q''(0)}\right)^{\frac{1}{2}} + O(\tau) = \left(\frac{4\tau}{V^{(2)}(r_\tau)}\right)^{\frac{1}{2}} + O(\tau) \tag{6.138}$$

or more simply

$$r_\tau = O(\tau^{\frac{1}{2}}) \tag{6.139}$$

as $\tau \to 0$.

Proof. This is a inverse function theorem with a Taylor expansion for equalizing powers. □

Note also that for $d \geq 1$, we can choose C_1 a constant uniformly for all τ and write

$$|V_\tau^{(d)}(r_\tau)| = |q^{(d)}(r_\tau) + (-1)^d 2\tau(d-1)! r_\tau^{-d}| \leq C_1(1 + \tau^{-\frac{d}{2}+1}). \tag{6.140}$$

We then start as usual dividing the integral into two parts

$$h_j = \int_0^\infty 2r^{2j+1} \mathrm{e}^{-Nq(r)} \,\mathrm{d}r = \int_0^\infty 2r \,\mathrm{e}^{-NV_{\tau(j)}(r)} \,\mathrm{d}r \tag{6.141}$$

$$= \int_{r_{\tau(j)}-\delta_N}^{r_{\tau(j)}+\delta_N} 2r \,\mathrm{e}^{-NV_{\tau(j)}(r)} \,\mathrm{d}r + \int_{|r-r_{\tau(j)}|>\delta_N} 2r \,\mathrm{e}^{-NV_{\tau(j)}(r)} \,\mathrm{d}r = (1) + (2) \tag{6.142}$$

and now we have two integrals to solve. Let's start with the outer integral, which we should be able to control the growth.

Lemma 8. *Consider the case $m_N \leq j \leq N-1$. For the error-bound integral we have the asymptotic*

$$(2) = \mathrm{e}^{-NV_{\tau(j)}(r_{\tau(j)})} O(\mathrm{e}^{c_2(\ln(N))^2}). \tag{6.143}$$

for some positive constant c_2.

Proof. For $m_N \leq j < N$, we have

$$r_{\tau(j+1)} - r_{\tau(j)} = O((jN)^{-\frac{1}{2}}). \tag{6.144}$$

Since the potential is increasing in (r_τ, ∞), the Taylor series expansion for V_τ is for all $r > r_{\tau(j)} + \delta_N$

$$V_{\tau(j+1)}(r) \geq V_{\tau(j+1)}(r_{\tau(j)} + \delta_N) \tag{6.145}$$

$$= V_{\tau(j+1)}(r_{\tau(j+1)}) + \frac{1}{2} V^{(2)}_{\tau(j+1)}(r_{\tau(j+1)})(r_{\tau(j)} + \delta_N - r_{\tau(j+1)})^2 + O(\tau(j)^{-1/2}\delta_N^3). \tag{6.146}$$

where $O(\tau(j)^{-1/2}\delta_N^3) = O(N^{-1}j^{-1/2}(\ln(N))^3)$. Similarly, since V_τ is decreasing in $(0, r_\tau)$, we have for all $r < r_{\tau(j)} - \delta_N$

$$V_{\tau(j+1)}(r) \geq V_{\tau(j+1)}(r_{\tau(j)} - \delta_N) \tag{6.147}$$

$$= V_{\tau(j+1)}(r_{\tau(j+1)}) + \frac{1}{2}V^{(2)}_{\tau(j+1)}(r_{\tau(j+1)})(r_{\tau(j)} - \delta_N - r_{\tau(j+1)})^2 + O(\tau(j)^{-1/2}\delta_N^3). \tag{6.148}$$

Using the Taylor series for V_τ again, we have

$$V_{\tau(j+1)}(r_{\tau(j+1)}) - V_{\tau(j)}(r_{\tau(j)}) = V_{\tau(j)}(r_{\tau(j+1)}) - V_{\tau(j)}(r_{\tau(j)}) - \frac{1}{N}\ln r_{\tau(j+1)} \tag{6.149}$$

$$= O((jN)^{-1}) - \frac{1}{N}\ln r_{\tau(j+1)}. \tag{6.150}$$

Thus, if follows from the last two inequalities that

$$e^{NV_{\tau(j)}(r_{\tau(j)})}\int_{|r-r_{\tau(j)}|>\delta_N} 2r^{2j+1}e^{-Nq(r)}\,dr = \int_{|r-r_{\tau(j)}|>\delta_N} 2\,e^{-N\left(V_{\tau(j+1/2)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,dr \tag{6.151}$$

$$\leq e^{-c_1(\ln(N))^2}\,e^{\frac{N-M}{N}\ln(r_{\tau(j+1/2)})}\int_{|r-r_{\tau(j)}|>\delta_N} 2\,e^{-M\left(V_{\tau(j+1/2)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,dr \tag{6.152}$$

$$= O(e^{c_2(\ln(N))^2}) \tag{6.153}$$

for some constants $c_1, c_2 > 0$. □

Lemma 9. *Consider the case $m_N \leq j \leq N-1$. For the value-contributing integral we have the asymptotic*

$$(1) = e^{-NV_{\tau(j)}}(r_{\tau(j)})\left[\frac{1}{\sqrt{N}}\left(\frac{8\pi r_{\tau(j)}^2}{V^{(2)}(r_{\tau(j)})}\right)^{\frac{1}{2}}\left(1 + \frac{1}{N}\mathfrak{B}_1(r_{\tau(j)}) + \epsilon_{N,1}\right)\right] \tag{6.154}$$

where

$$\epsilon_{N,1} = O(j^{-\frac{3}{2}}(\ln N)^\alpha) \tag{6.155}$$

for some $\alpha > 0$.

Proof. We now need to examine the integral near the critical point. That is,

$$\int_{r_{\tau(j)}-\delta_N}^{r_{\tau(j)}+\delta_N} 2r\,e^{-NV_{\tau(j)}}\,dr \tag{6.156}$$

$$= e^{-NV_{\tau(j)}(r_{\tau(j)})}\int_{-\delta_N}^{\delta_N} 2(r_{\tau(j)}+t)\,e^{-N\left(\frac{1}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})(t)^2+\frac{1}{3!}V^{(3)}_{\tau(j)}(r_{\tau(j)})(t)^3+\frac{1}{4!}V^{(4)}_{\tau(j)}(r_{\tau(j)})(t)^4+O(\tau(j)^{-\frac{3}{2}}t^5)\right)}\,dt \tag{6.157}$$

$$= e^{-NV_{\tau(j)}(r_{\tau(j)})}\int_{-\sqrt{N}\delta_N}^{\sqrt{N}\delta_N} 2\,e^{-\frac{1}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})t^2}(r_{\tau(j)} + \frac{t}{\sqrt{N}})\Bigg(1 - \frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})(t)^3}{6\sqrt{N}} - \frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})(t)^4}{24N} + \frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})^2(t)^6}{72N} + \epsilon'_{N,1}\Bigg)dt \tag{6.158}$$

where $\epsilon_{N,1} = O(j^{-\frac{3}{2}}(\ln N)^{\alpha})$ for some $\alpha > 0$. From here, the expression can be directly identified with Equation 6.72 and the techniques for getting the result are one-to-one with the ones already done in that case. □

Proof. (Theorem 21) This is now only a matter to notice that (2) is absolved in to the error term of (1) and the result follows. □

So far, nothing different has happened, we are dealing with point distant from the origin. Naturally then one might assume, and would correct, that nothing is that different in the two regimes if the points are distant from r_0.

We must now deal with the more delicate case, for $j \leq m_N$, we have the Theorem.

Theorem 22. *For $j < m_N$, we will have the following asymptotic for the norming constant*

$$h_j = \mathrm{e}^{-Nq(0)}\left[N^{-(j+1)}\Gamma(j+1)\left(\frac{1}{2}q''(0)\right)^{-(j+1)}\left(1+O(N^{-\frac{1}{2}}(j+1/2)^{\frac{3}{2}}(\ln N)^3)\right)+\epsilon_N(j)\right] \tag{6.159}$$

where Γ is the usual Gamma function and

$$\epsilon_N(j) = O\left(\mathrm{e}^{-c_2(j+1)(\ln N)^2}\left(\frac{j+1}{N}(\ln(N))^2\right)^{(j+1)\frac{N-M}{N}}\right) \tag{6.160}$$

with $c_2 > 0$ a constant.

Proof. For $j < m_N$, we need to define some $r_\tau^* := r_\tau \ln(N)$ from which we separate the integral in a important part close to the origin and another part which has a lesser order. Consider then the decomposition

$$h_j = \int_0^\infty 2r^{2j+1}\,\mathrm{e}^{-Nq(r)}\,\mathrm{d}r = \int_0^{r^*_{\tau(j+1)}} 2r^{2j+1}\,\mathrm{e}^{-Nq(r)}\,\mathrm{d}r + \int_{r^*_{\tau(j+1)}}^\infty 2r^{2j+1}\,\mathrm{e}^{-Nq(r)}\,\mathrm{d}r. \tag{6.161}$$

Due to strict subharmonicity of Q in a neighborhood of the droplet, r_τ satisfies $r_\tau = O(\tau^{\frac{1}{2}})$ as $\tau \longrightarrow 0$, this is given by Proposition 9. Since V_τ has a global minimum at $r = r_\tau$ and increases for larger values, for all $r > r^*_{\tau(j+1)}$ we have

$$V_{\tau(j+1/2)}(r) \geq V_{\tau(j+1)}(r)(r^*_{\tau(j+1)}) = q(r^*_{\tau(j+1)}) - \frac{2j+1}{N}\ln(r^*_{\tau(j+1)}) \tag{6.162}$$

$$= q(0) + \frac{1}{2}q''(0)(r^*_{\tau(j+1)})^2 - \frac{2j+1}{N}\ln(r^*_{\tau(j+1)}) + O(r^*_{\tau(j+1)})^3. \tag{6.163}$$

Note that $q'(0) = 0$ since $\Delta Q(z) \in (0,\infty)$ near the origin and $\Delta Q(z) \propto q'(0)/r + q''(0)$. Take

$$\mathrm{e}^{-Nq(0)} \int_{r^*_{\tau(j+1)}}^{\infty} 2r^{2j+1}\,\mathrm{e}^{-Nq(r)+Nq(0)}\,\mathrm{d}r = \mathrm{e}^{-Nq(0)} \int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{(2j+1)\ln(r)-Nq(r)+Nq(0)}\,\mathrm{d}r, \quad (6.164)$$

$$= \mathrm{e}^{-Nq(0)} \int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-N\left(-\frac{2}{N}\left(j+\frac{1}{2}\right)\ln(r)-q(r)+q(0)\right)}\,\mathrm{d}r, \quad (6.165)$$

$$= \int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-N(V_{\tau(j+1/2)}(r)-q(0))}\,\mathrm{d}r \quad (6.166)$$

and, using the inequality explicited above

$$\mathrm{e}^{-NV_{\tau(j+1)}(r)+Nq(0)} = \mathrm{e}^{-NV_{\tau(j+1)}(r)+Nq(0)+MV_{\tau(j+1)}(r)-MV_{\tau(j+1)}(r)} \quad (6.167)$$

$$= \mathrm{e}^{-(N-M)V_{\tau(j+1)}(r)+Nq(0)-MV_{\tau(j+1)}(r)} \quad (6.168)$$

$$\leq \mathrm{e}^{-(N-M)V_{\tau(j+1)}(r^*)+Nq(0)-MV_{\tau(j+1)}(r)} \quad (6.169)$$

$$= \mathrm{e}^{-(N-M)\left(V_{\tau(j+1)}(r^*)-q(0)\right)}\,\mathrm{e}^{-M\left(V_{\tau(j+1)}(r)-q(0)\right)} \quad (6.170)$$

then

$$\int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-N(V_{\tau(j+1)}(r)-q(0))}\,\mathrm{d}r \quad (6.171)$$

$$\leq \mathrm{e}^{-(N-M)V_{\tau(j+1)}(r^*)}\,\mathrm{e}^{(N-M)q(0)} \int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-MV_{\tau(j+1)}(r)}\,\mathrm{d}r \quad (6.172)$$

$$= \mathrm{e}^{-(N-M)\left[q(r^*_{\tau(j+1)})-\frac{2j+2}{N}\ln(r^*_{\tau(j+1)})\right]}\,\mathrm{e}^{(N-M)q(0)} \int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-MV_{\tau(j+1)}(r)}\,\mathrm{d}r \quad (6.173)$$

$$= \mathrm{e}^{-(N-M)q(r^*_{\tau(j+1)})}\, {r^*_{\tau(j+1)}}^{(2j+2)\frac{N-M}{N}}\,\mathrm{e}^{(N-M)q(0)} \int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-MV_{\tau(j+1)}(r)}\,\mathrm{d}r \quad (6.174)$$

$$= \mathrm{e}^{-(N-M)\left[\frac{q''(0)}{2}(r^*_{\tau(j+1)})^2+O(r^*_{\tau(j+1)})^3\right]}\, {r^*_{\tau(j+1)}}^{(2j+2)\frac{N-M}{N}} \int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-MV_{\tau(j+1)}(r)}\,\mathrm{d}r \quad (6.175)$$

$$\leq \mathrm{e}^{-\frac{(N-M)}{2}q''(0)(r^*_{\tau(j+1)})^2)}\, {r^*_{\tau(j+1)}}^{(2j+2)\frac{N-M}{N}} \int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-MV_{\tau(j+1)}(r)}\,\mathrm{d}r \quad (6.176)$$

$$= \mathrm{e}^{-c_1\frac{(N-M)}{2}r^2_{\tau(j+1)}\ln(N)^2}\, {r^*_{\tau(j+1)}}^{(2j+2)\frac{N-M}{N}} \int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-MV_{\tau(j+1)}(r)}\,\mathrm{d}r := \epsilon_N(j) \quad (6.177)$$

$$(6.178)$$

for some $c_1 > 0$ and M as given in the other proof such that the integral converges. Since $r_\tau = O(\tau^{\frac{1}{2}})$, there is a constant $c_2 > 0$ such that as N goes to infinity

$$\epsilon_N(j) = O\left(\mathrm{e}^{-c_1\frac{(N-M)}{2}\tau(j+1)\ln(N)^2}\, {r^*_{\tau(j+1)}}^{(2j+2)\frac{N-M}{N}}\right) \quad (6.179)$$

$$= O\left(\mathrm{e}^{-c_2(j+1)\ln(N)^2}\, {r^*_{\tau(j+1)}}^{(2j+2)\frac{N-M}{N}}\right) \quad (6.180)$$

$$= O\left(\mathrm{e}^{-c_2(j+1)(\ln N)^2}\left(\frac{j+1}{N}(\ln(N))^2\right)^{(j+1)\frac{N-M}{N}}\right) \quad (6.181)$$

Therefore we obtain

$$h_j = \int_0^\infty 2r^{2j+1}\, \mathrm{e}^{-Nq(r)}\, \mathrm{d}r = \int_0^{r^*_{\tau(j+1)}} 2r^{2j+1}\, \mathrm{e}^{-Nq(r)}\, \mathrm{d} + \mathrm{e}^{-Nq(0)}\, \epsilon_N(j) \tag{6.182}$$

$$= \int_0^{r^*_{\tau(j+1)}} 2r^{2j+1}\, \mathrm{e}^{-N\left(q(0)+\frac{1}{2}q''(0)r^2\right)+O(N(r^*_{\tau(j+1)})^3)}\, \mathrm{d}r + \mathrm{e}^{-Nq(0)}\, \epsilon_N(j) \tag{6.183}$$

$$= \mathrm{e}^{-Nq(0)} \left[\int_0^{r^*_{\tau(j+1)}} 2r^{2j+1}\, \mathrm{e}^{-\frac{1}{2}q''(0)r^2 N}\, \mathrm{d}r \left(1 + O(N(r^*_{\tau(j+1)})^3)\right) + \epsilon_N(j)\right] \tag{6.184}$$

$$= \mathrm{e}^{-Nq(0)} \left[N^{-(j+1)}\Gamma(j+1)\left(\frac{1}{2}q''(0)\right)^{-(j+1)} \left(1 + O(N^{-\frac{1}{2}}(j+1/2)^{\frac{3}{2}}(\ln N)^3)\right) + \epsilon_N(j)\right] \tag{6.185}$$

where Γ is the usual gamma function with integral representation

$$\Gamma(j+1) = \int_0^\infty t^j\, \mathrm{e}^{-t}\, \mathrm{d}t \tag{6.186}$$

and we used $t = 1/2q''(0)r^2N$.

□

6.3.2.2 Asymptotics on the Summation

So now we need to return to (6.21) and calculate each term for $\sum_{j=0}^{N-1} \ln h_j$ using the expansion in terms of Theorem 21 and 22 for the expression

$$\sum_{j=0}^{N} \ln(h_j) = \sum_{j=0}^{m_N-1} \ln(h_j) + \sum_{j=m_N}^{N-1} \ln(h_j). \tag{6.187}$$

Theorem 23.

$$\sum_{j=0}^{N} \ln(h_j) \tag{6.188}$$

To prove this theorem, we must give the asymptotics for both summations. This means we need both Lemmas 10 and 11. We start by the lower power terms.

Lemma 10. *We have the following asymptotic for the summation of* $\ln(h_j)$ *from* 0 *to* $m_N - 1$*,*

$$\sum_{j=0}^{m_N-1} \ln(h_j) = -m_N N q(0) - \frac{m_N(m_N+1)}{2}\left(\ln(N) + \ln\left(\frac{1}{2}q''(0)\right)\right) + \frac{m_N^2 \ln(m_N)}{2} - \frac{3}{4}m_N^2$$
$$+ \frac{\ln(2\pi)m_N}{2} - \frac{\ln(m_N)}{12} + \zeta'(-1) + O(m_N^{-2} + N^{-\frac{1}{2}(1-5\epsilon)}(\ln(N))^3). \tag{6.189}$$

Proof. The first term in the right is straightforwardly determined, we need only to write, using Theorem 22,

$$\begin{aligned}\sum_{j=0}^{m_N-1}\ln(h_j) &= -m_N Nq(0)-\frac{m_N(m_N+1)}{2}\left(\ln(N)+\ln\left(\frac{1}{2}q''(0)\right)\right)\\ &\quad+\ln(G(m_N+1))+O(N^{-\frac{1}{2}(1-5\epsilon)}(\ln(N))^3)\end{aligned} \tag{6.190}$$

$$\begin{aligned}&= -m_N Nq(0)-\frac{m_N(m_N+1)}{2}\left(\ln(N)+\ln\left(\frac{1}{2}q^{(2)}(0)\right)\right)+\frac{m_N^2\ln(m_N)}{2}-\frac{3}{4}m_N^2\\ &\quad+\frac{\ln(2\pi)m_N}{2}-\frac{\ln(m_N)}{12}+\zeta'(-1)+O(m_N^{-2}+N^{-\frac{1}{2}(1-5\epsilon)}(\ln(N))^3)\end{aligned} \tag{6.191}$$

where G is the Barnes G-function defined recursively as

$$G(z+1)=\Gamma(z)G(z) \tag{6.192}$$

and we have used the asymptotic for the G-Barnes function

$$\ln(G(z+1))=\frac{z^2\ln(z)}{2}-\frac{3}{4}z^2+\frac{\ln(2\pi)z}{2}-\frac{\ln(z)}{12}+\zeta'(-1)+O(\frac{1}{z^2}). \tag{6.193}$$

□

Lemma 11. *We have the following asymptotic for the summation of* $\ln(h_j)$ *from* m_N *to* $N-1$,

$$\sum_{j=m_N}^{N-1}\ln(h_j)= \tag{6.194}$$

The second summation is a bit more detailed, we use the previously defined asymptotics in Theorem 21,

$$\ln(h_j)=-NV_{\tau(j)}(r_{\tau(j)})+\frac{1}{2}\ln\left(\frac{8\pi r_{\tau(j)}^2}{NV^{(2)}(r_{\tau(j)})}\right)+\frac{1}{N}\mathfrak{B}_1(r_{\tau(j)})+O(j^{-\frac{3}{2}}(\ln(N))^{\alpha}) \tag{6.195}$$

and need to determine three things in order to get the full summation,

$$\sum_{j=m_N}^{N-1}V_{\tau(j)}(r_{\tau(j)}),\qquad \sum_{j=m_N}^{N-1}\ln V^{(2)}(r_{\tau(j)})\quad\text{and}\quad \sum_{j=m_N}^{N-1}\ln(r_{\tau(j)}). \tag{6.196}$$

Proposition 10. *The following asymptotic holds under the same context as Lemma 11.*

$$\begin{aligned}\sum_{j=m_N}^{N-1}V_{\tau(j)}(r_{\tau(j)}) &= NI_Q[\mu_Q]-U_{\mu_Q}(0)-\frac{1}{6N}\ln(r_1)-m_Nq(0)-\frac{3}{4}\frac{m_N^2}{N}\\ &\quad+\frac{1}{2N}\left(m_N^2-m_N+\frac{1}{6}\right)\ln\left(\frac{4m_N}{NV^{(2)}(0)}\right)+\frac{m_N}{2N}+O(N^{-1/2(3-5\epsilon)}).\end{aligned} \tag{6.197}$$

Here,

$$I_Q[\mu_Q] = \left[\frac{1}{4}\int_{r_0}^{r_1} q(s)sV^{(2)}_{\tau(j)}(s)\mathrm{d}s - \ln(r_1) + \frac{1}{2}q(r_1)\right] \tag{6.198}$$

and

$$2U_{\mu_Q}(0) = q(r_1) - 2\ln(r_1) - q(0). \tag{6.199}$$

Proof. For the first we proceed as before and perform the Euler-Maclaurin formula.

$$\sum_{j=m_N}^{N-1} V_{\tau(j)}(r_{\tau(j)}) = \int_{m_N}^{N} V_{\tau(j)}(r_{\tau(j)})\mathrm{d}t - \frac{1}{2}\left(V_{\tau(j)}(r_{\tau(j)}) - V_{\tau(m_N)}(r_{\tau(m_N)})\right) + \\ \frac{1}{12}\left[\partial_t V_{\tau(j)}(r_{\tau(j)})|_{t=N} - \partial_t V_{\tau(j)}(r_{\tau(j)})|_{t=m_N}\right] + O(N^{-3}). \tag{6.200}$$

We then are able to rewrite, setting $s = r_{\tau(j)}$,

$$\begin{aligned} \int_{m_N}^{N} V_{\tau(t)}(r_{\tau(t)})\mathrm{d}t &= \int_{m_N}^{N} (q(r_{\tau(t)}) - 2\tau(t)\ln(r_{\tau(t)}))\mathrm{d}t && (6.201)\\ &= \frac{1}{2}\int_{r_{\tau(m_N)}}^{r_1} (q(s) - sq'(s)\ln(s))sV^{(2)}_{\tau}(s)N\mathrm{d}s && (6.202)\\ &= N\left[\frac{1}{2}\int_{r_{\tau(m_N)}}^{r_1} q(s)sV^{(2)}_{\tau}(s)\mathrm{d}s - \frac{1}{2}\int_{r_{\tau(m_N)}}^{r_1} s^2q'(s)\ln(s)V^{(2)}_{\tau}(s)\mathrm{d}s\right] && (6.203) \end{aligned}$$

and finally using that $r_1q'(r_1) = 2$ we can apply the integration by parts on the second term and get

$$\begin{aligned} \frac{1}{2}\int_{r_{\tau(m_N)}}^{r_1} s^2q'(s)\ln(s)V^{(2)}_{\tau}(s)\mathrm{d}s &= \frac{1}{2}\int_{r_{\tau(m_N)}}^{r_1} s^2q'(s)\ln(s)\frac{1}{s}(sq'(s))'\mathrm{d}s && (6.204)\\ &= \frac{1}{4}\int_{r_{\tau(m_N)}}^{r_1} ((sq'(s))^2)'\ln(s)\mathrm{d}s && (6.205) \end{aligned}$$

and then, by integration by parts

$$\begin{aligned} &= \frac{1}{4}(sq'(s))^2\ln(s)|_{r_{\tau(m_N)}}^{r_1} - \frac{1}{4}\int_{r_{\tau(m_N)}}^{r_1} (sq'(s))^2\frac{1}{s}\mathrm{d}s && (6.206)\\ &= \ln(r_1) - \frac{1}{4}\int_{r_{\tau(m_N)}}^{r_1} sq'(s)^2\mathrm{d}s - \frac{m_N^2}{N^2}\ln(r_{\tau(m_N)}) && (6.207)\\ &= \ln(r_1) - \frac{1}{4}(sq'(s)q(s))|_{r_{\tau(m_N)}}^{r_1} + \frac{1}{4}\int_{r_{\tau(m_N)}}^{r_1} q(s)(sq'(s))'\mathrm{d}s - \frac{m_N^2}{N^2}\ln(r_{\tau(m_N)}) && (6.208)\\ &= \ln(r_1) - \frac{1}{2}q(r_1) + \frac{1}{4}\int_{r_{\tau(m_N)}}^{r_1} q(s)(sq'(s))'\mathrm{d}s - \frac{m_N^2}{N^2}\ln(r_{\tau(m_N)}) + \frac{m_N}{2N}q(r_{\tau(m_N)}) && (6.209) \end{aligned}$$

Leaving us with

$$\int_{m_N}^{N} V_{\tau(j)}(r_{\tau(j)})\mathrm{d}t = N\left[\frac{1}{4}\int_{r_{\tau(m_N)}}^{r_1} q(s)sV_\tau^{(2)}(s)\mathrm{d}s - \ln(r_1) + \frac{1}{2}q(r_1) - \frac{m_N^2}{N^2}\ln(r_{\tau(m_N)}) + \frac{m_N}{2N}q(r_{\tau(m_N)})\right] \tag{6.210}$$

$$:= NI_Q[\mu_Q] - \frac{1}{4}\int_{r_0}^{r_{\tau(m_N)}} q(s)sV_\tau^{(2)}(s)\mathrm{d}s - \frac{m_N^2}{N}\ln(r_{\tau(m_N)}) + \frac{m_N}{2}q(r_{\tau(m_N)}) \tag{6.211}$$

where, because of Proposition 9,

$$\ln(r_{\tau(m_N)}) = \frac{1}{2}\ln\left(\frac{4\tau(m_N)}{V^{(2)}(0)}\right) + O(\tau(m_N)^{\frac{1}{2}}) = \frac{1}{2}\ln\left(\frac{4\tau(m_N)}{V^{(2)}(0)}\right) + O(N^{-\frac{1}{2}(1-\epsilon)}), \tag{6.212}$$

$$q(r_{\tau(m_N)}) = q(0) + \frac{1}{2}q''(0)(r_{\tau(m_N)})^2 + O((\tau(m_N)^{\frac{3}{2}})) = q(0) + \tau(m_N) + O(N^{-\frac{3}{2}(1-\epsilon)}) \tag{6.213}$$

and, using again integration by parts

$$\frac{1}{4}\int_{r_0}^{r_{\tau(m_N)}} q(s)sV_\tau^{(2)}(s)\mathrm{d}s = \frac{1}{4}\int_{r_0}^{r_{\tau(m_N)}} q(s)(sq'(s))'\mathrm{d}s \tag{6.214}$$

$$= \frac{1}{2}\tau(m_N)q(r_{\tau(m_N)}) - \frac{1}{4}\int_0^{r_{\tau(m_N)}} s(q'(s))^2\mathrm{d}s \tag{6.215}$$

$$= \frac{1}{2}\tau(m_N)q(r_{\tau(m_N)}) - \frac{1}{4}(q''(0))^2\int_0^{r_{\tau(m_N)}} s^3\mathrm{d}s + O(r_{\tau(m_N)}^5) \tag{6.216}$$

$$= \frac{1}{2}\tau(m_N)q(r_{\tau(m_N)}) - \frac{1}{4}\tau(m_N)^2 + O(N^{-5/2(1-\epsilon)}). \tag{6.217}$$

Finally,

$$\begin{aligned}\int_{m_N}^{N} V_{\tau(t)}(r_{\tau(t)})\mathrm{d}t &= NI_Q[\mu_Q] - \frac{1}{2}\tau(m_N)q(r_{\tau(m_N)}) - \frac{1}{4}\tau(m_N)^2 + O(N^{-5/2(1-\epsilon)}) \\ &\quad - \frac{m_N^2}{2N}\ln\left(\frac{4\tau(m_N)}{V^{(2)}(0)}\right) + O(N^{-\frac{3}{2}(1-\epsilon)}) + \frac{m_N}{2}(q(0)+\tau(m_N)) + O(N^{-\frac{1}{2}(1-\epsilon)})\end{aligned} \tag{6.218}$$

$$= NI_Q[\mu_Q] - m_Nq(0) - \frac{3}{4}\frac{m_N^2}{N} + \frac{m_N^2}{2N}\ln\left(\frac{4m_N}{NV^{(2)}(0)}\right) + O\left(N^{-\frac{1}{2}(3-5\epsilon)}\right) \tag{6.219}$$

For the other terms in the Euler-Maclaurin formula we have by the results above

$$V_{\tau(N)}(r_{\tau(N)}) - V_{\tau(m_N)}(r_{\tau(m_N)}) = q(r_1) - 2\ln(r_1) - q(r_{\tau(m_N)}) + 2\tau(m_N)\ln(r_{\tau(m_N)}) \tag{6.220}$$

$$= q(r_1) - 2\ln(r_1) - q(0) - \frac{m_N}{N} + \frac{m_N}{N}\ln\left(\frac{4m_N}{NV^{(2)}(0)}\right) + O(N^{-3/2(1-\epsilon)}) \tag{6.221}$$

$$= 2U_{\mu_Q}(0) - \frac{m_N}{N} + \frac{m_N}{N}\ln\left(\frac{4m_N}{NV^{(2)}(0)}\right) + O(N^{-3/2(1-\epsilon)}) \tag{6.222}$$

Furthermore,

$$\partial_t(V_{\tau(t)}(r_{\tau(t)}))|_{t=N} - \partial_t(V_{\tau(t)}(r_{\tau(t)}))|_{t=m_N} = \frac{2}{N}(\ln(r_{\tau(m_N)}) - \ln(r_1)) \tag{6.223}$$

$$= \frac{1}{N}\ln\left(\frac{4m_N}{NV^{(2)}(0)}\right) - \frac{2}{N}\ln(r_1) + O(N^{-1/2(3-\epsilon)}). \tag{6.224}$$

And, gathering terms, we get the lemma. □

Proposition 11. *The following asymptotic holds under the same context as Lemma 11.*

$$\sum_{j=m_N}^{N-1} \ln V^{(2)}(r_{\tau(j)}) = \frac{N}{2}\int_{r_{\tau(m_N)}}^{r_1} \ln(V^{(2)}(s))(sq'(s))'\mathrm{d}s - \frac{1}{2}\ln\left(\frac{V_1^{(2)}(r_1)}{V_{r_{\tau(m_N)}}^{(2)}(r_{\tau(m_N)})}\right) + O(N^{-1}) \tag{6.225}$$

Proof. Again, we start with the Euler-Maclaurin formula

$$\sum_{j=m_N}^{N-1} \ln V_{\tau(j)}^{(2)}(r_{\tau(j)}) = \int_{m_N}^{N} \ln V_{\tau(t)}^{(2)}(r_{\tau(t)})\mathrm{d}t - \frac{1}{2}\ln\left(\frac{V_1^{(2)}(r_1)}{V_{\tau(m_N)}^{(2)}(r_{\tau(m_N)})}\right) + \frac{1}{12}\left[\partial_t \ln V_{\tau(t)}^{(2)}(r_{\tau(t)})|_{t=N} - \partial_t \ln V_{\tau(t)}^{(2)}(r_{\tau(t)})|_{t=m_N}\right] + O(N^{-3}) \tag{6.226}$$

We first tackle the integral on the right hand side. Changing variables $s = r_{\tau(t)}$ and noting that $\ln(V^{(2)}(r_{\tau(m_N)})) = \ln(V^{(2)}(0)) + O(r_{\tau(m_N)})$, we would have

$$\int_{m_N}^{N} \ln V_{\tau(t)}^{(2)}(r_{\tau(t)})\mathrm{d}t = \frac{N}{2}\int_{r_{\tau(m_N)}}^{r_1} \ln(V^{(2)}(s))(sq'(s))'\mathrm{d}s. \tag{6.227}$$

And, noticing that

$$\partial_t \ln V_{\tau(t)}^{(2)}(r_{\tau(t)}) = \partial_V\left(\ln V_{\tau(t)}^{(2)}(r_{\tau(t)})\right)\partial_r\left(V_{\tau(t)}^{(2)}(r_{\tau(t)})\right)\partial_\tau\left(r_{\tau(t)}\right) = \frac{\left(\partial_r V^{(2)}\right)(r_{\tau(t)})}{r_{\tau(t)}\left(V^{(2)}(r_{\tau(t)})\right)^2}, \tag{6.228}$$

we have for the third term on the right hand side,

$$\partial_t \ln V_{\tau(t)}^{(2)}(r_{\tau(t)})|_{t=N} - \partial_t \ln V_{\tau(t)}^{(2)}(r_{\tau(t)})|_{t=m_N} = \frac{1}{2N}\left(\frac{\left(\partial_r V^{(2)}\right)(r_1)}{r_1\left(V^{(2)}(r_1)\right)^2} - \frac{\left(\partial_r V^{(2)}\right)(r_{\tau(m_N)})}{r_{\tau(m_N)}\left(V^{(2)}(r_{\tau(m_N)})\right)^2}\right) \tag{6.229}$$

$$= O(N^{-1}) \tag{6.230}$$

then,

$$\sum_{j=0}^{N-1} \ln V_{\tau(j)}^{(2)}(r_{\tau(j)}) = \frac{N}{2}\int_{r_{\tau(m_N)}}^{r_1} \ln(V^{(2)}(s))(sq'(s))'\mathrm{d}s - \frac{1}{2}\ln\left(\frac{V_1^{(2)}(r_1)}{V_{r_{\tau(m_N)}}^{(2)}(r_{\tau(m_N)})}\right) + O(N^{-1}) \tag{6.231}$$

□

Proposition 12. *The following asymptotic holds under the same context as Lemma 11.*

$$\sum_{j=m_N}^{N-1} \ln(r_{\tau(j)}) \tag{6.232}$$

Proof. content... □

6.3.2.3 Free Energy Disk

We can finally prove Theorem 20.

Proof. (Theorem 20) From

$$\sum_{j=0}^{N-1} \ln(h_j) = \tag{6.233}$$

We can get the asymptotic expansion for the Free energy using that

$$\ln(N!) = N\ln(N) - N + \frac{1}{2}\ln(N) + \frac{1}{2}\ln(2\pi) + O(N^{-1}) \tag{6.234}$$

that is,

$$\ln \mathcal{Z}_N = \ln N! - 2\sum_{j=0}^{N-1} \ln h_j \tag{6.235}$$

$$= \tag{6.236}$$

$$= \tag{6.237}$$

□

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